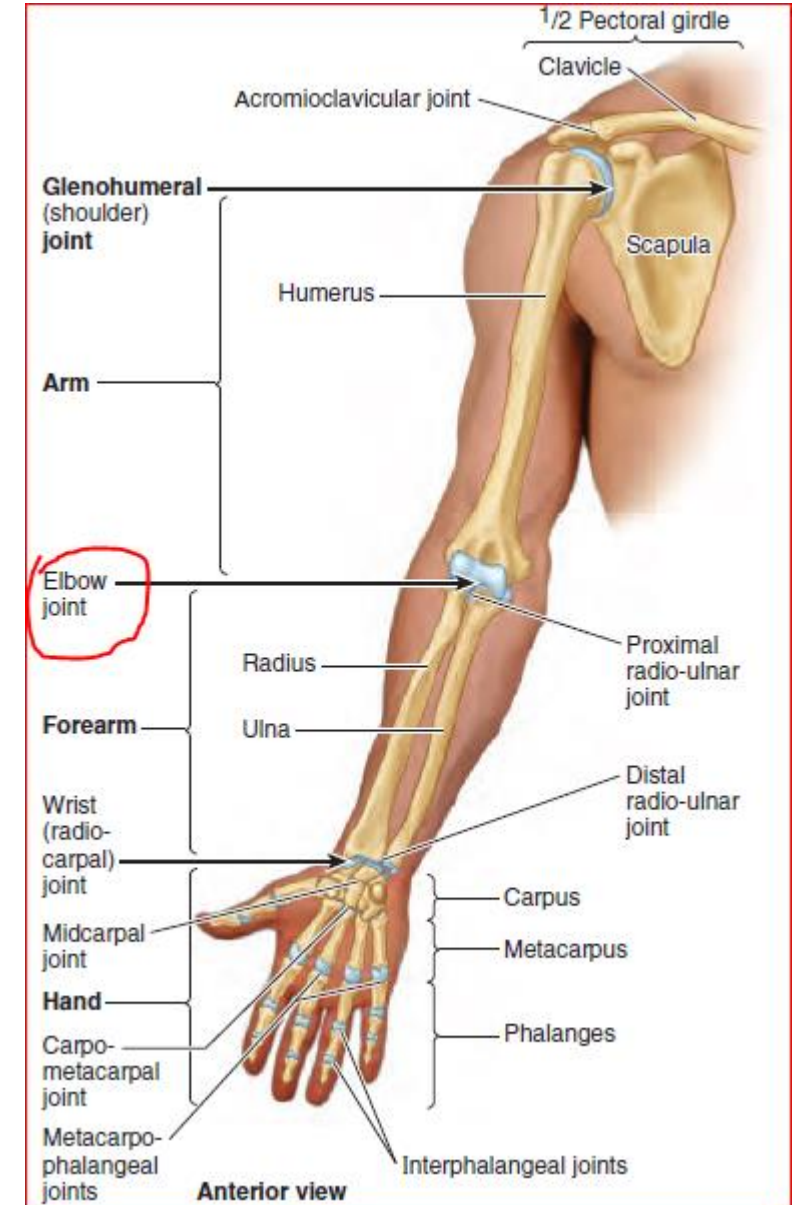


• **Anatomy of upper limb:**

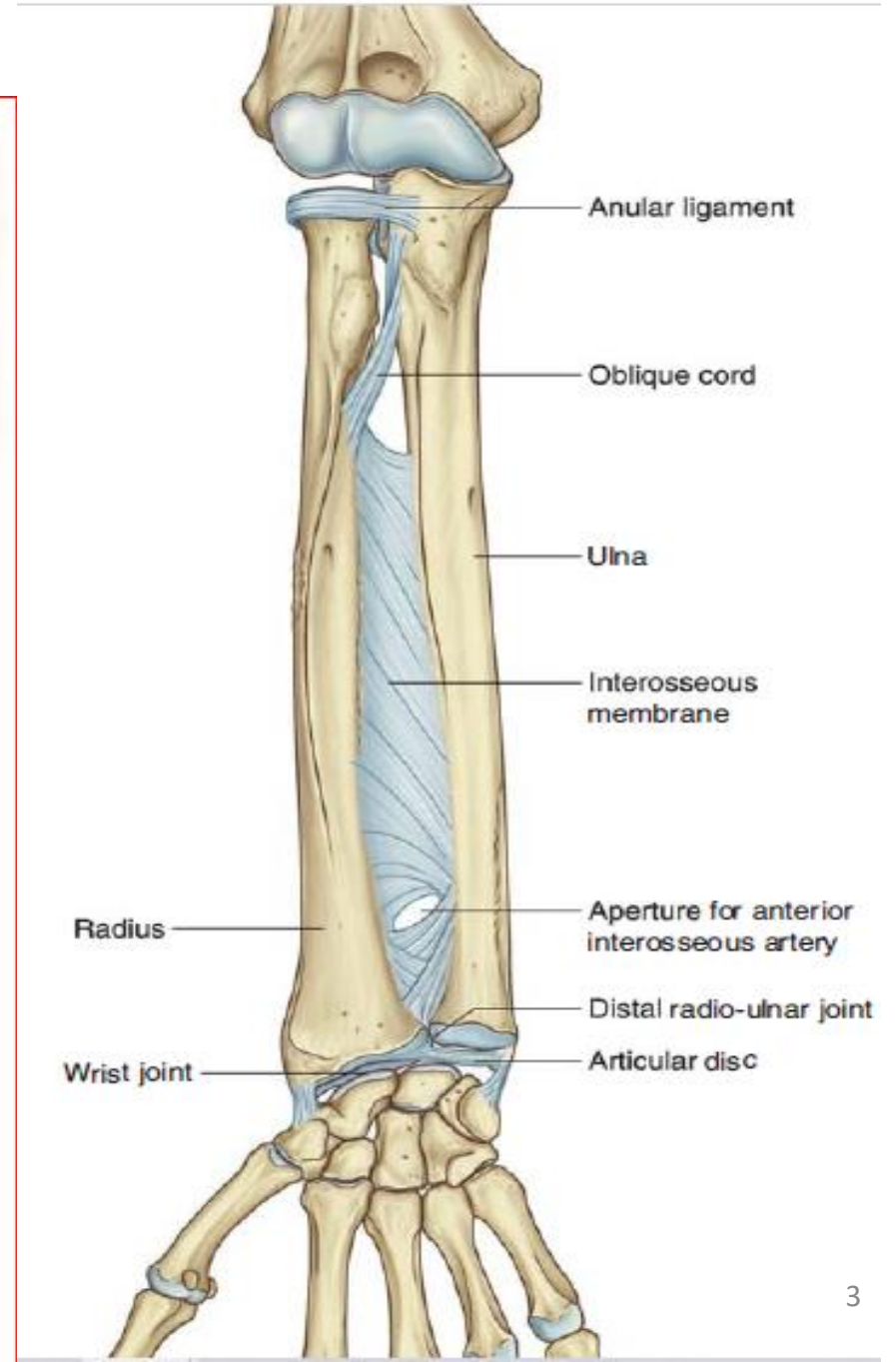
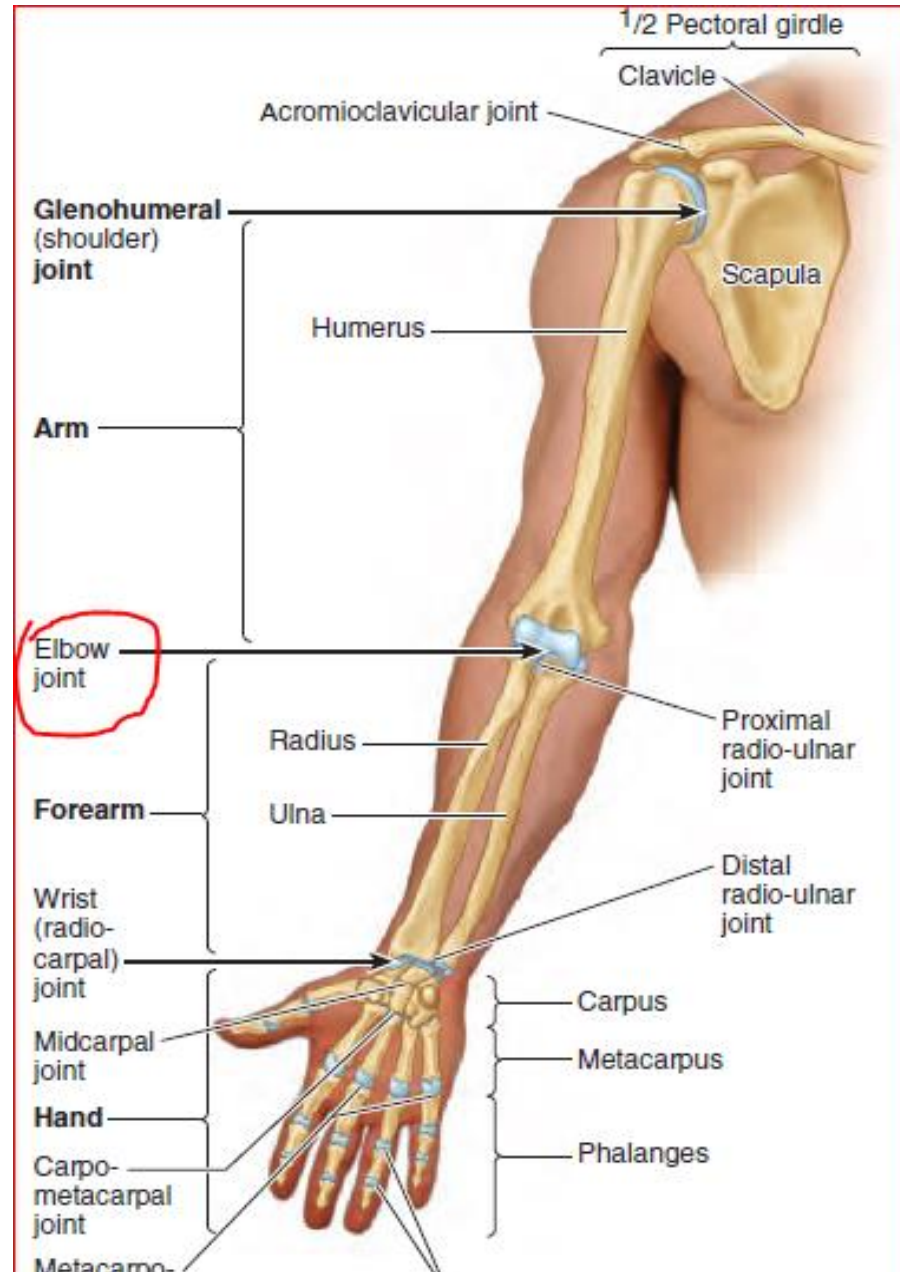
- ✓ Introduction to the upper limb
- ✓ Bones of the upper limb
- ✓ Superficial structures of upper limb
- ✓ Axioappendicular Muscles
- ✓ Axilla
- ✓ Brachial plexus
- ✓ Arm
- ✓ Cubital fossa
- ✓ Forearm
- ✓ Hand
- ✓ Surface anatomy the upper limb
- ✓ Joints of the upper limb

Forearm

- The forearm is the part of the upper limb between the elbow and wrist joints.
- It contains two bones, the *radius* and *ulna* are connected by the **interosseous** membrane, **superior radioulnar joint** and **inferior radioulnar joint**.
- The interosseous membrane also provides attachments for muscles



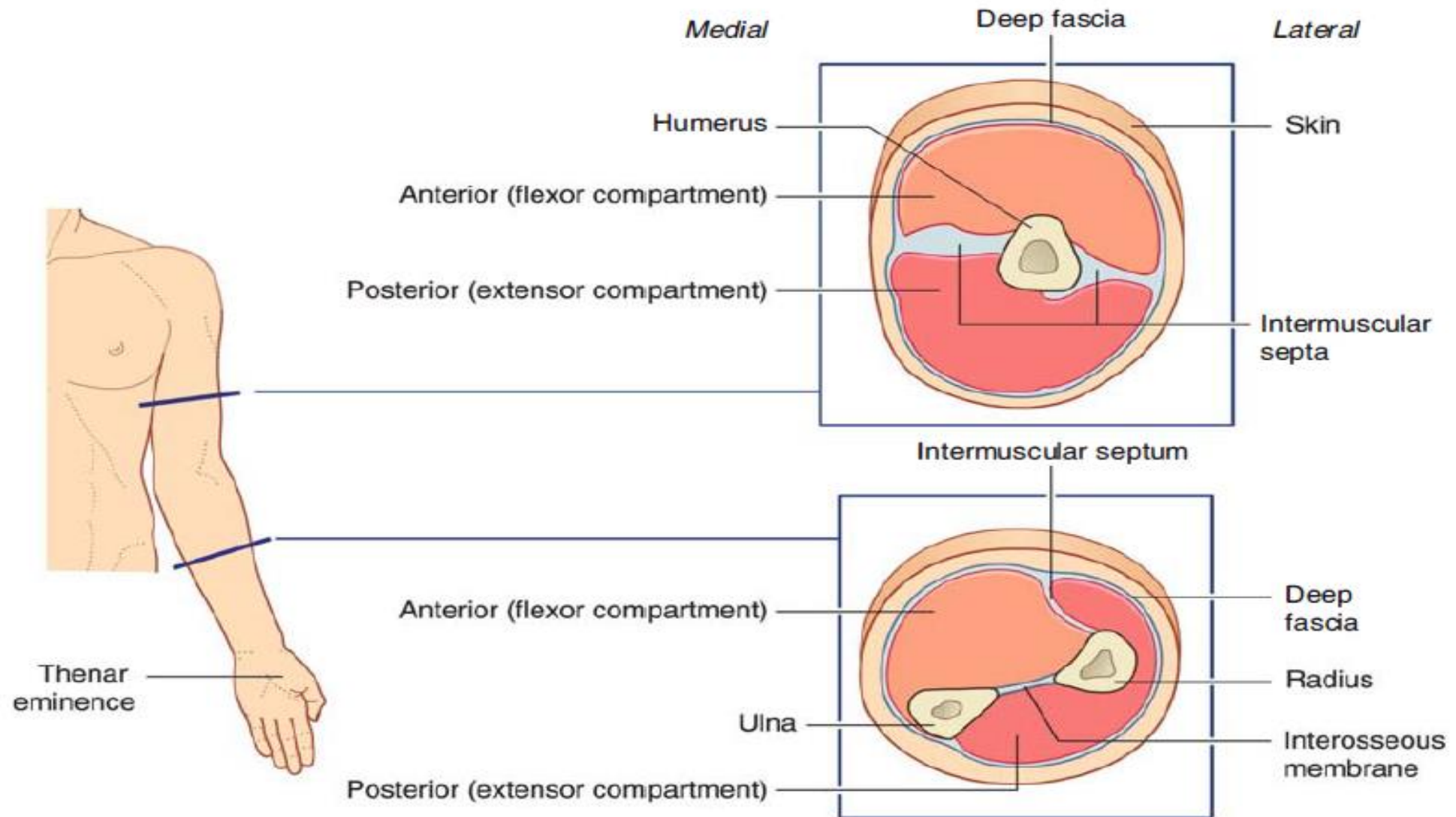
Forearm,,



Muscles of Forearm

- The forearm is divided into *anterior & posterior* compartments separated by *radius, ulna & interosseus membrane*.
- ➔ The **flexors** and **pronators** of the forearm are in the anterior compartment and are served **mainly** by the **median nerve** the one and a half exceptions are innervated by the *ulnar nerve*.
- ✚ The **extensor** and **supinators** are in posterior compartment and are **all** innervated by the **radial nerve**.
- Generally, flexors lie anteriorly and extensors posteriorly; however, the anterior and posterior aspects of the distal humerus are occupied by the chief flexors and extensors of the elbow.
- **These act on the elbow & wrist joints and those of the fingers.**

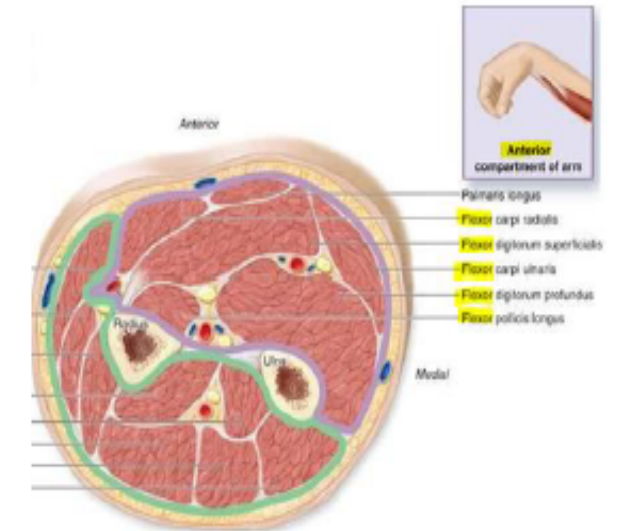
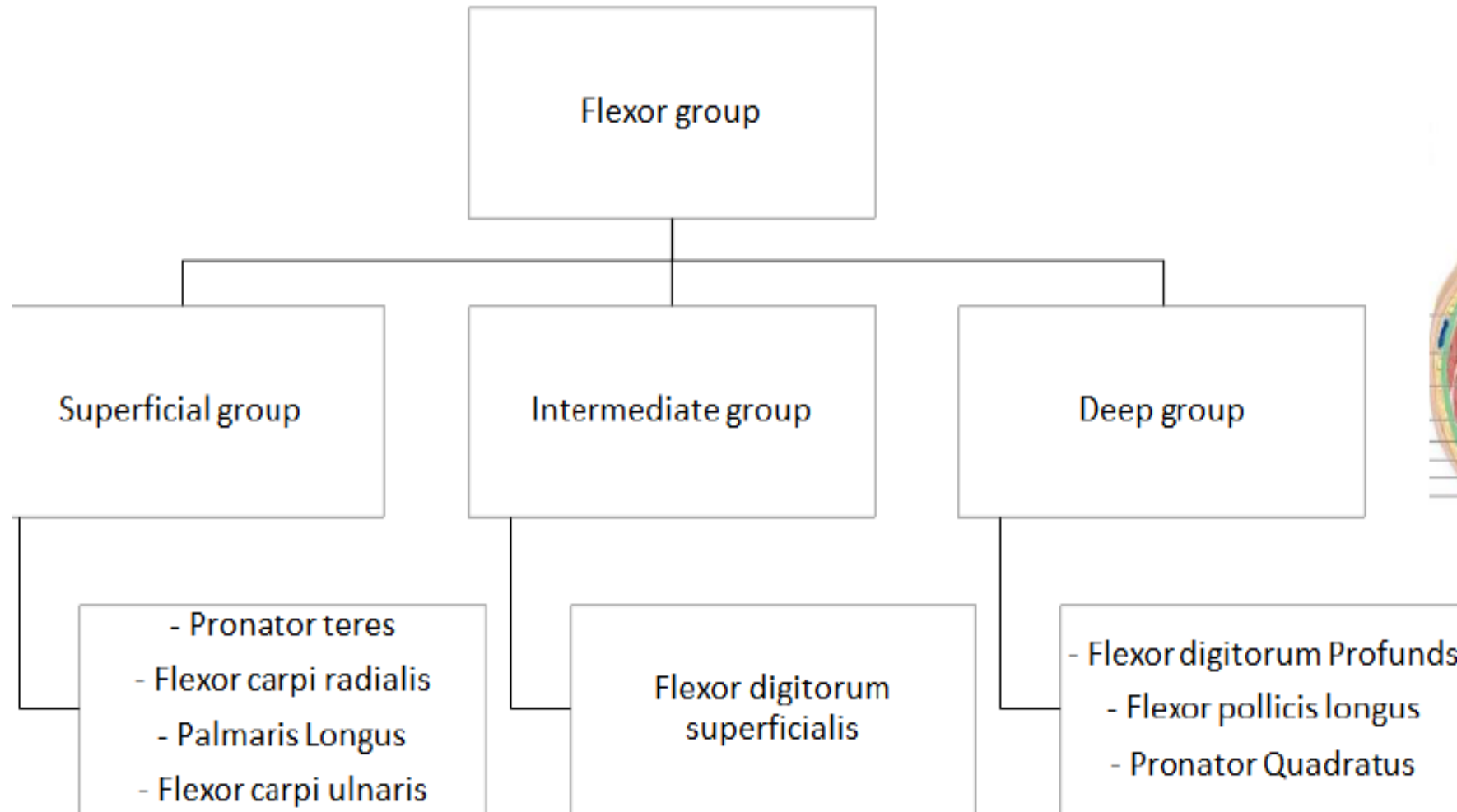
Muscles of Forearm,,



Muscle components in the arm and forearm.

Muscles on anterior compartment of forearm

- Act on the elbow & wrist joints and those of the fingers
- Form **fleshy** masses in the **proximal** part and become **tendons** in the **distal** part of the forearm.



Muscles on anterior compartment of forearm,,,

- **Superficial Muscles:**

- Have a common origin by **common flexor tendon** from the medial epicondyle of the humerus

- They are arranged from lateral to medial:

1. **Pronator teres**

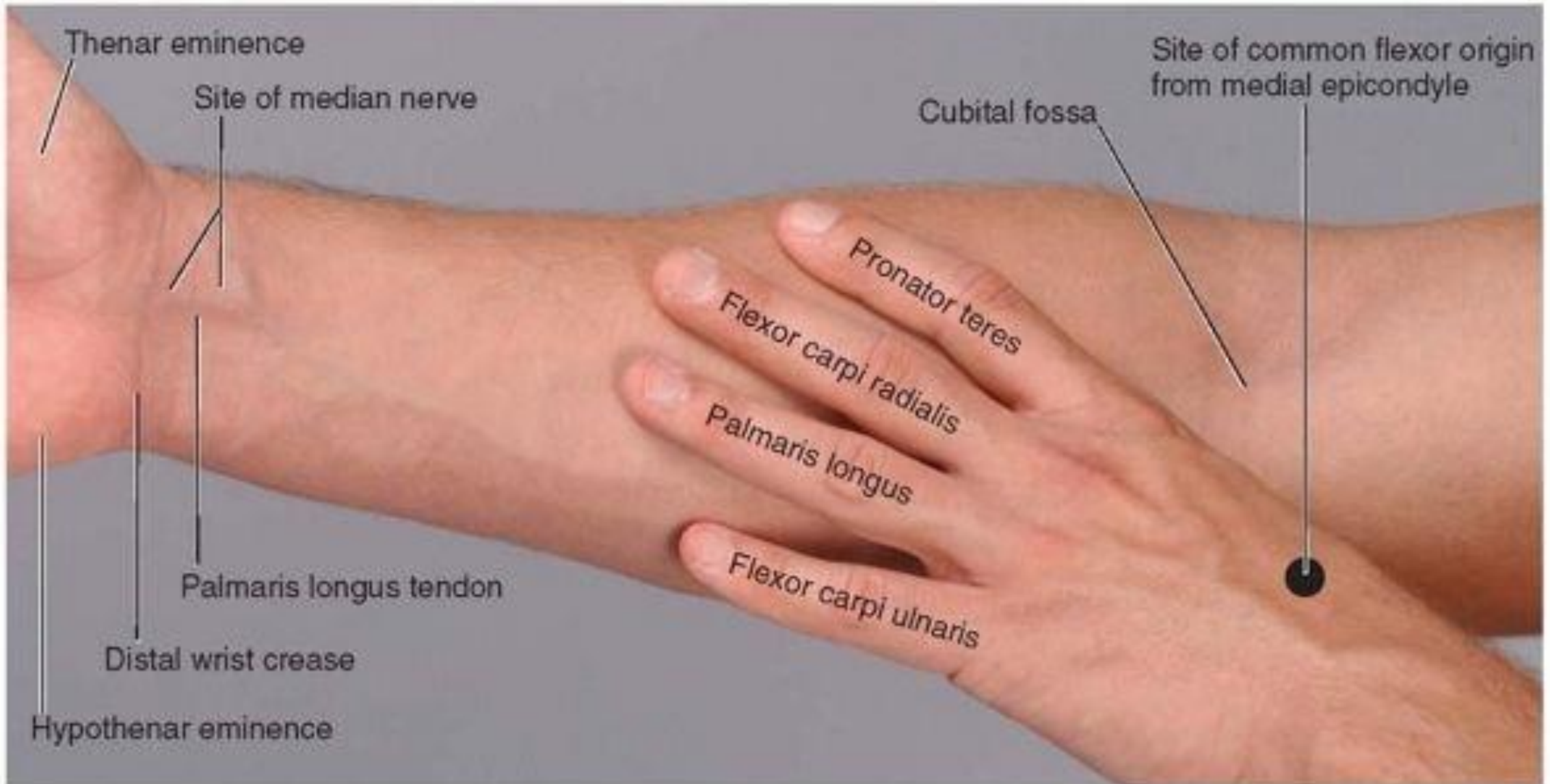
2. **Flexor Carpi radialis**

3. **Palmaris longus**

4. **Flexor Carpi ulnaris**

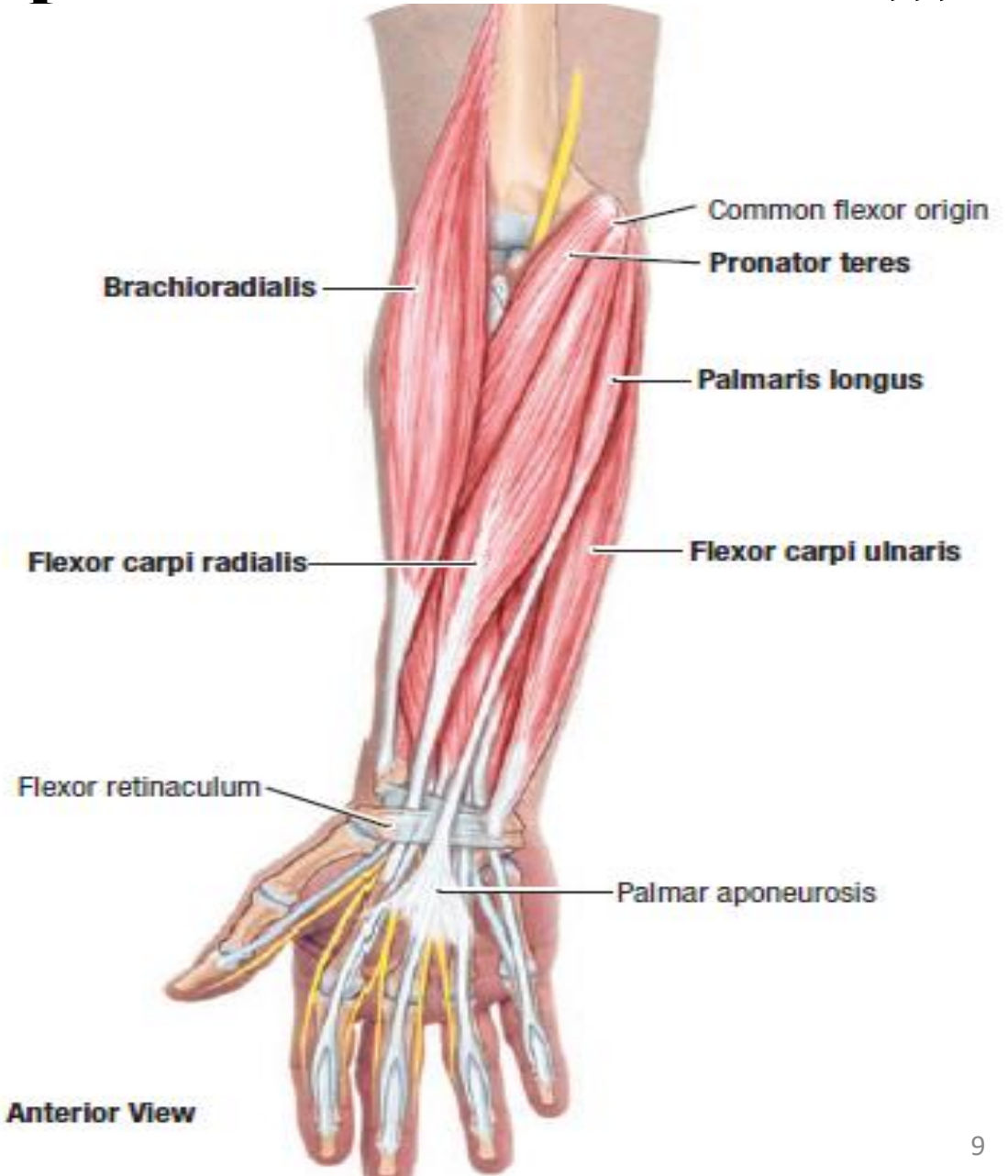
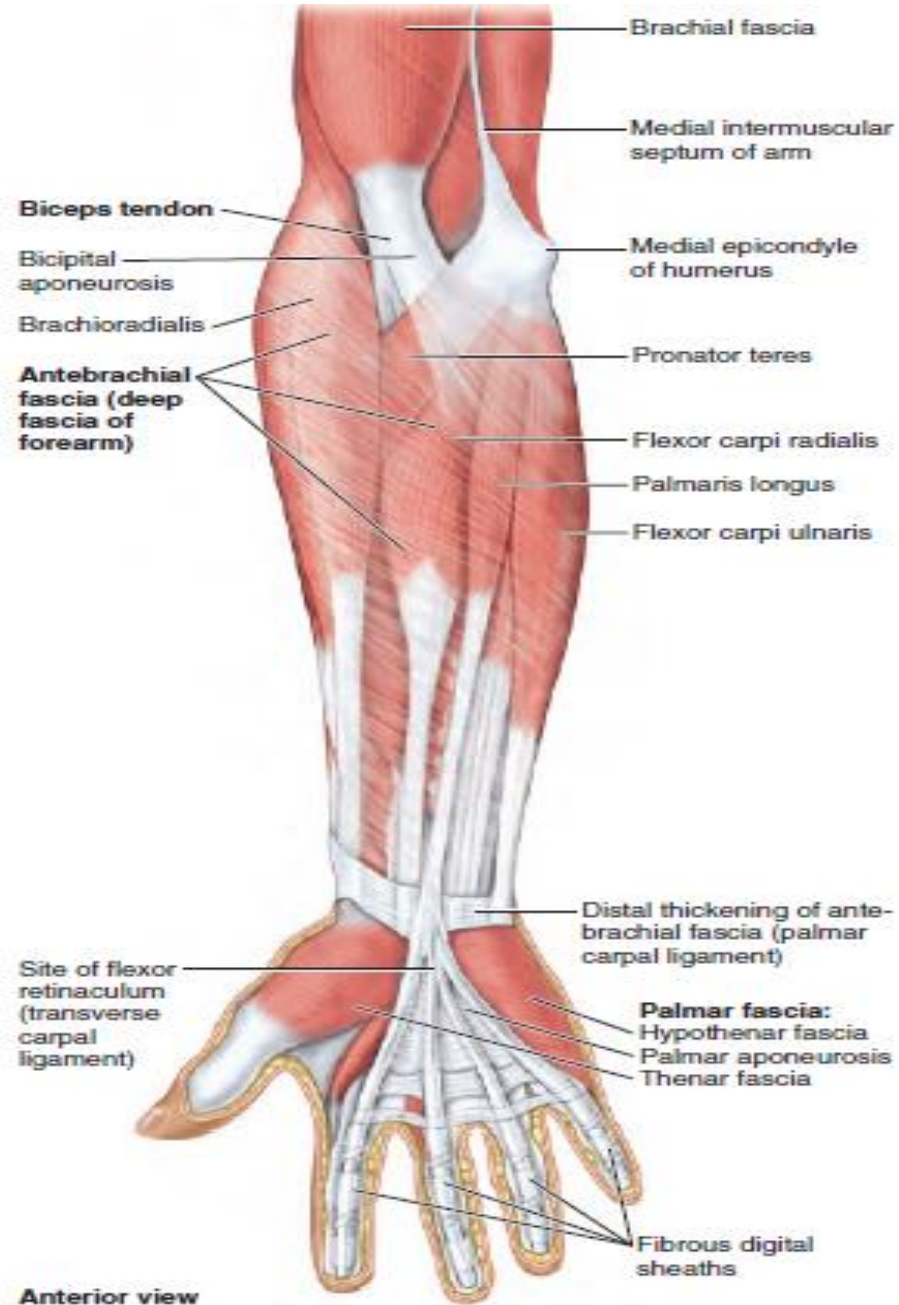
❖ All these muscles are supplied by *the median nerve* and its branches, except the flexor carpi **ulnaris** and the **medial part of the flexor digitorum profundus**, which are supplied by the ulnar nerve.

Muscles on anterior compartment of forearm,,



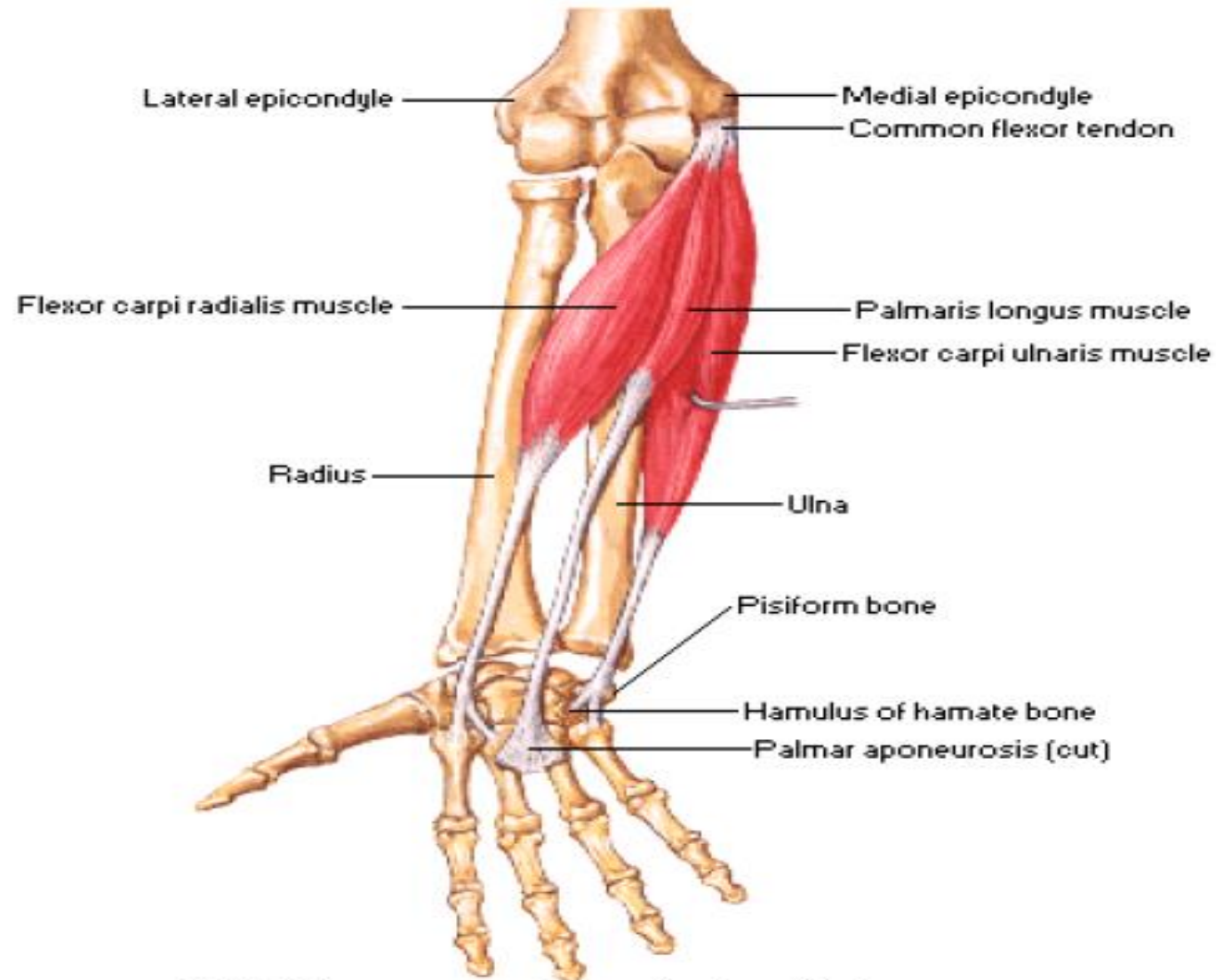
(A) Anterior view of supinated forearm

Muscles on anterior compartment of forearm,,



Muscles on anterior compartment of forearm,,

Flexors of Wrist



Right forearm: anterior (palmar) view

Muscles on anterior compartment of forearm,,,

Muscle	Proximal Attachment	Distal Attachment	Innervation ^a	Main Action
Superficial (first) layer				
Pronator teres				
Ulnar head	Coronoid process	Middle of convexity of lateral surface of radius	Median nerve (C6, C7)	Pronates and flexes forearm (at elbow)
Humeral head	Medial epicondyle of humerus (common flexor origin)			
Flexor carpi radialis (FCR)		Base of 2nd metacarpal		Flexes and abducts hand (at wrist)
Palmaris longus		Distal half of flexor reti-naculum and apex of palmar aponeurosis		Flexes hand (at wrist) and tenses palmar aponeurosis
Flexor carpi ulnaris (FCU)				
Humeral head				
Ulnar head	Olecranon and posterior border of ulna (via aponeurosis)	Pisiform, hook of hamate, 5th metacarpal	Ulnar nerve (C7, C8)	Flexes and adducts hand (at wrist)

Pronator Teres

- Is the most lateral of the superficial forearm flexors
- Its lateral border forms the medial boundary of the cubital fossa

ORIGIN:

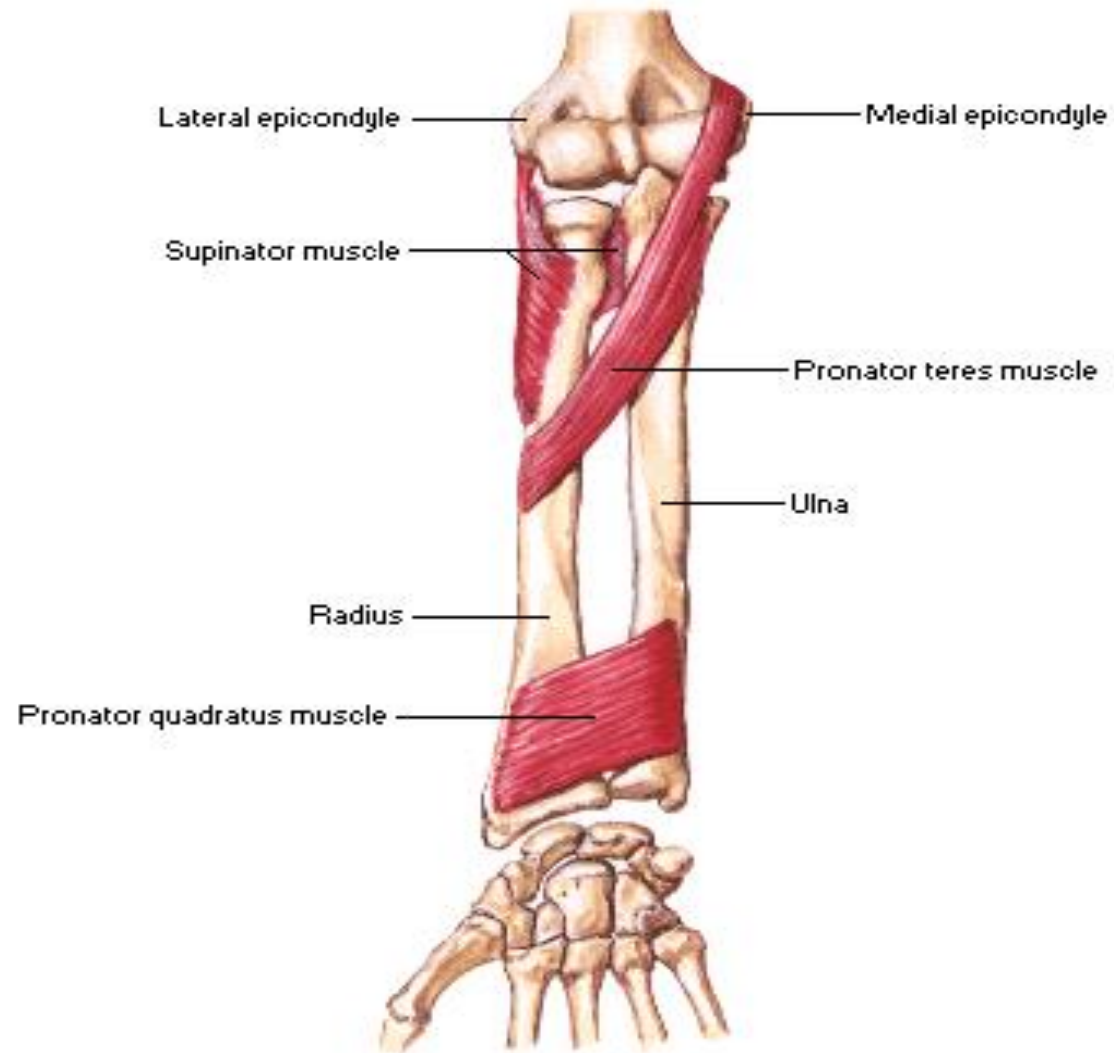
- **humeral head:** from common flexor tendon
- **ulnar head:** from medial side of ulnar coronoid process

INSERTION:

- Into the middle of the lateral surface of the shaft of the radius
- **Nerve Supply:** the median N.
- **ACTION:** pronates the forearm, flexion of the elbow joint

Pronator Teres,,,

Rotators of Radius - Supination



Flexor Carpi Radialis

- Is a long fusiform muscle, medial to the pronator teres
- In the middle of the forearm, its fleshy belly is replaced by a long, flattened tendon that becomes cord-like as it approaches the wrist
- To reach its distal attachment, the FCR tendon passes through a canal in the lateral part of the flexor retinaculum and through a vertical groove in the trapezium in its own synovial tendinous sheath of the flexor carpi radialis
- Its tendon is a good guide to the radial artery, which lies just lateral to it

Flexor Carpi Radialis,,,

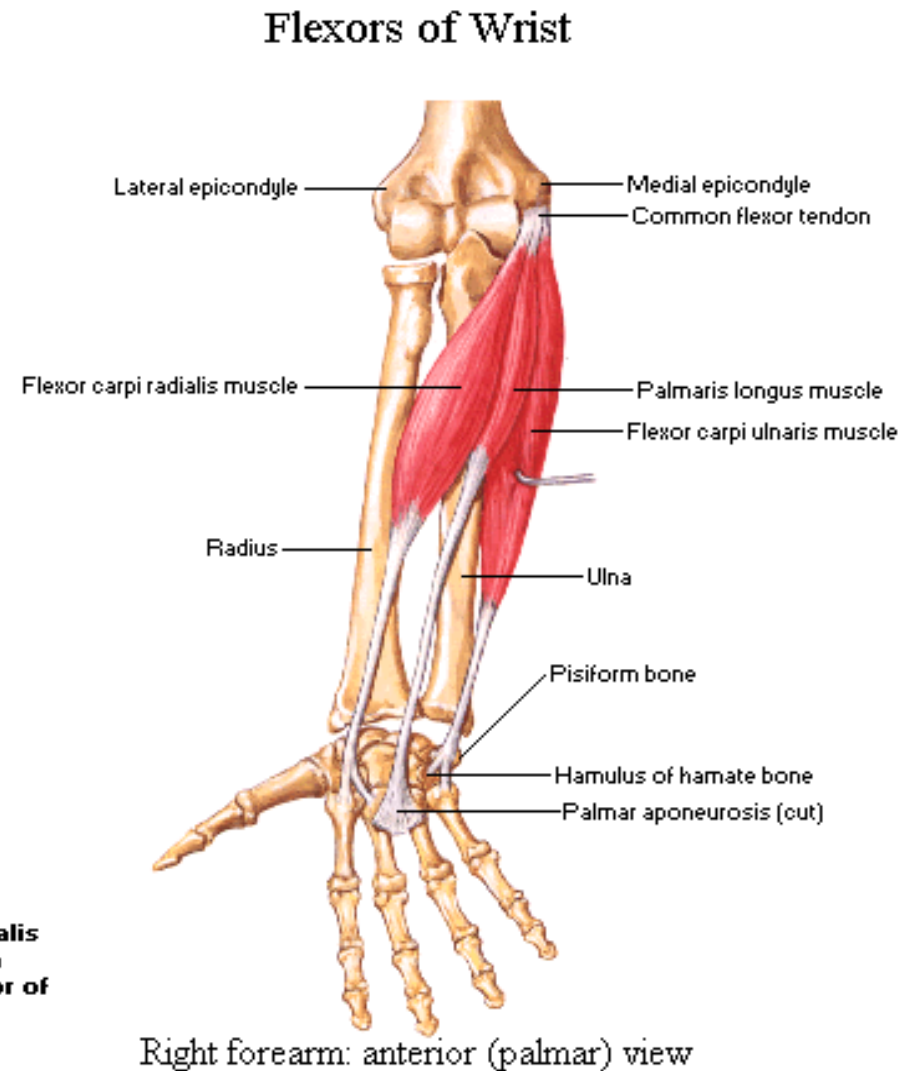
❑ **Origin:** Medial epicondyle of humerus (common flexor origin)

❑ **Insertion:** Base of 2nd metacarpal

❑ **Main action:** Flexes and abducts hand (at wrist)

❑ **Innervation:** Median nerve (C6, C7)

Note: brachioradialis muscle not shown because it is flexor of

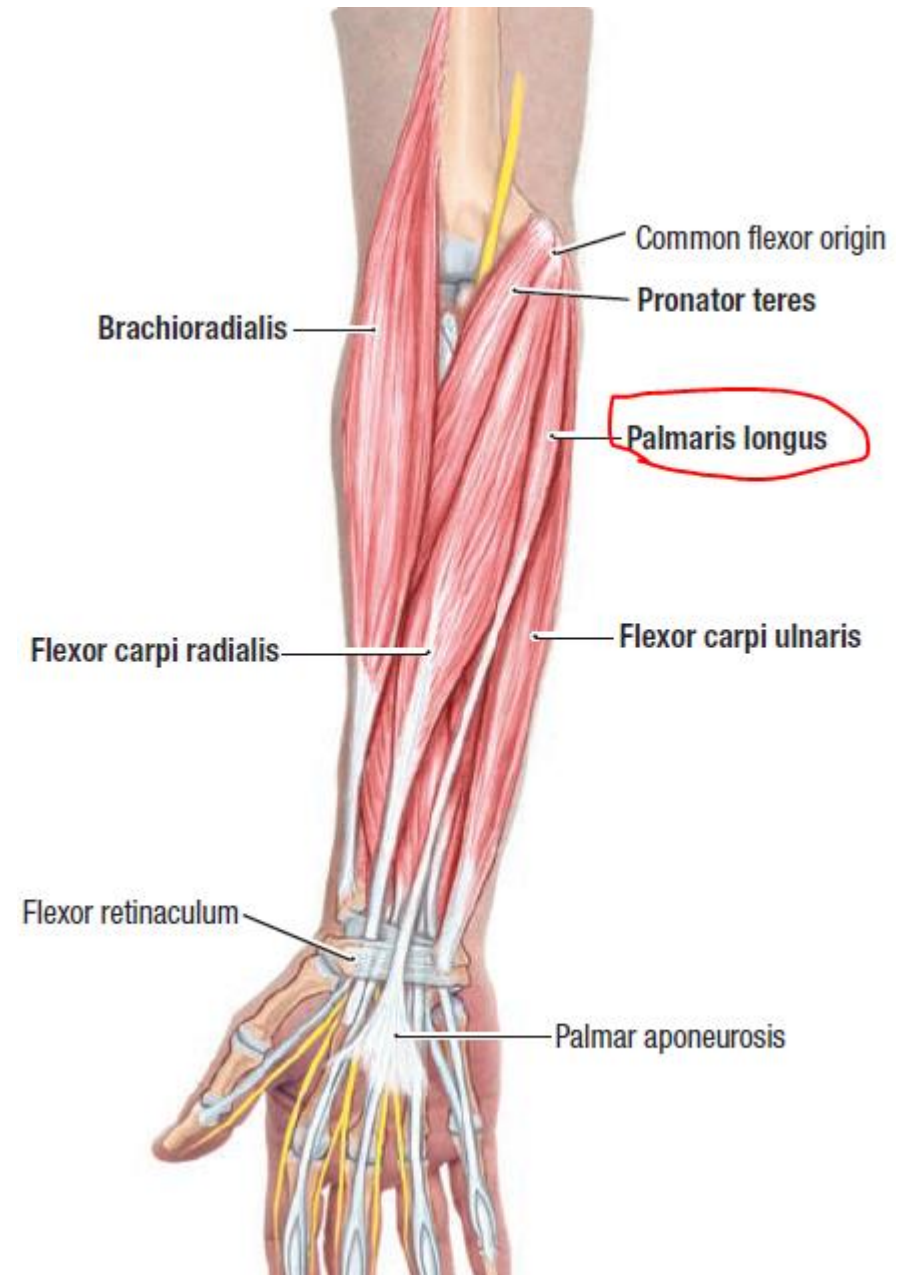


Palmaris Longus

- Is a small fusiform muscle
- Is absent on one or both sides (usually the left) in approximately 14% of people
- It has a short belly and a long, cord-like tendon that passes superficial to the flexor retinaculum
- Its tendon is a useful guide to the median nerve at the wrist
- The tendon lies deep and slightly medial to this nerve before it passes deep to the flexor retinaculum.

Palmaris Longus,,,

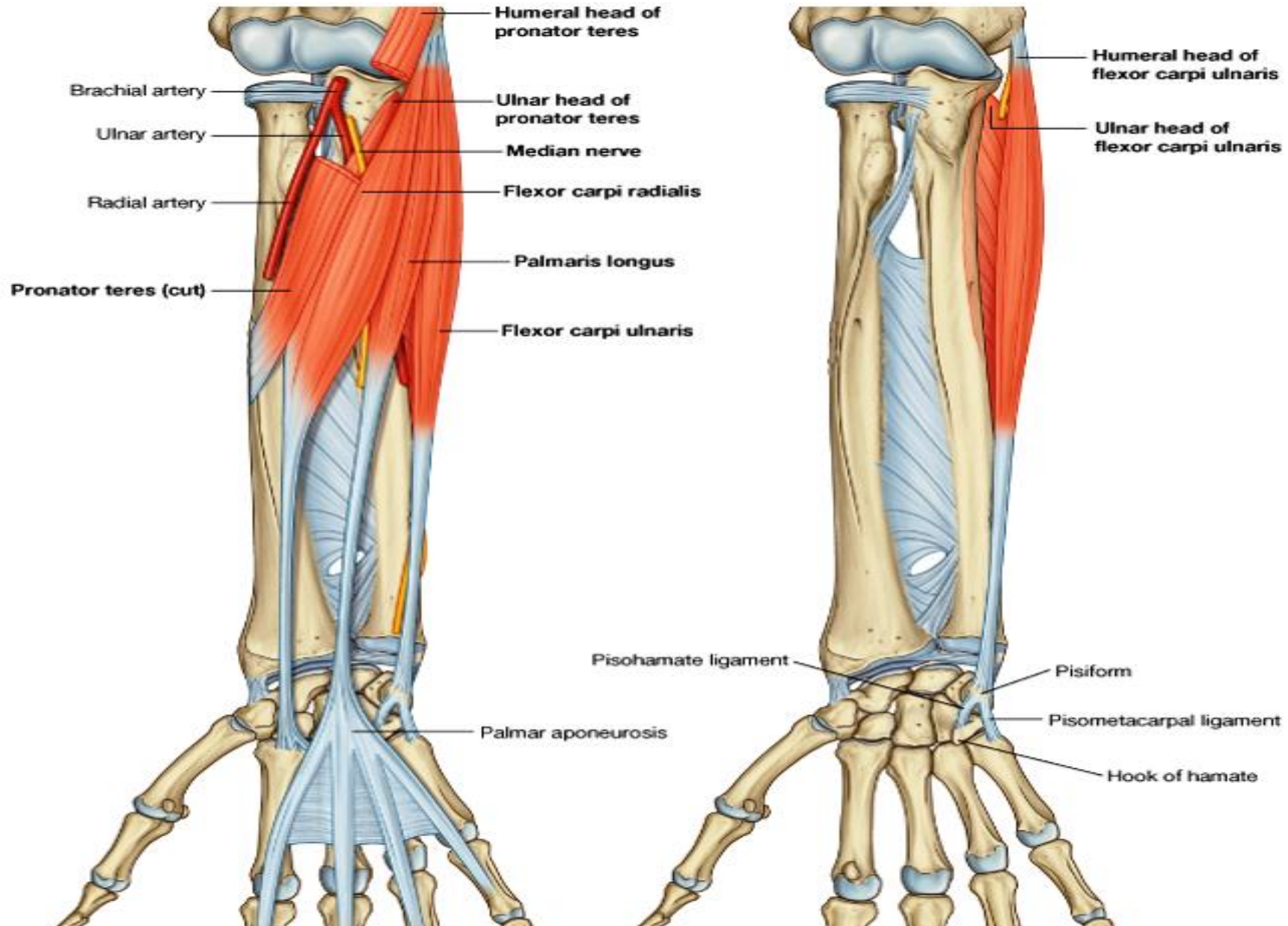
- ❑ **Origin:** Medial epicondyle of humerus (common flexor origin)
- ❑ **Insertion:** Distal half of flexor retinaculum and apex of palmar aponeurosis
- ❑ **Nerve supply:** Median nerve (C7, C8)
- ❑ **Action:** Flexes hand (at wrist) and tenses palmar aponeurosis



Flexor Carpi Ulnaris

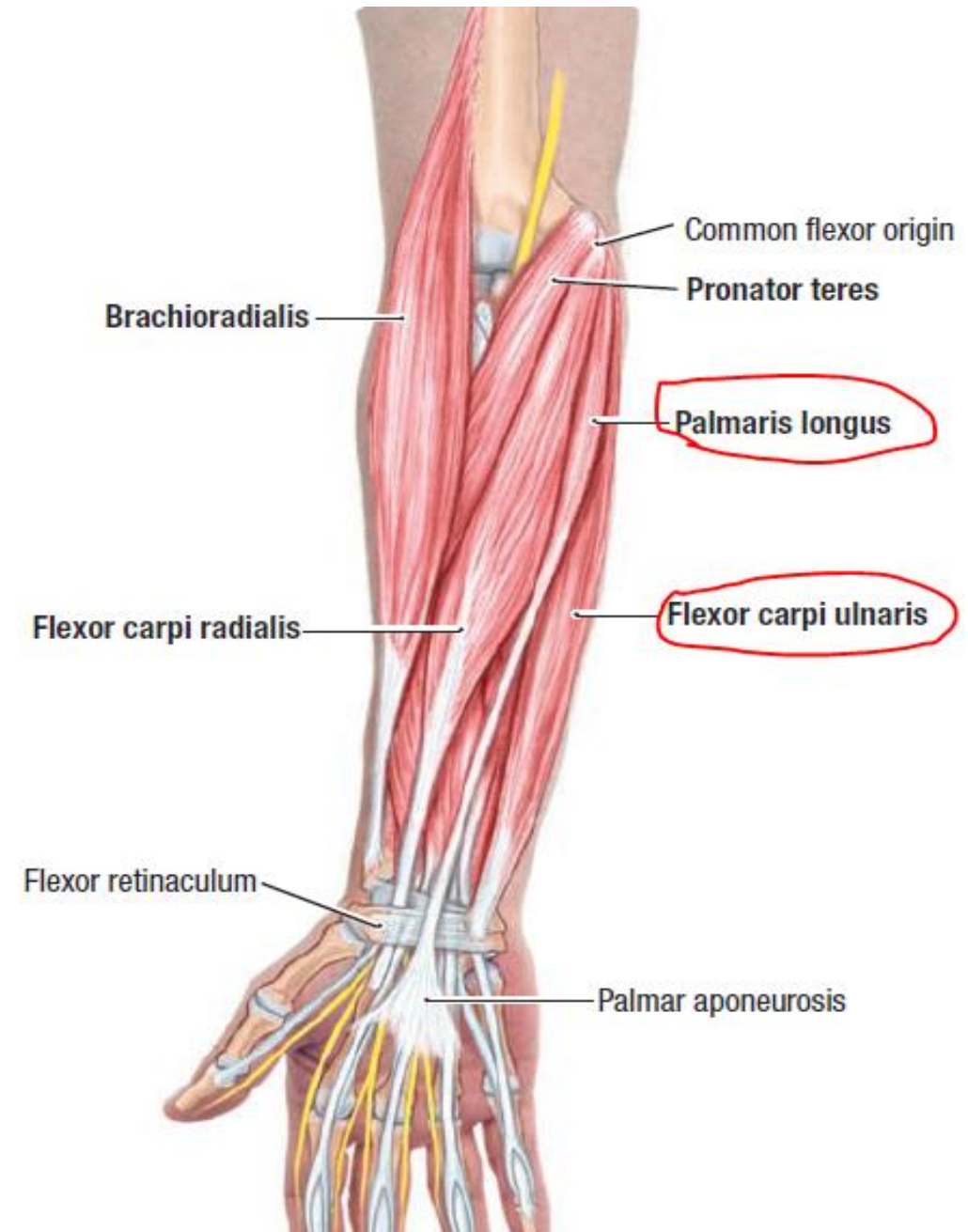
- Is the most medial of the superficial flexor muscles
- ***The ulnar nerve*** enters the forearm by passing between the humeral and the ulnar heads of its proximal attachment.
- This muscle is exceptional among muscles of the anterior compartment, being **fully innervated by the ulnar nerve**
- The tendon of the FCU is a guide to the ulnar nerve and artery, which are on its lateral side at the wrist

Flexor Carpi Ulnaris,,,



Flexor Carpi Ulnaris,,,

- ❑ **Origin:** *Humeral head:* Medial epicondyle of humerus
- ❑ *Ulnar head:* Medial aspect of olecranon process and posterior border of ulna
- ❑ **Insertion:** Pisiform bone, hook of the hamate, base at fifth metacarpal bone
- ❑ **Nerve supply:** Ulnar nerve
- ❑ **Action:** Flexes and adducts hand at wrist joint

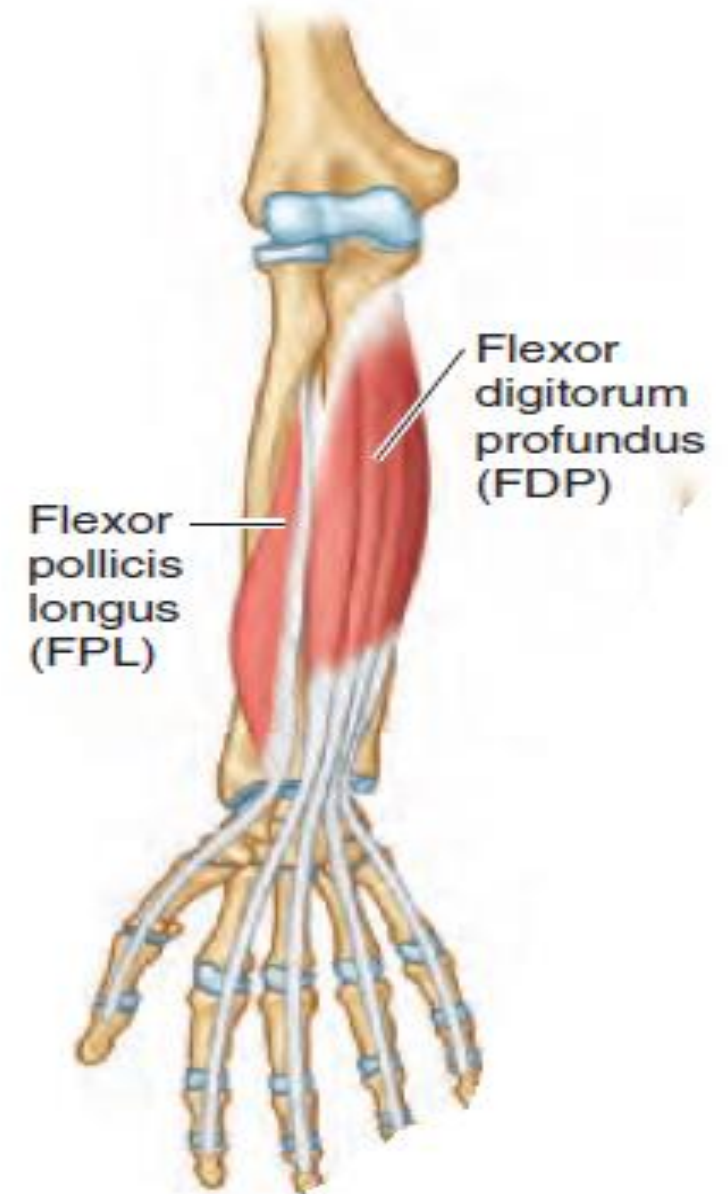


Muscles on anterior compartment of forearm,,

❑ Intermediate Layer

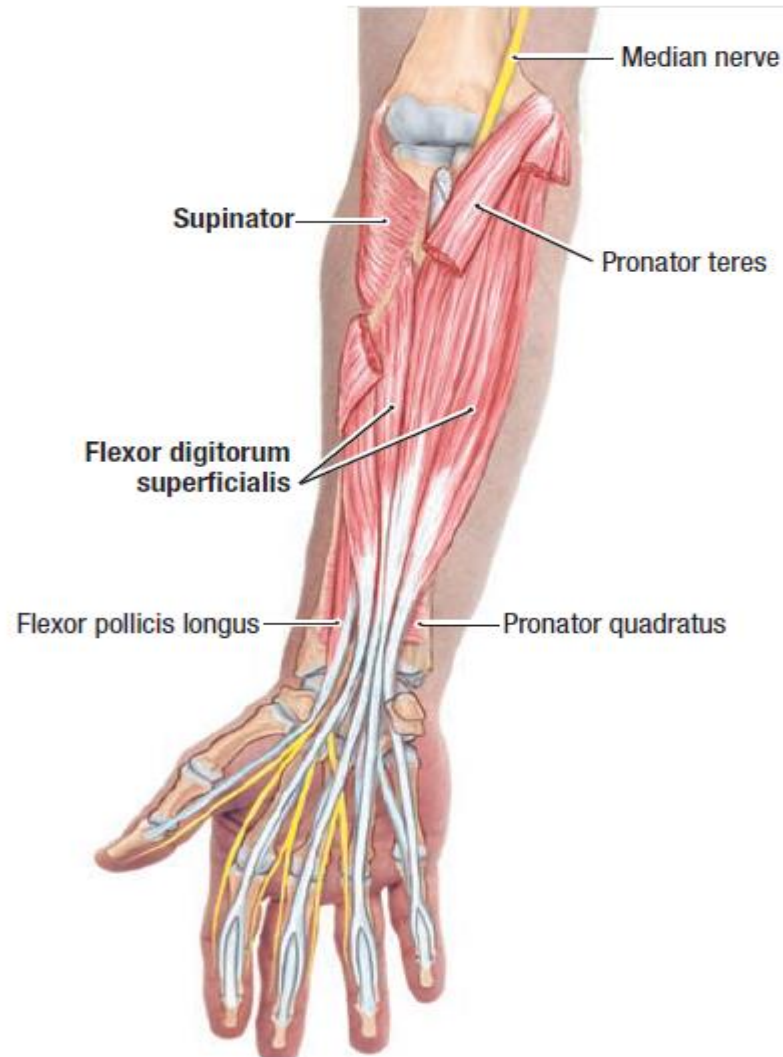
❖ Flexor Digitorum Superficialis(FDS).

- Is the largest superficial muscle in the forearm
- Forms an intermediate layer between the superficial and the deep groups of forearm muscles
- Near the wrist, the FDS gives rise to four tendons, which pass deep to the flexor retinaculum through the carpal tunnel to the fingers

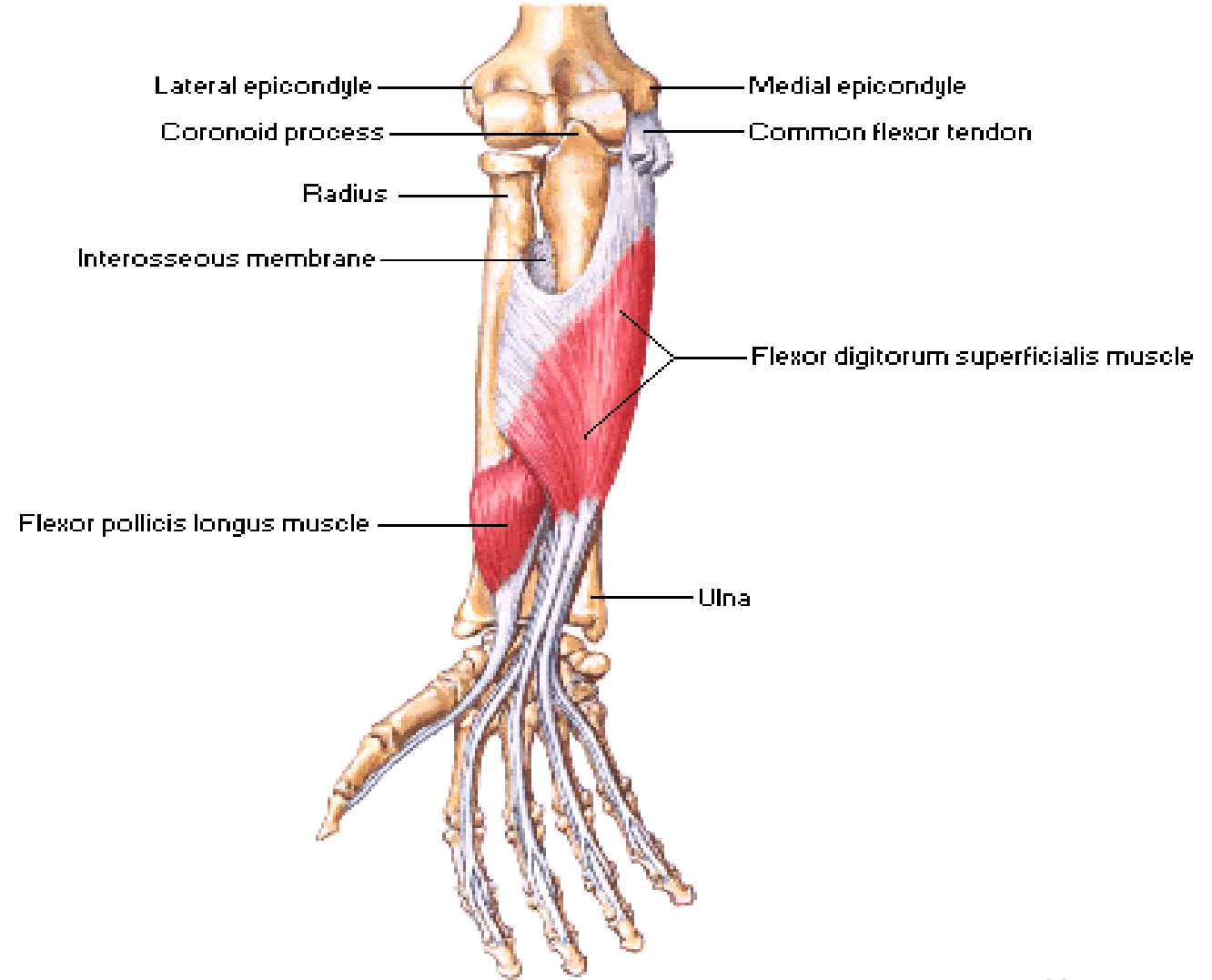


Muscles on anterior compartment of forearm,,,

□ Intermediate Layer,,



Flexors of Digits



Right forearm: anterior (palmar) view

Muscles on anterior compartment of forearm,,

□ Intermediate Layer,,

❖ Flexor Digitorum Superficialis

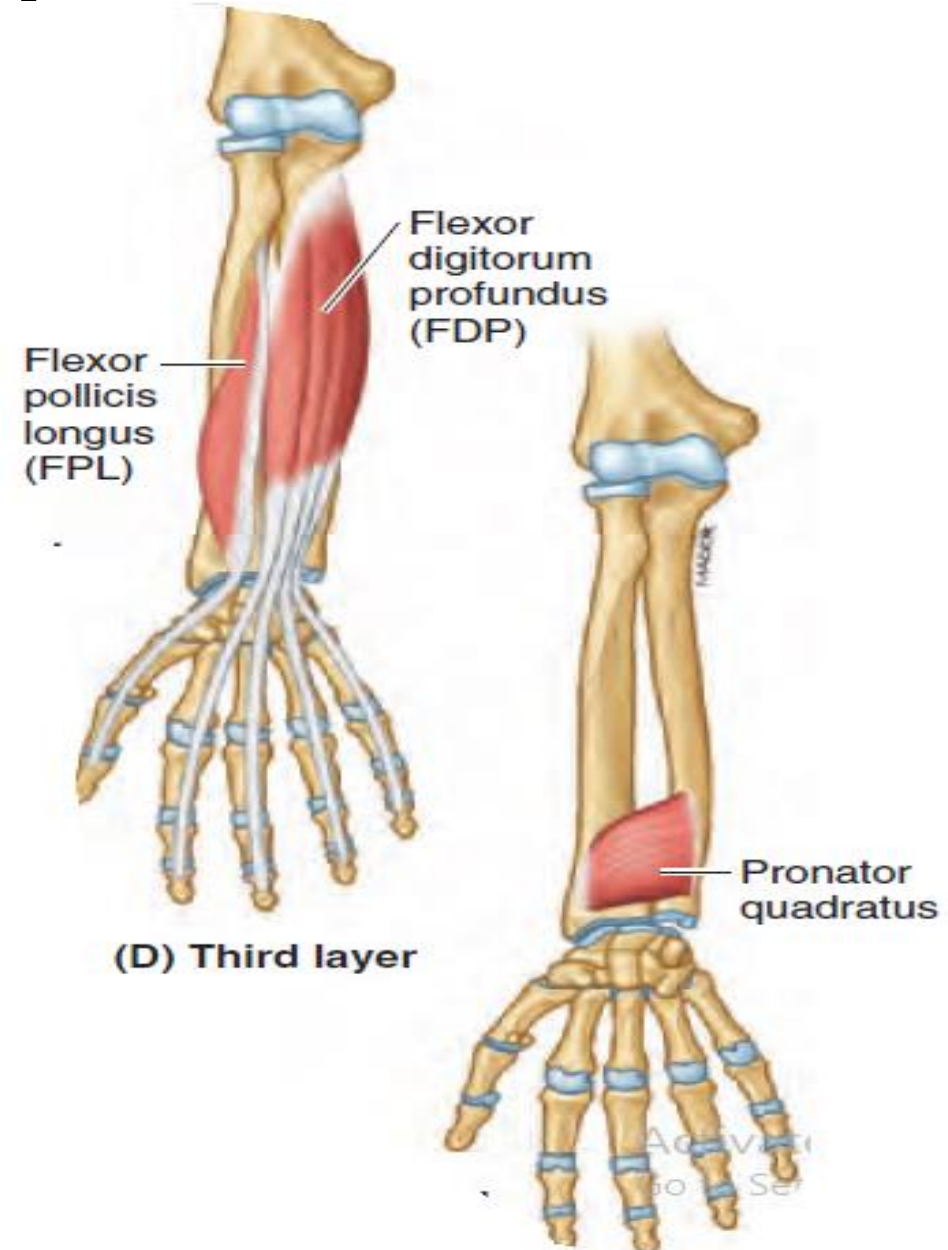
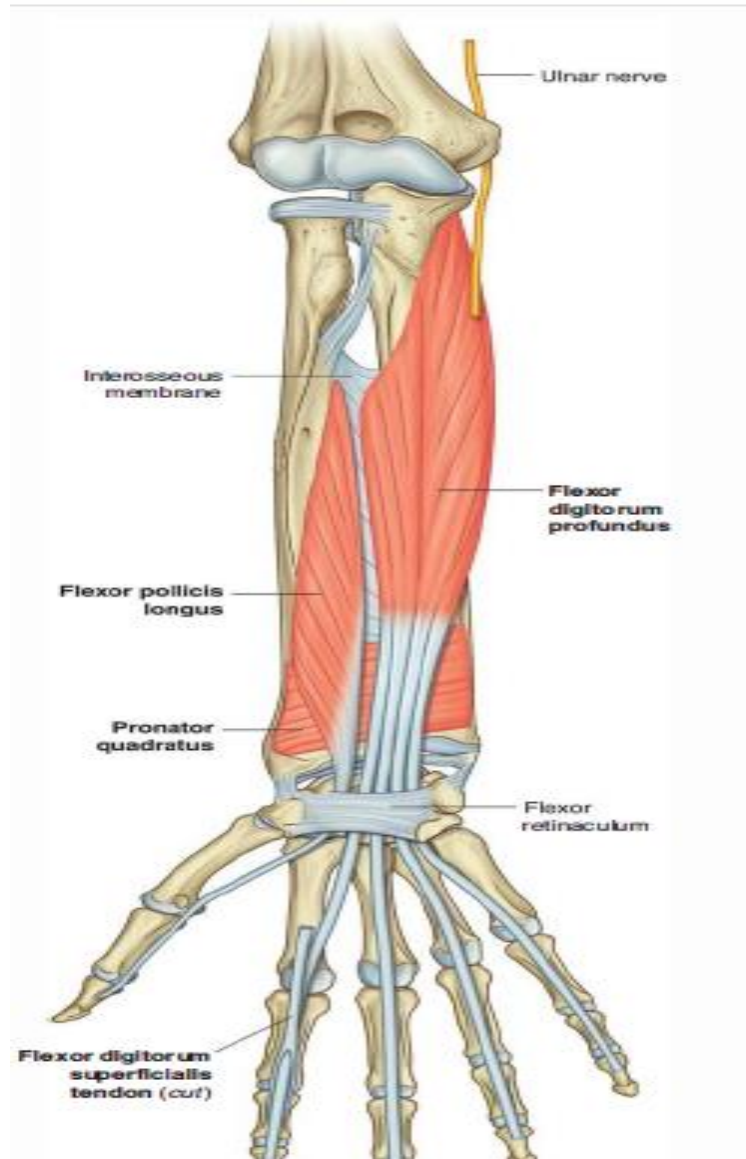
- ✓ **Origin:** Humeroulnar head: Medial epicondyle (common flexor origin and coronoid process)
- **Radial head:** Superior half of anterior border of radius
- **Insertion:** Shafts of middle phalanges of medial four fingers
- **Nerve supply:** Median nerve
 - ✚ **Action:** Flexion at proximal interphalangeal joints of middle four fingers acting more strongly; and it also flexes proximal phalanges at metacarpophalangeal joints.

Muscles on anterior compartment of forearm,,

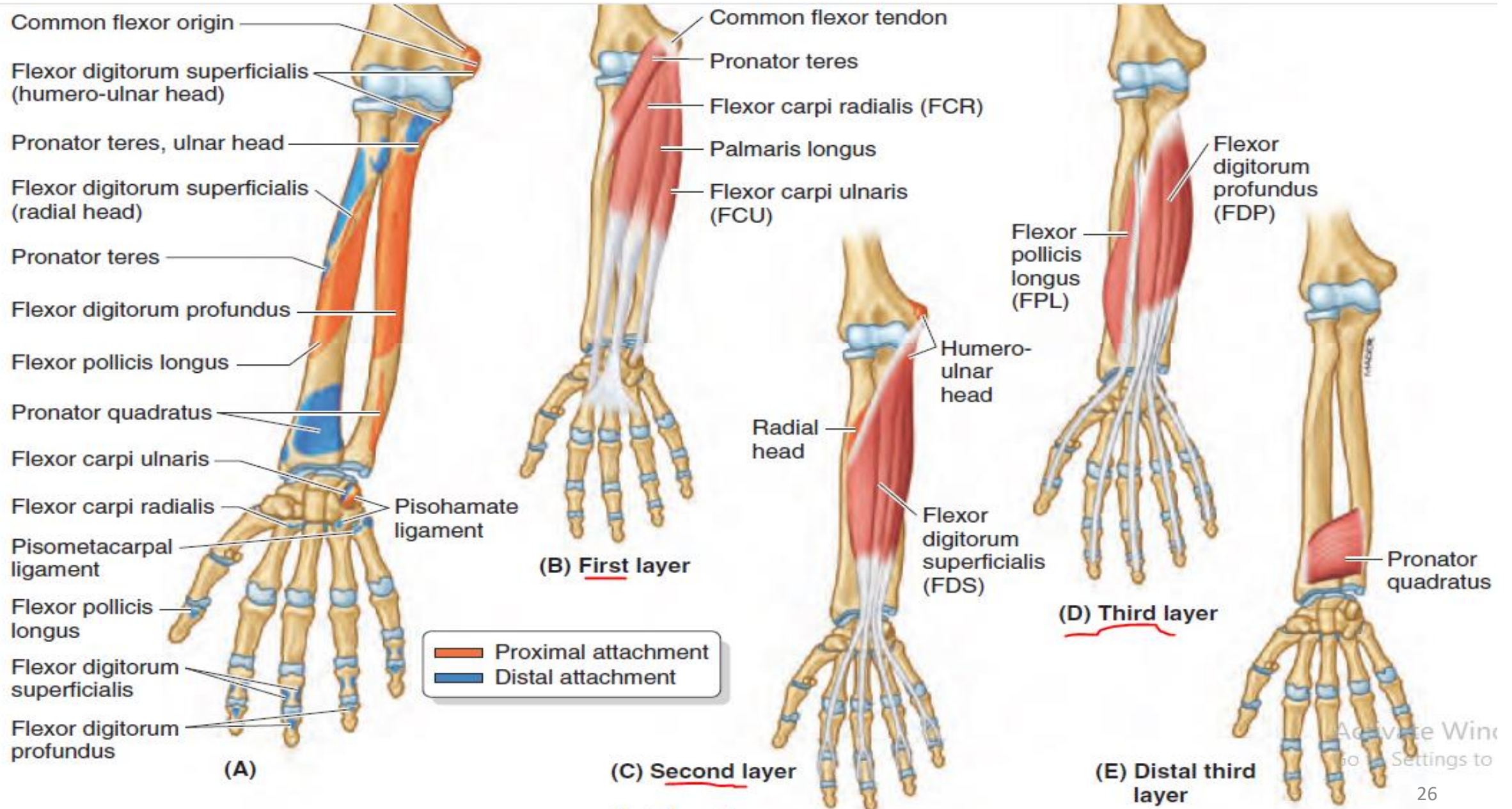
□ Deep muscles of the forearm

1. Flexor pollicis longus
 2. Flexor digitorum profundus
 3. Pronator quadratus
- Always remember that:
 - ✓ The deep muscles of the front of the forearm are supplied by the **anterior interosseous nerve** (branch of median n.)
 - ✓ Except the medial half of the flexor digitorum profundus which is supplied by the ulnar nerve

Muscles on anterior compartment of forearm,,



Muscles on anterior compartment of forearm,,,



Muscles on anterior compartment of forearm,,,

Muscle	Origin	Insertion	Action	Nerve supply
pronator teres	medial epicondyle of humerus; coronoid process of ulna	lateral aspect of shaft of radius	pronates hand, flexes forearm	median
flexor carpi radialis	medial epicondyle of humerus	bases of 2nd and 3rd metacarpal bones	flex forearm and hand aid in pronation and Abduction of hand	median
palmaris longus	medial epicondyle of humerus	flexor retinaculum of palm of hand	flexes hand and wrinkles skin of palm of hand	median
flexor digitorum superficialis	medial epicondyle of humerus; coronoid process of ulna; anterior border of radius	tendons split to attach to lateral sides of middle phalanges	flexes phalanges, wrist, and forearm	median

Muscles on anterior compartment of forearm,,,

Muscle	Origin	Insertion	Action	Nerve supply
flexor carpi ulnaris	medial epicondyle of humerus; olecranon process & posterior border of ulna	pisiform, hamate base of 5th metacarpal	flexes forearm and hand, adducts hand	ulnar
flexor pollicis longus	radius and interosseous membrane	base of distal phalanx of thumb	flexes thumb	median
flexor digitorum profundus	upper anterior 3/4ths aspect of ulna & adjacent interosseous membrane	base of distal phalanges of fingers	flexes phalanges	radial half by median ulnar half by ulnar
pronator quadratus	distal anterior aspect of ulna	distal anterior aspect of radius	pronates hand	Median

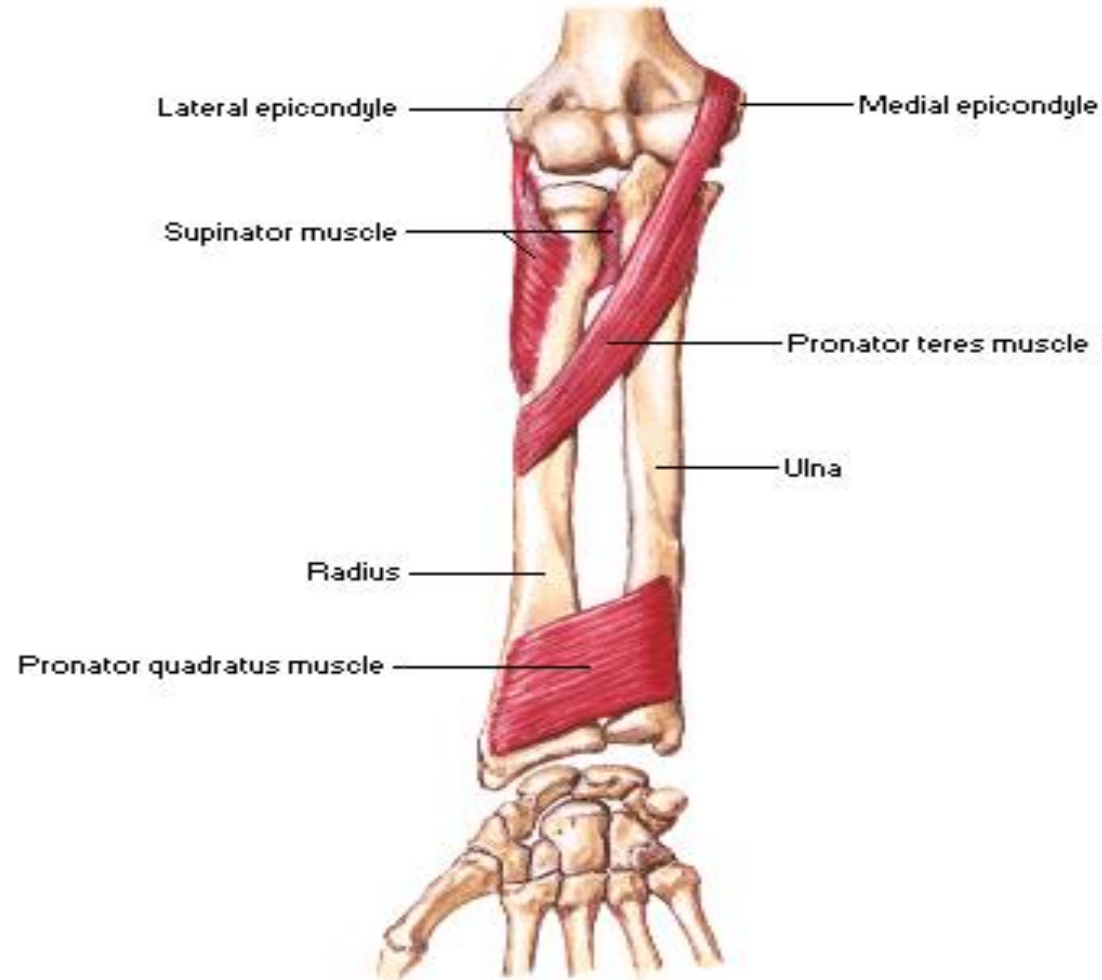
Muscles on anterior compartment of forearm,,

❑Pronator Quadratus

- Is quadrangular muscle and pronates the forearm
- Is the deepest muscle in the anterior aspect of the forearm
- Sometimes it is considered to constitute a fourth muscle layer
- Clothes the distal fourth of the radius and ulna and the interosseous membrane between them
- Is the only muscle that attaches only to the ulna at one end and only to the radius at the other end
- Is the prime mover for pronation

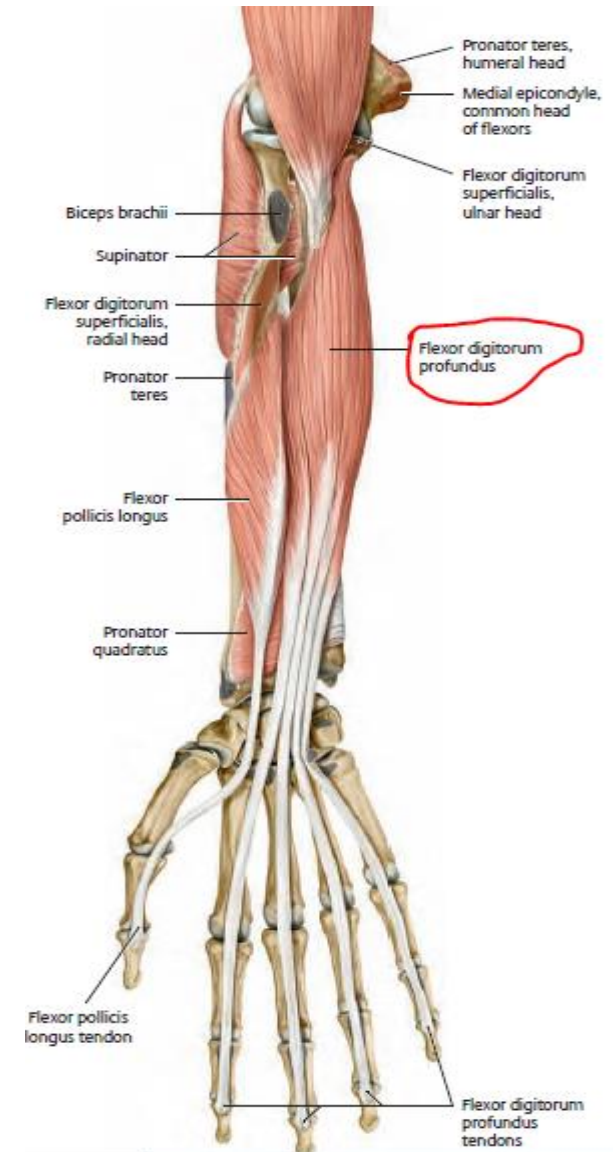
Muscles on anterior compartment of forearm,,

Rotators of Radius - Supination

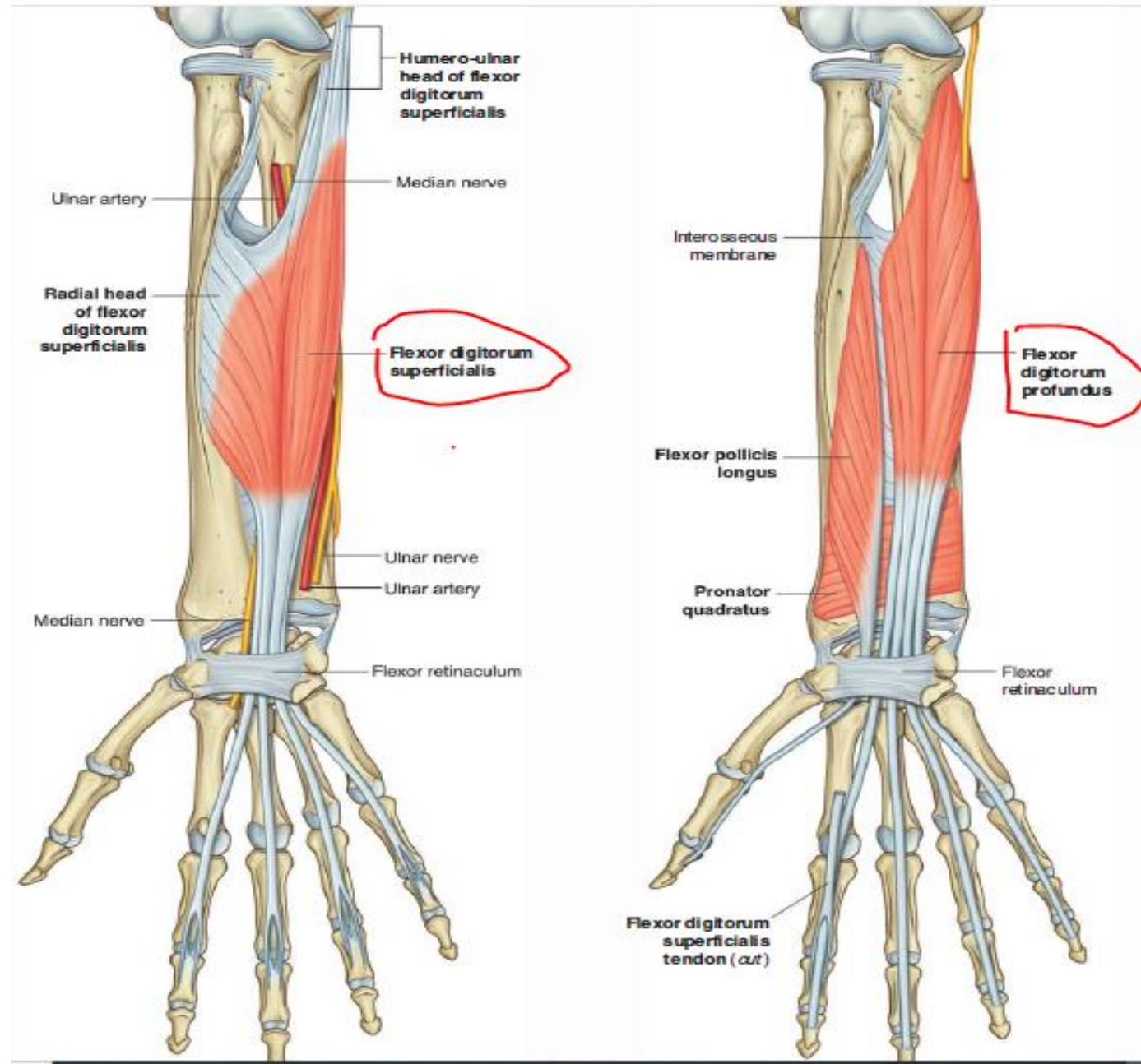


Flexor Digitorum Profundus

- Is the only muscle that can flex the distal interphalangeal joints of the fingers
- Clothes the anterior aspect of the ulna
- The FDP flexes the distal phalanges of the medial four fingers after the FDS has flexed their middle phalanges



Flexor Digitorum Profundus,,



Flexor Digitorum Profundus,,

❑ **Origin:** Proximal three quarters of medial and anterior surfaces of ulna and interosseous membrane

❑ **Insertion:** Bases of the Distal phalanges of medial four fingers

❑ **Nerve supply:**

- **Medial part:** Ulnar nerve
- **Lateral part:** Anterior interosseous nerve, from median nerve

❑ **Main action:**

- Flexes distal phalanges at distal interphalangeal joints

Flexor Pollicis Longus

- The long flexor of the thumb (L. Pollex, thumb).
- Lies lateral to the FDP, where it clothes the anterior aspect of the radius distal to the attachment of the supinator.
- The flat FPL tendon passes deep to the flexor retinaculum, enveloped in its own synovial tendinous sheath.

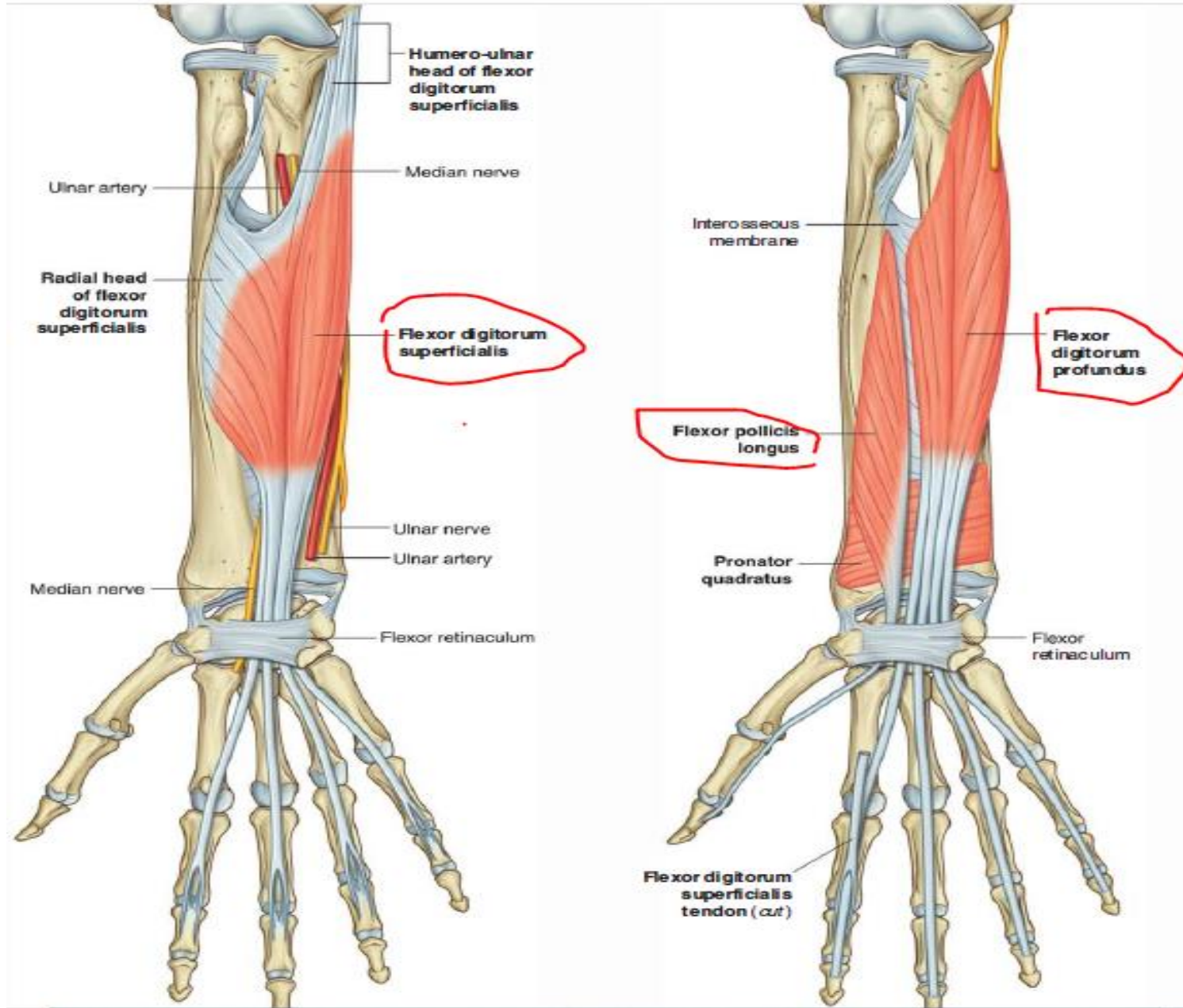
❑ **Origin:** Anterior surface of radius and adjacent interosseous membrane

❑ **Insertion:** Base of distal phalanx of thumb

❑ **Nerve supply:** Anterior interosseous nerve, from median nerve.

❑ **Main action:** Flexes phalanges of 1st digit (thumb)

Flexor Pollicis Longus,,

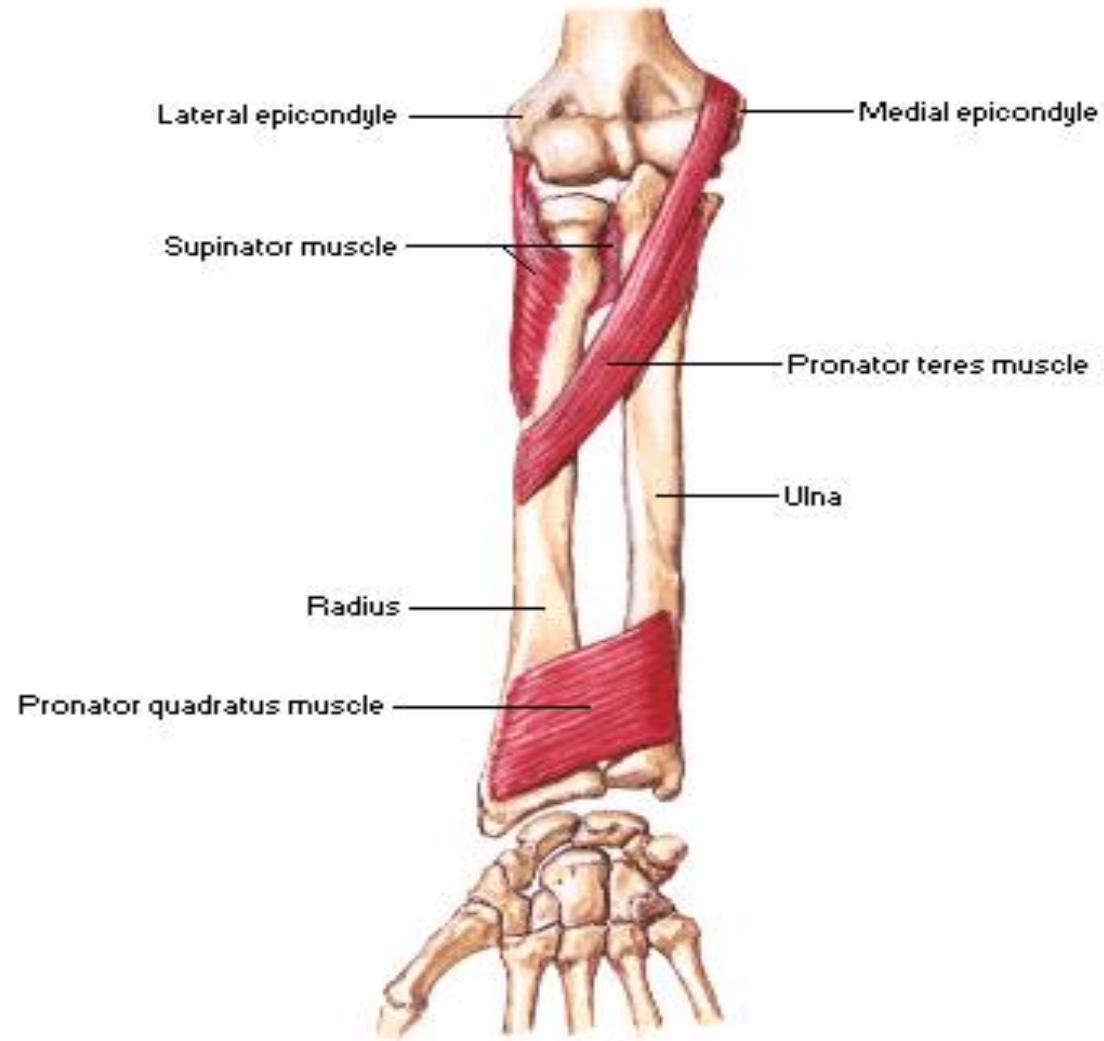


Pronator Quadratus

- Is quadrangular muscle and pronates the forearm
- Is the deepest muscle in the anterior aspect of the forearm
- Sometimes it is considered to constitute a fourth muscle layer
- Clothes the distal fourth of the radius and ulna and the interosseous membrane between them
- Is the only muscle that attaches only to the ulna at one end and only to the radius at the other end
- Is the prime mover for pronation

Pronator Quadratus,,

Rotators of Radius - Supination



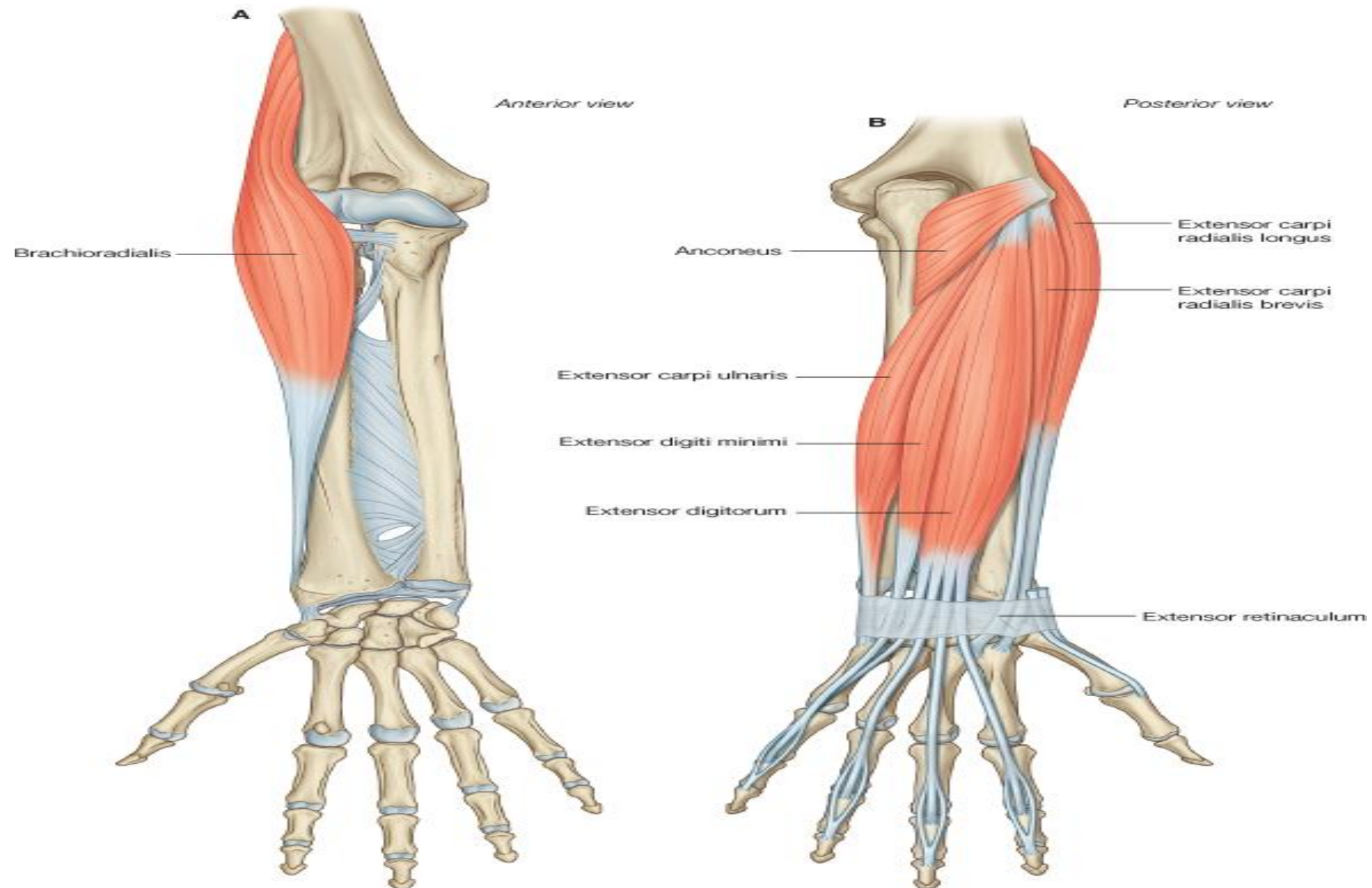
Pronator Quadratus,,

- ❑ **Origin:** Distal quarter of anterior surface of ulna
- ❑ **Insertion:** Distal quarter of anterior surface of radius
- ❑ **Nerve supply:** Anterior interosseous nerve, from median nerve
- ❑ **Main action:**
 - Pronates forearm; deep fibers bind radius and ulna together

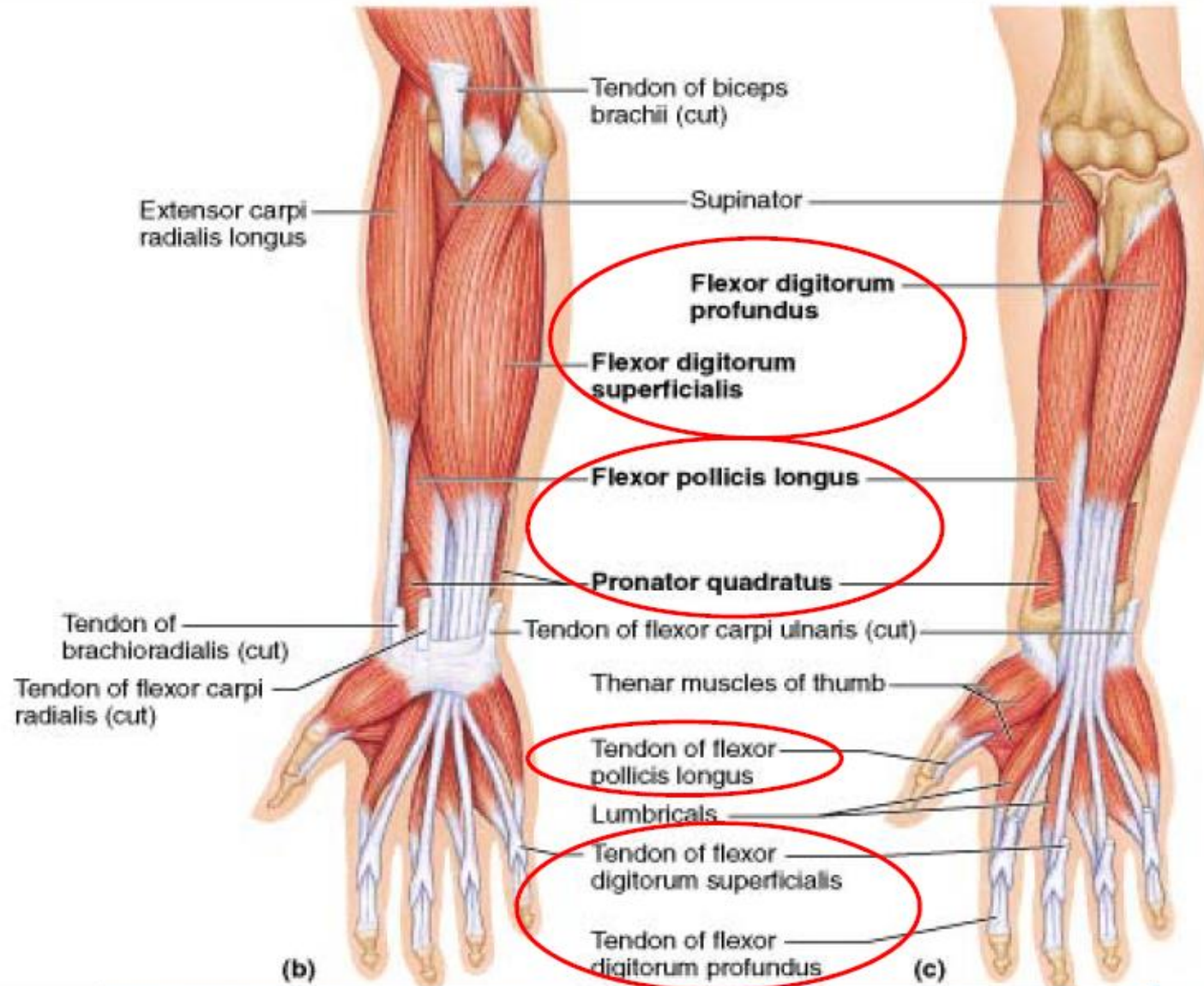
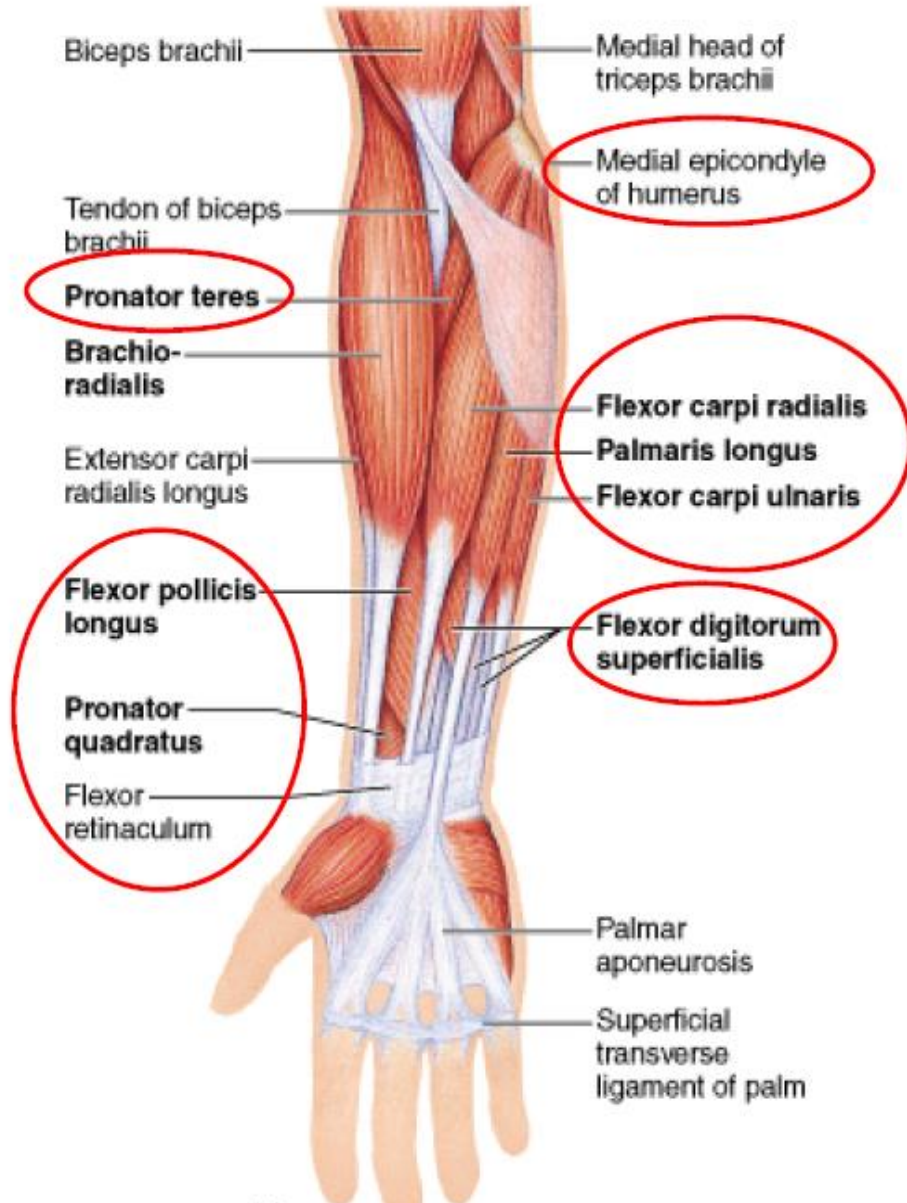
Muscles on anterior compartment of forearm,,

- The 8 muscles in the anterior (flexor-pronator) compartment of the forearm act on the elbow , wrist joints and the fingers.
- Their fleshy masses is in the proximal part and their tendons parts are located on the wrist and are held in place by the **palmar carpal ligament** & the *flexor retinaculum*.
- Arise More or less from the Common Flexor origin (front of medial epicondyle).
- All are supplied by median nerve except one, Flexor Carpi Ulnaris.
- All cross the wrist joint except one, pronator teres
- Functionally, the **brachioradialis** is a flexor of the forearm, but it is located in the posterior (posterolateral) compartment & is thus supplied by the radial nerve

Muscles on anterior compartment of forearm,,,



Muscles on anterior compartment of forearm,,



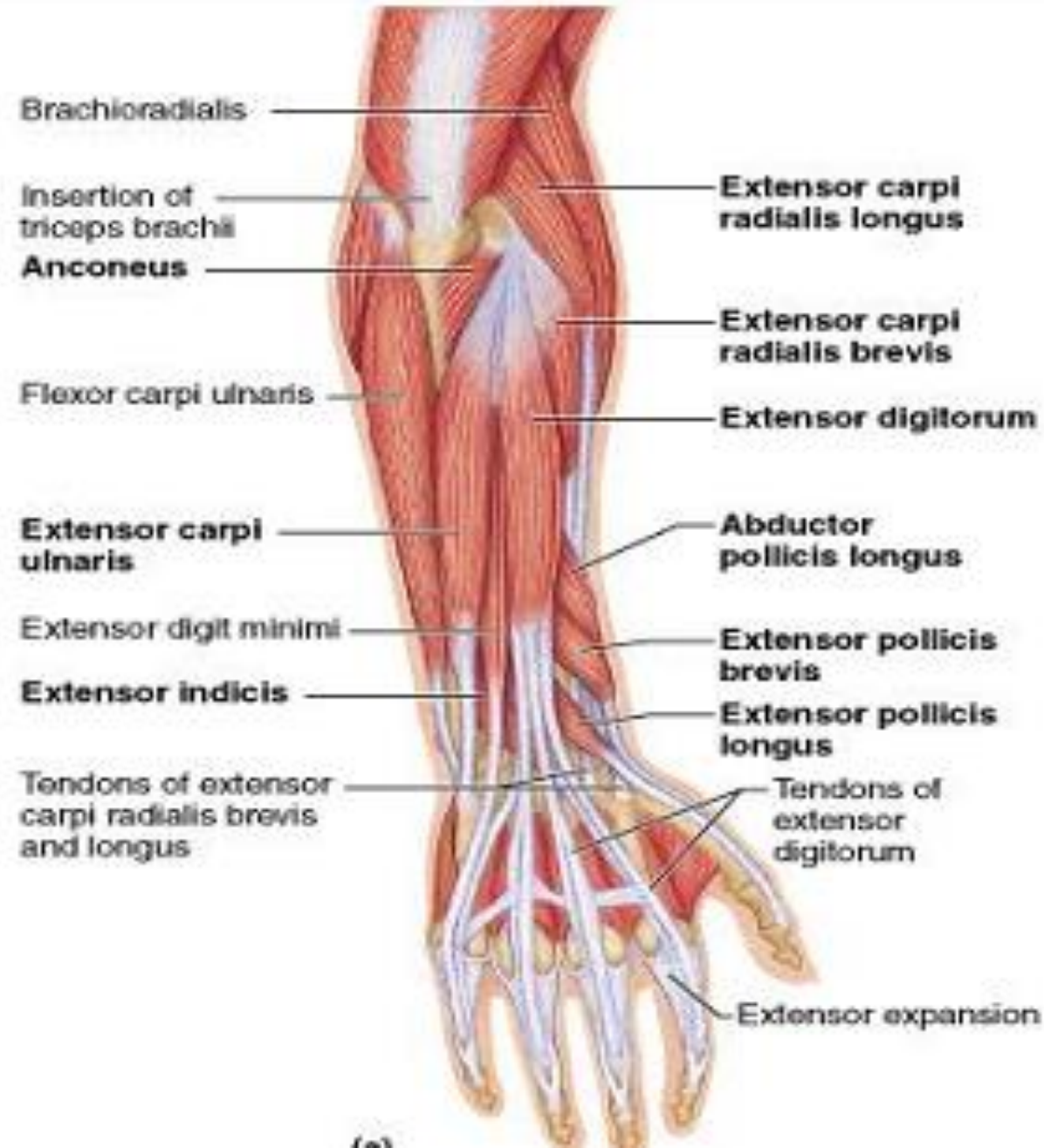
Muscles of the Posterior Compartment of the Forearm

- Also known as *Extensor - Supinator Muscles of the Forearm*

□ *These muscles can be organized physiologically into 3 functional groups:*

- ✓ **Muscles that extend and abduct or adduct the hand at the wrist joint** (extensor carpi radialis longus, extensor carpi radialis brevis, and extensor carpi ulnaris).
- ✓ **Muscles that extend the medial four fingers** (extensor digitorum, extensor indicis, and extensor digiti minimi).
- ✓ **Muscles that extend or abduct the thumb** (abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus).

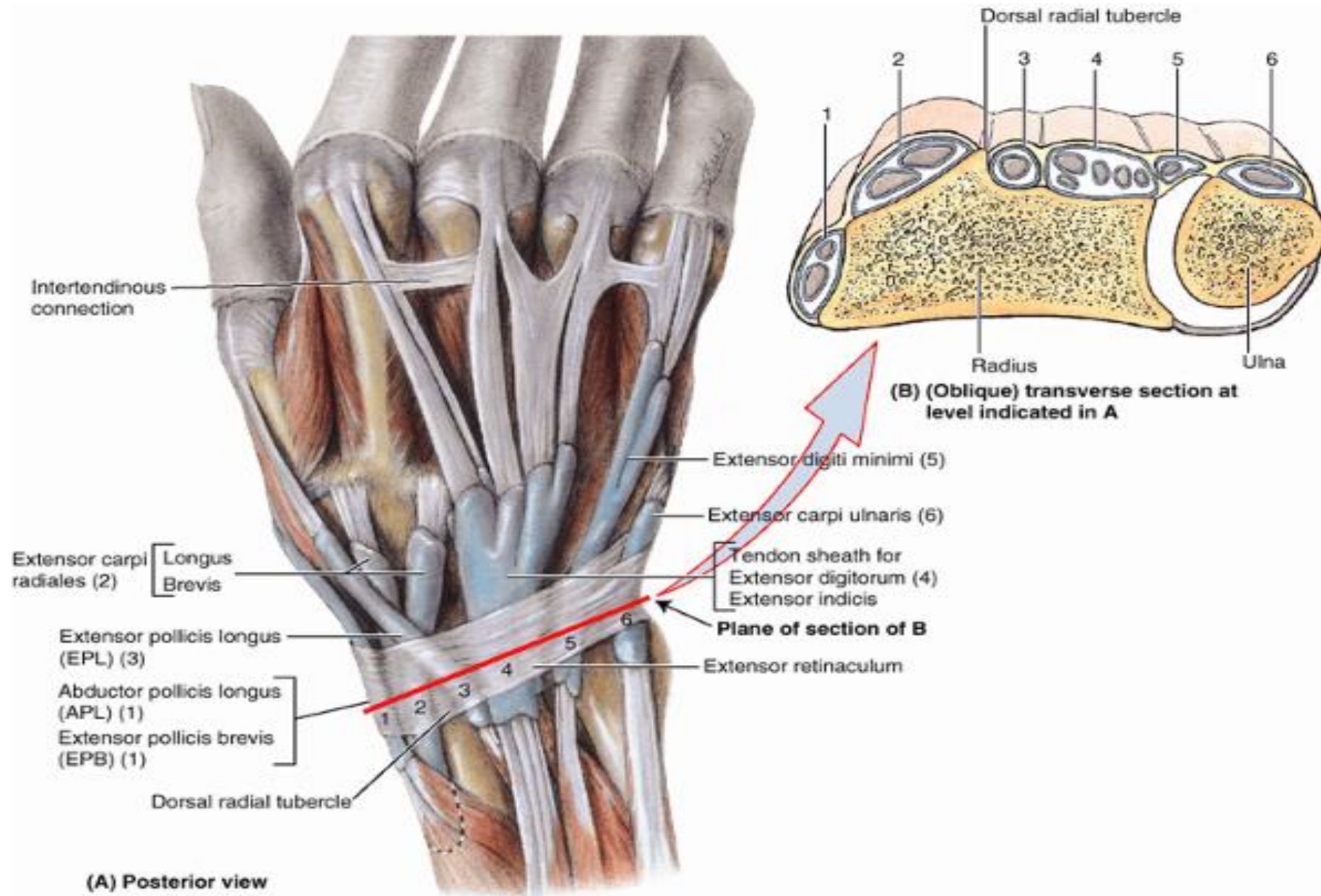
Muscles of the Posterior Compartment of the Forearm,,



Muscles of the Posterior Compartment of the Forearm,,

- The extensor tendons are held in place in the wrist region by the **extensor retinaculum**, which prevents bowstringing of the tendons when the hand is extended at the wrist joint
- As the tendons pass over the dorsum of the wrist, they are provided with synovial tendon sheaths that reduce friction for the extensor tendons as they traverse the osseofibrous tunnels formed by the attachment of the extensor retinaculum to the distal radius and ulna.

Muscles of the Posterior Compartment of the Forearm,,,



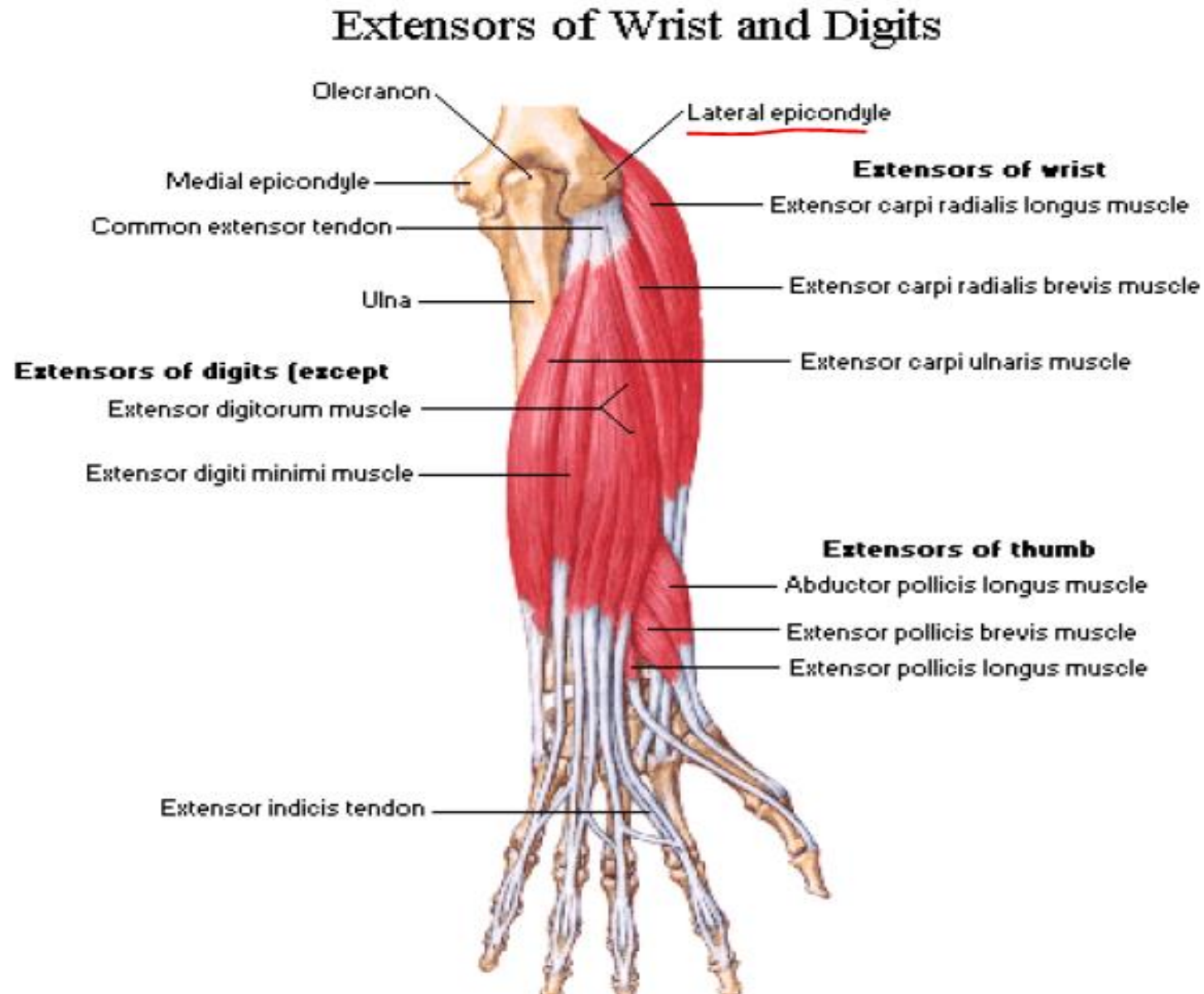
Muscles of the Posterior Compartment of the Forearm,,

- The extensor muscles of the forearm are organized **anatomically** into superficial and deep layers

❖ **Superficial extensors:** extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, and extensor carpi ulnaris- are attached proximally by a common extensor tendon to the **lateral epicondyle**.

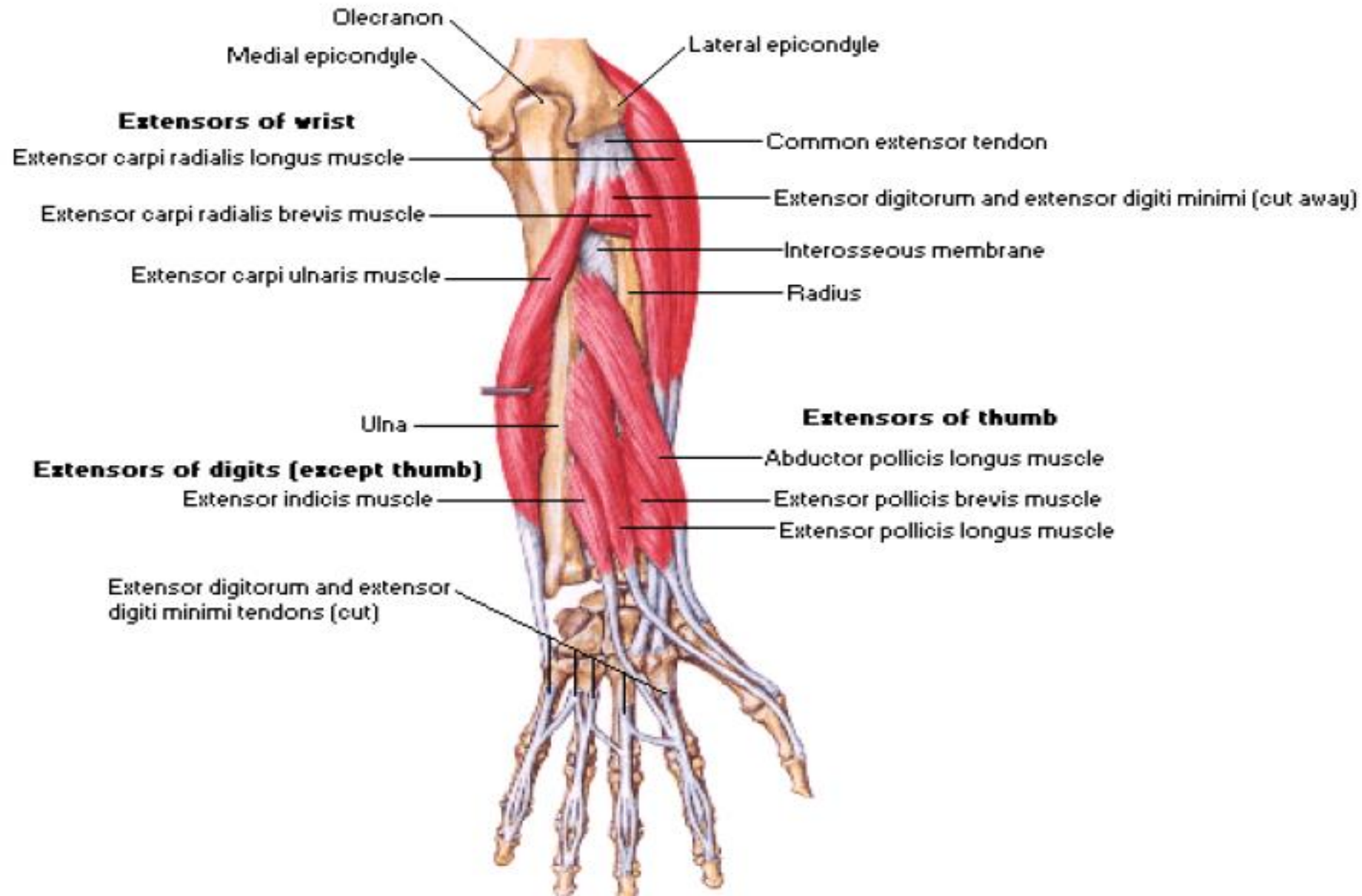
❖ **Deep extensors:** Supinator, Extensor indicis, Abductor pollicis longus (APL), Extensor pollicis longus (EPL), Extensor pollicis brevis (EPB) - the latter three are termed as **Outcropping muscles of deep layer**.

Muscles of the Posterior Compartment of the Forearm,,

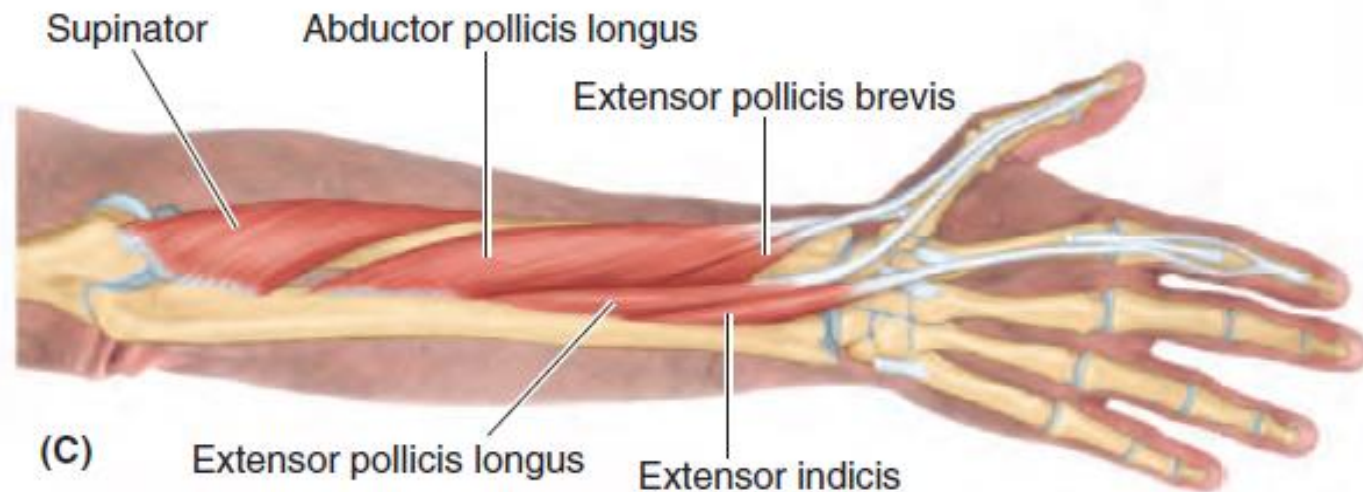
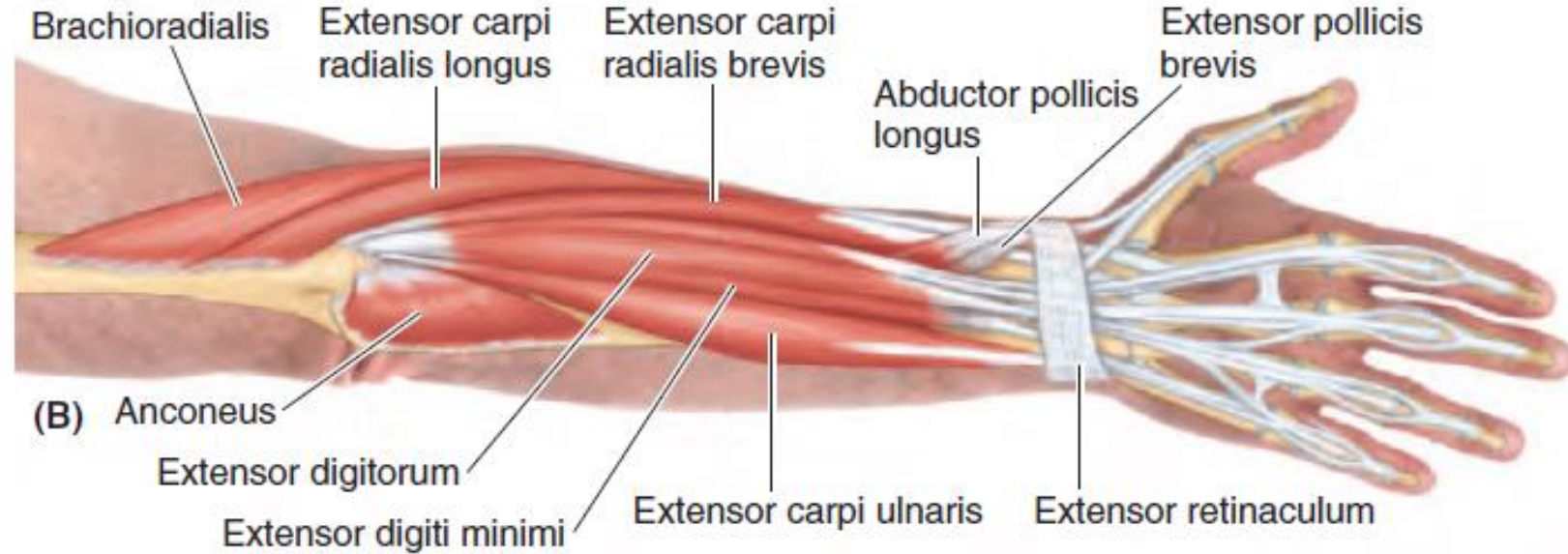
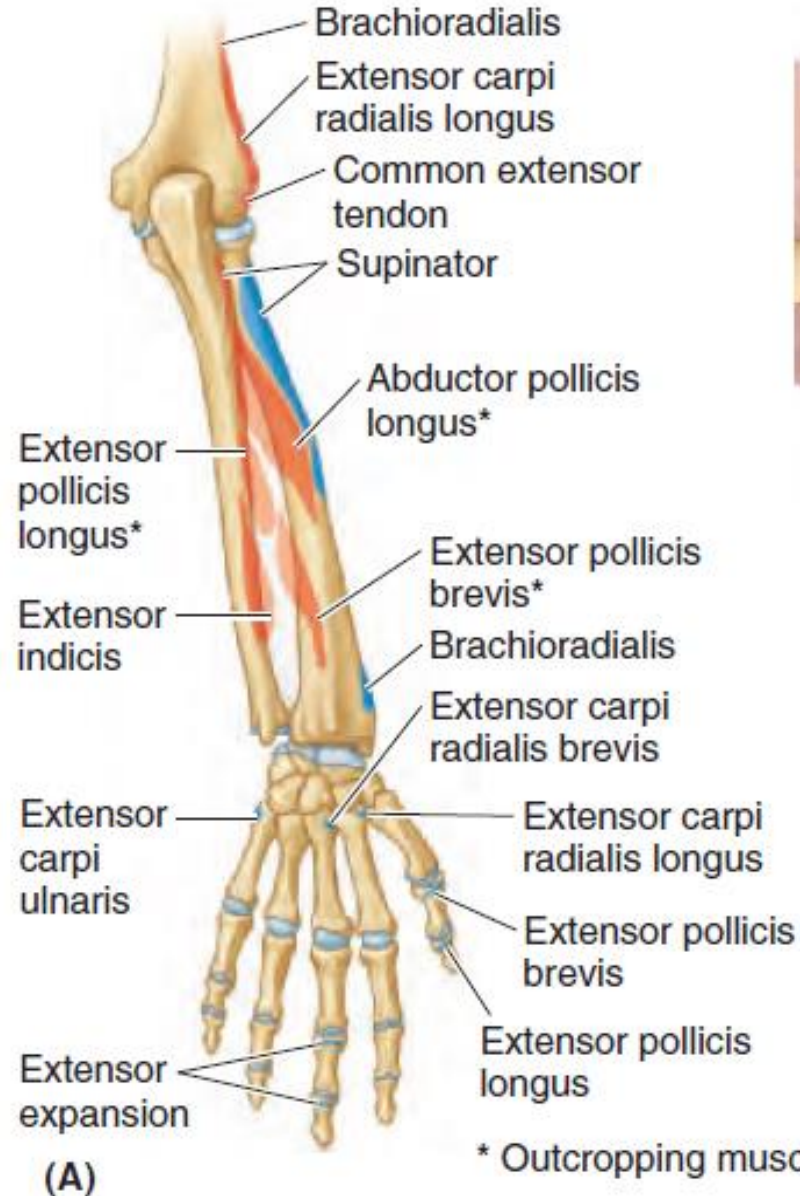


Muscles of the Posterior Compartment of the Forearm,,

Extensors of Wrist and Digits

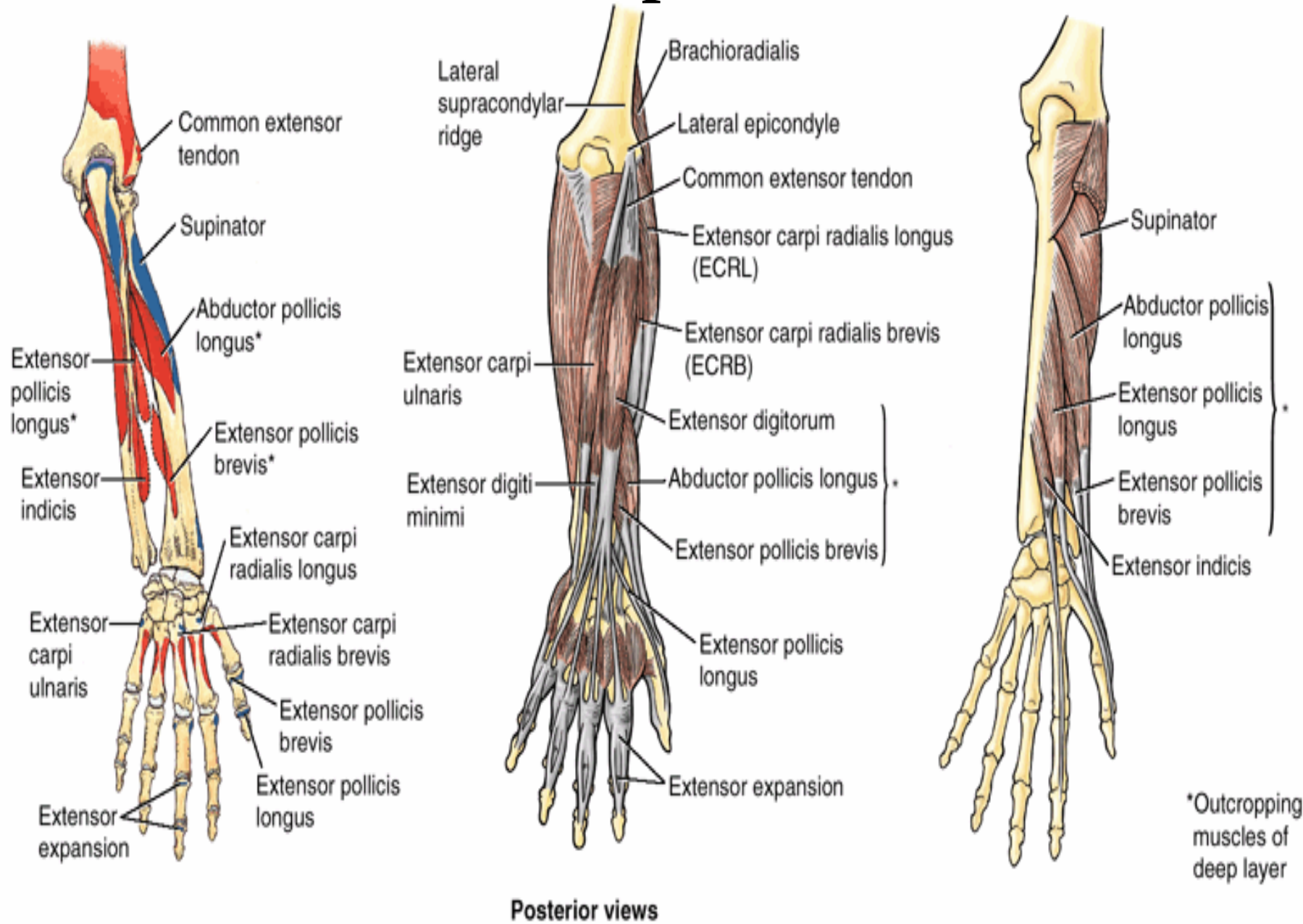


Muscles of the Posterior Compartment of the Forearm,,



Posterior views

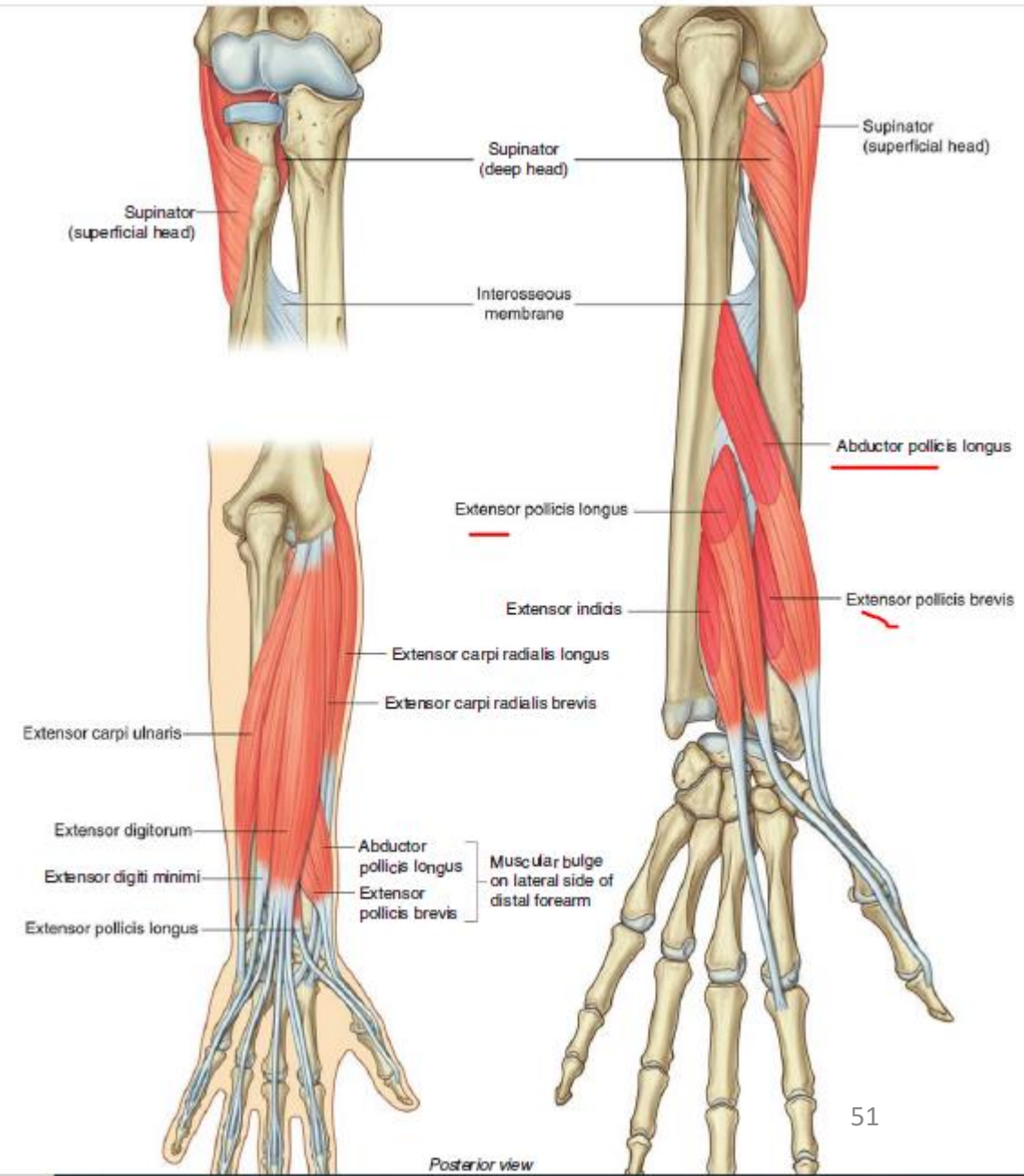
Muscles of the Posterior Compartment of the Forearm,,



Muscles of the Posterior Compartment of the Forearm,,

Deep extensors

- **Three act on the thumb**
 - deep to the superficial extensors and emerge ("crop out") from a furrow in the lateral part of the forearm = outcropping muscles
 - ✓ Abductor pollicis longus
 - ✓ Extensor pollicis brevis
 - ✓ Extensor pollicis longus
- **One act on index finger**
 - Extensor indicis
 - confers independence to the index finger



Superficial Extensor Muscles of Forearm

Muscle	Origin	Insertion	Nerve supply	Action
brachioradialis	lateral supracondylar ridge of humerus	styloid process of radius	radial	flexes forearm & supinates hand
extensor carpi radialis longus	lateral supracondylar ridge of humerus	base of 2nd metacarpal	radial	extends & abducts the wrist
extensor carpi radialis brevis	lateral epicondyle of humerus	base of 3rd metacarpal	radial	extends & abducts the wrist
extensor carpi ulnaris	lateral epicondyle of humerus posterior border of ulna	base of 5th metacarpal	radial	extends & adducts the hand
extensor digitorum	lateral epicondyle of humerus	extensor expansion over fingers	radial	extends fingers, hand & forearm
extensor digiti minimi	lateral epicondyle of humerus	extensor expansion of little finger	radial	extends little finger

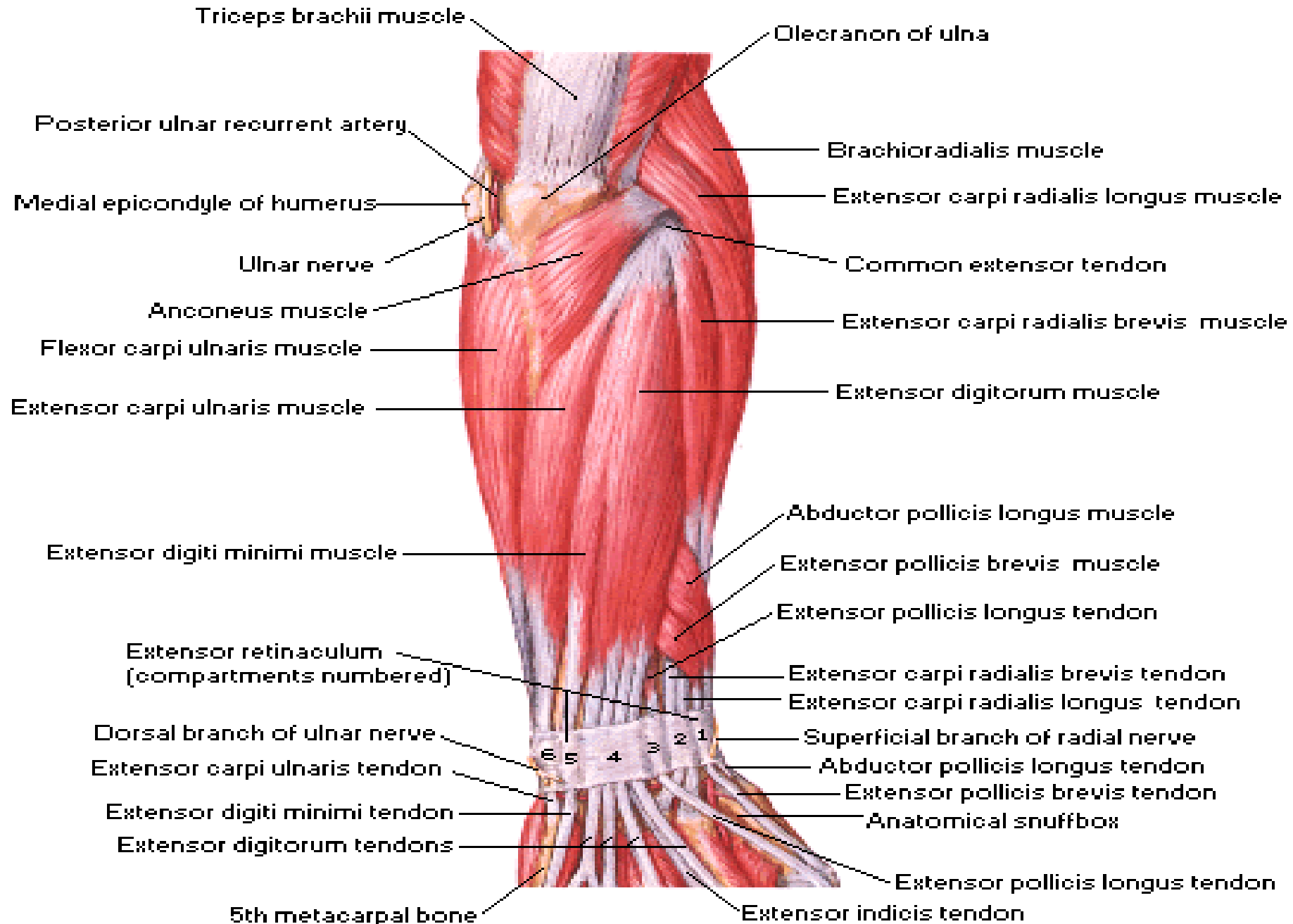
Deep Extensor Muscles of Forearm

Muscle	Origin	Insertion	Nerve supply	Action
anconeus	back of lateral epicondyle of humerus	olecranon process poster surface of ulna	radial	extends forearm
supinator	lateral epicondyle of humerus crest of ulna	upper third of radius	radial	supinates the forearm
abductor pollicis longus	posterior surface of ulnar middle aspect of radius	base of 1st metacarpal	radial	abducts thumb
extensor pollicis brevis	middle 1/3rd of radius	base of proximal phalanx of thumb	radial	extends thumb
extensor pollicis longus	middle 1/3rd ulna & interosseous membrane	base of distal phalanx of thumb	radial	extends thumb
extensor indicis	posterior surface of ulna	extensor expansion of index finger	radial	extends index finger

Extensor Carpi Radialis Brevis

- As its name indicates, is a shorter muscle than the ECRL
- Along much of its course, extensor carpi radialis brevis lies deep to extensor carpi radialis longus
- As it passes distally, it is covered by the **ECRL**.
- ❖ **Origin:** Common extensor origin-lateral epicondyle of humerus
- ❖ **Insertion:** Dorsal aspect of base of 3rd metacarpal
- ❖ **Nerve supply:** Deep branch of radial nerve
- ❖ **Main action:** Extend and abduct hand at the wrist joint

Posterior View



Extensor Digitorum

- Is the principal extensor of the medial four fingers
- Occupies much of the posterior surface of the forearm
- On the dorsum of the hand, the tendons spread out as they run toward the fingers
- On the distal ends of the metacarpals and along the phalanges, the four tendons flatten to form extensor expansions.
- Adjacent tendons are linked proximal to the metacarpophalangeal joints by three oblique intertendinous connections that restrict independent extension of the fingers (especially the ring finger).

Extensor Digitorum,,,

- Consequently, normally no finger can remain fully flexed as the other ones are fully extended.
- ❑ **Origin:** Lateral epicondyle of humerus (common extensor origin)
- ❑ **Insertion:** Extensor expansions of medial four fingers
- ❑ **Nerve supply:** Posterior interosseous nerve
- ❑ **Main action:** Extends medial four fingers primarily at metacarpophalangeal joint

Extensor Digiti Minimi

- Is a partially detached part of the extensor digitorum
- Is medial to extensor digitorum in the forearm
- The tendon of this extensor of the little finger runs through a separate compartment of the extensor retinaculum, within the tendinous sheath of the extensor digiti minimi

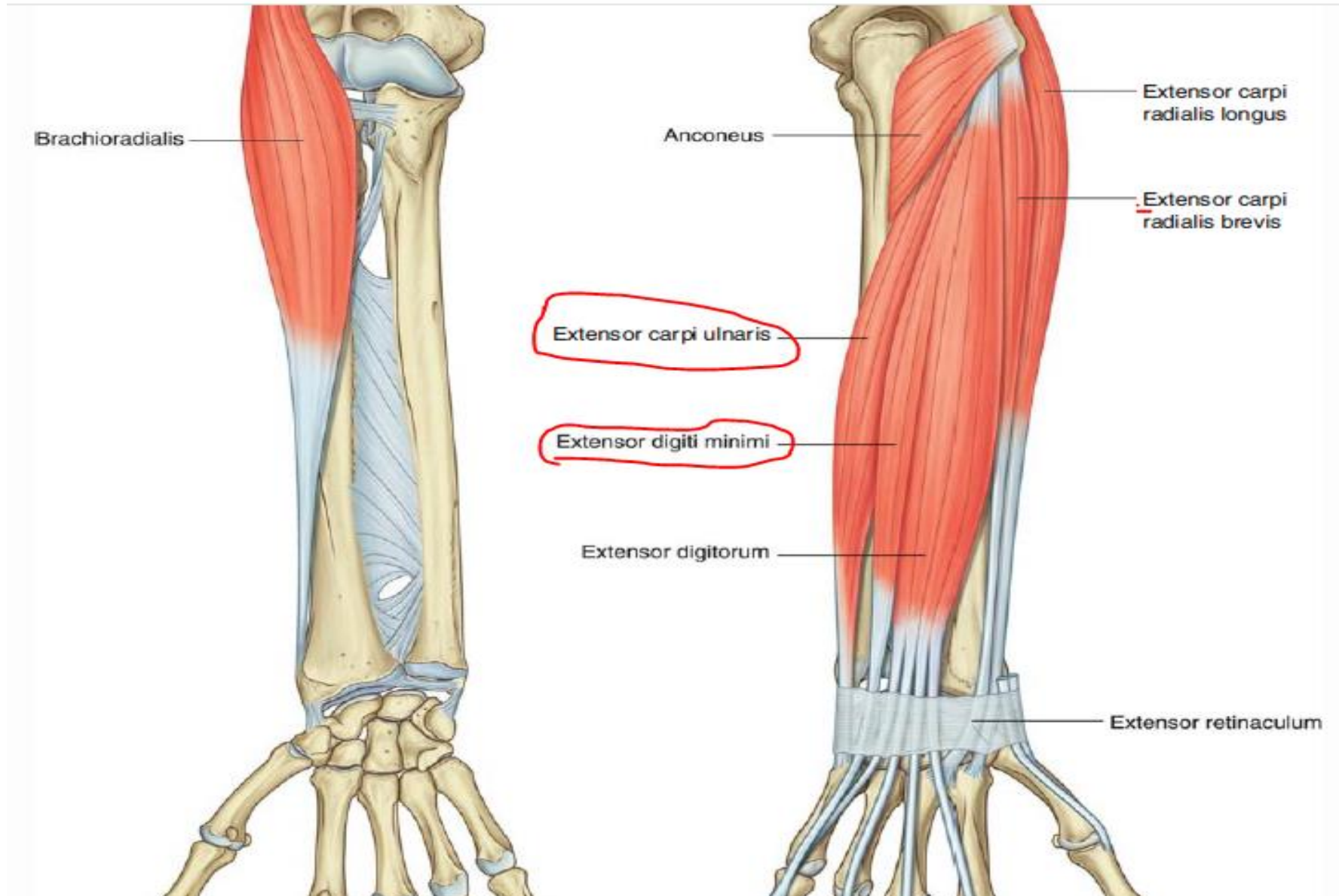
❑ **Origin:** Lateral epicondyle of humerus

❑ **Insertion:** Extensor expansion of 5th finger

❑ **Nerve supply:** Posterior interosseous nerve

❑ **Action:** Extends 5th finger primarily at metacarpophalangeal joint, secondarily at interphalangeal joint

Extensor Digiti Minimi,,,



Extensor Carpi Ulnaris

- Is a long fusiform muscle located on the medial border of the forearm

❖ **Has two heads:**

- ✓ **A humeral head:** from the common extensor tendon and
- ✓ **An ulnar head:** that arises by a common aponeurosis attached to the posterior border of the ulna and deep fascia of the forearm.

❑ **Origin:** Lateral epicondyle of humerus; posterior border of ulna

❑ **Insertion:** Dorsal aspect of base of 5th metacarpal

❑ **Innervation:** Posterior interosseous nerve

❑ **Main Action:** Extends and adducts hand at wrist joint.

Supinator

- The supinator lies deep in the cubital fossa
- Along with the brachialis, forms its floor of the cubital fossa
- Envelops the neck and proximal part of the shaft of the radius
- The deep branch of the radial nerve passes between its muscle fibers, separating them into superficial and deep parts, as it passes from the cubital fossa to the posterior part of the arm.
- **Has two heads of origin**, which insert together on the proximal aspect of the radius
- The supinator is the prime mover for slow, unopposed supination, especially when the forearm is extended.

Supinator,,

- The biceps brachii also supinates the forearm and is the prime mover during rapid and forceful supination against resistance when the forearm is flexed (e.g., when a right-handed person drives a screw).

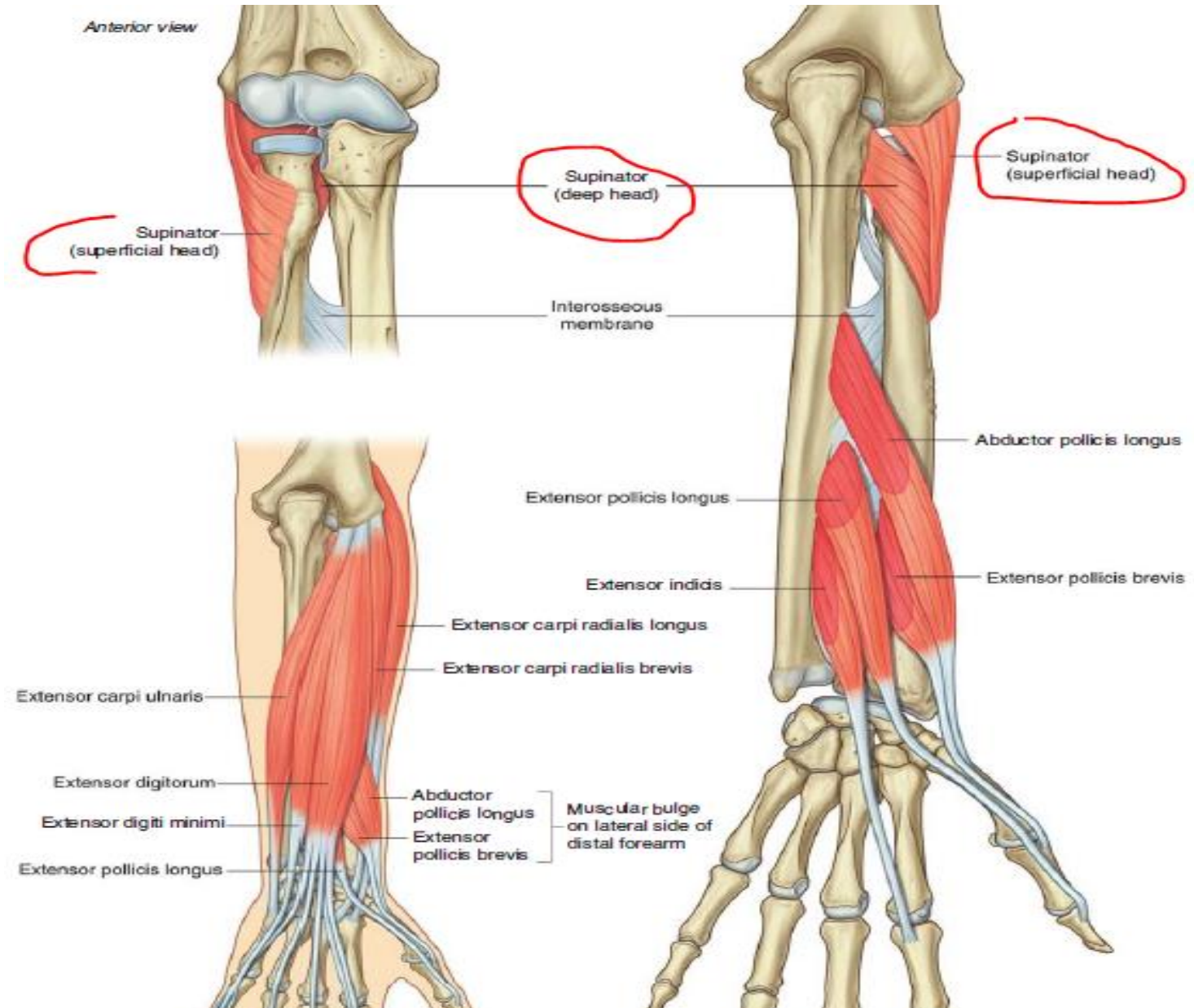
❑ **Origin:** Lateral epicondyle of humerus; radial collateral and anular ligaments & crest of ulna

❑ **Insertion:** Lateral posterior, and anterior surfaces of proximal third of radius

❑ **Nerve supply:** Deep branch of radial nerve

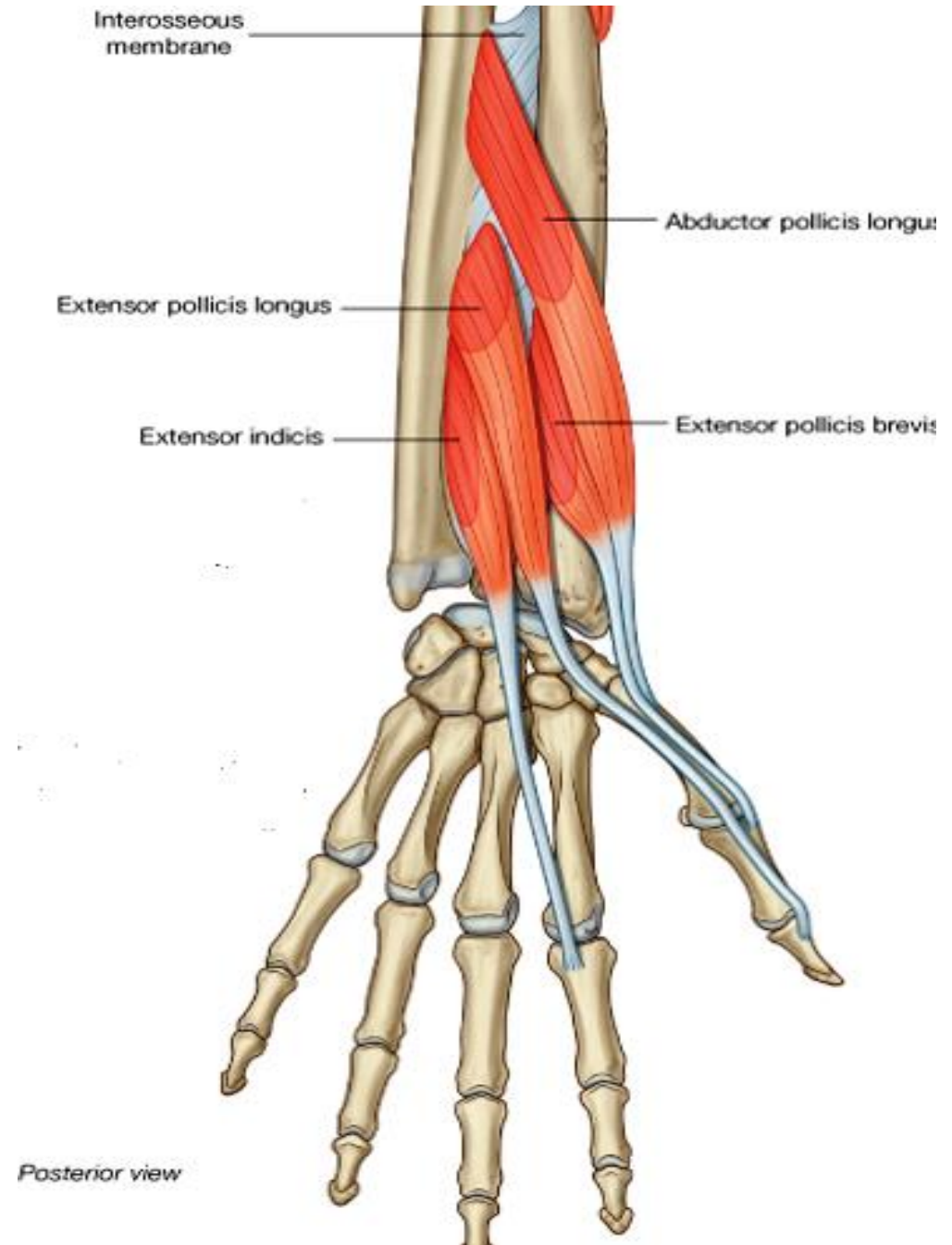
❑ **Main action:** Supinates forearm; rotates radius to turn palm anteriorly or superiorly.

Supinator,,,



Abductor Pollicis Longus

- Has a long, fusiform belly
- Its tendon, and sometimes its belly, is commonly split into two parts
- Its tendon passes deep to the extensor retinaculum.
- **Origin:** Posterior surface of proximal halves of ulna, radius, and interosseous membrane
- **Insertion:** Base of 1st metacarpal
- **Nerve supply:** Posterior interosseous nerve, continuation of deep branch of radial nerve
- **Main action:** Abducts thumb and extends it at carpometacarpal joint



Extensor Pollicis Brevis

- A short fusiform extensor of the thumb
- **Origin:** Posterior surface of distal third of radius and interosseous membrane
- **Insertion:** Dorsal aspect of base of proximal phalanx of thumb
- **Nerve supply:** Posterior interosseous nerve, continuation of deep branch of radial nerve
- **Main action:** Extends proximal phalanx of thumb at metacarpophalangeal joint; extends carpometacarpal joint

Extensor Pollicis Longus

- Is larger and its tendon is longer than that of the EPB
- The tendon passes under the extensor retinaculum in its own tunnel
- The tendons of the APL and EPB bound the anatomical snuff box anteriorly, and the tendon of the EPL bounds it posteriorly
- **Origin:** Posterior surface of middle third of ulna and interosseous membrane
- **Insertion:** Dorsal aspect of base of distal phalanx of thumb
- **Nerve supply:** Posterior interosseous nerve
- **Main action:** Extends distal phalanx of thumb at interphalangeal joint; extends metacarpophalangeal and carpometacarpal joints.

Extensor Indicis

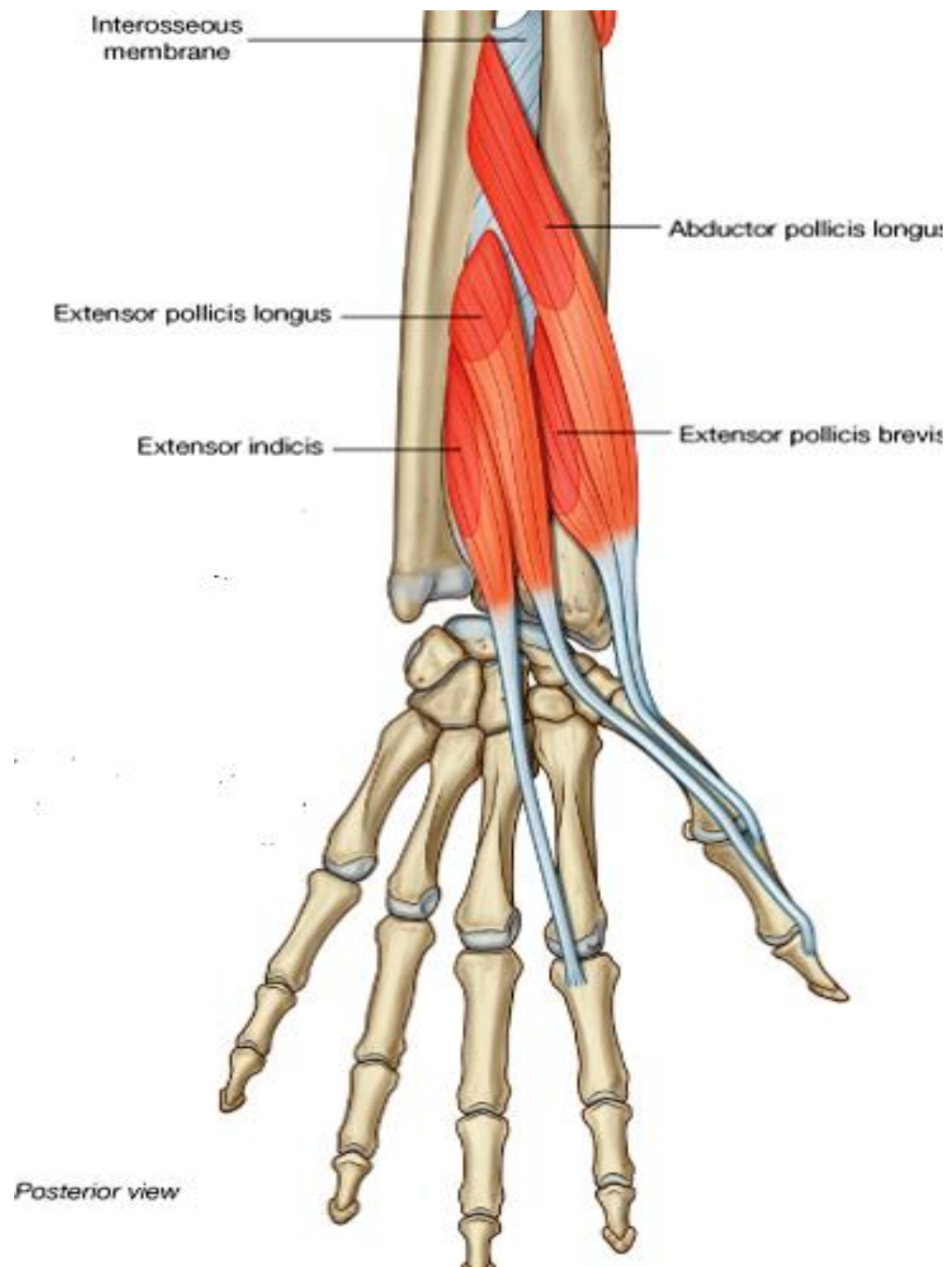
- Has a narrow, elongated belly that lies medial to and alongside that of the EPL
- May act alone or together with the extensor digitorum to extend the index finger at the proximal interphalangeal joint, as in pointing
- It also helps extend the hand

❑ **Origin:** Posterior surface of distal third of ulna and interosseous membrane

❑ **Insertion:** Extensor expansion of 2nd finger

❑ **Nerve supply:** Posterior interosseous nerve

❑ **Main action:** Extends 2nd finger (enabling its independent extension).



The nerves of the forearm

□ The major nerves of the forearm are the **median**, **ulnar**, and **radial**

❖ **Median nerve and its Branches include:**

1. **Muscular branch:** It innervates; FPL, 1/2FDP and pronator quadratus
2. **Anterior interosseous nerve**
3. **Palmar cutaneous branch of median nerve:** Passes superficial to flexor reticulum to reach skin of lateral two- third of the palm

❖ **Ulnar nerve: it has Muscular branch and cutaneous branch of ulnar nerve**

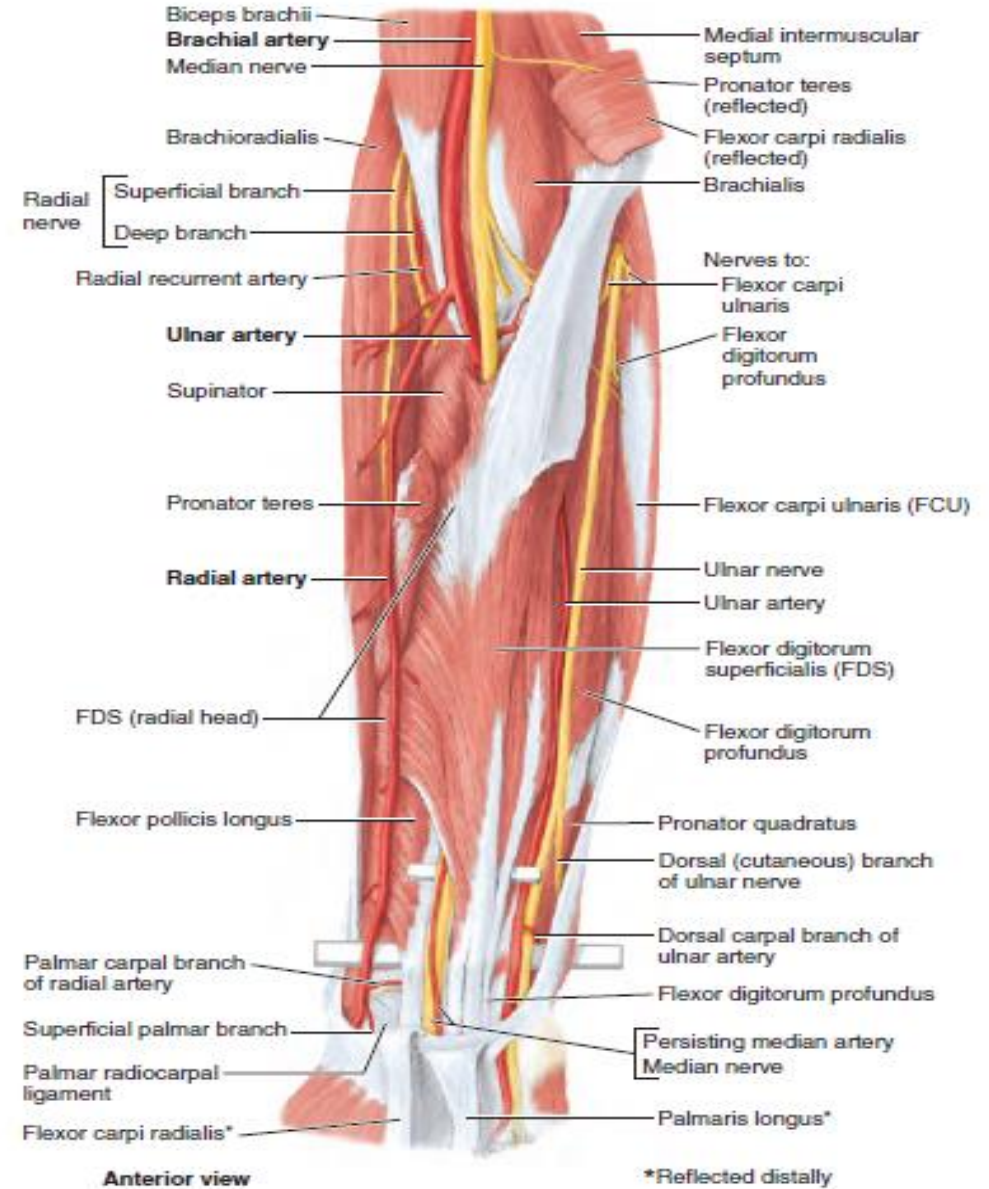
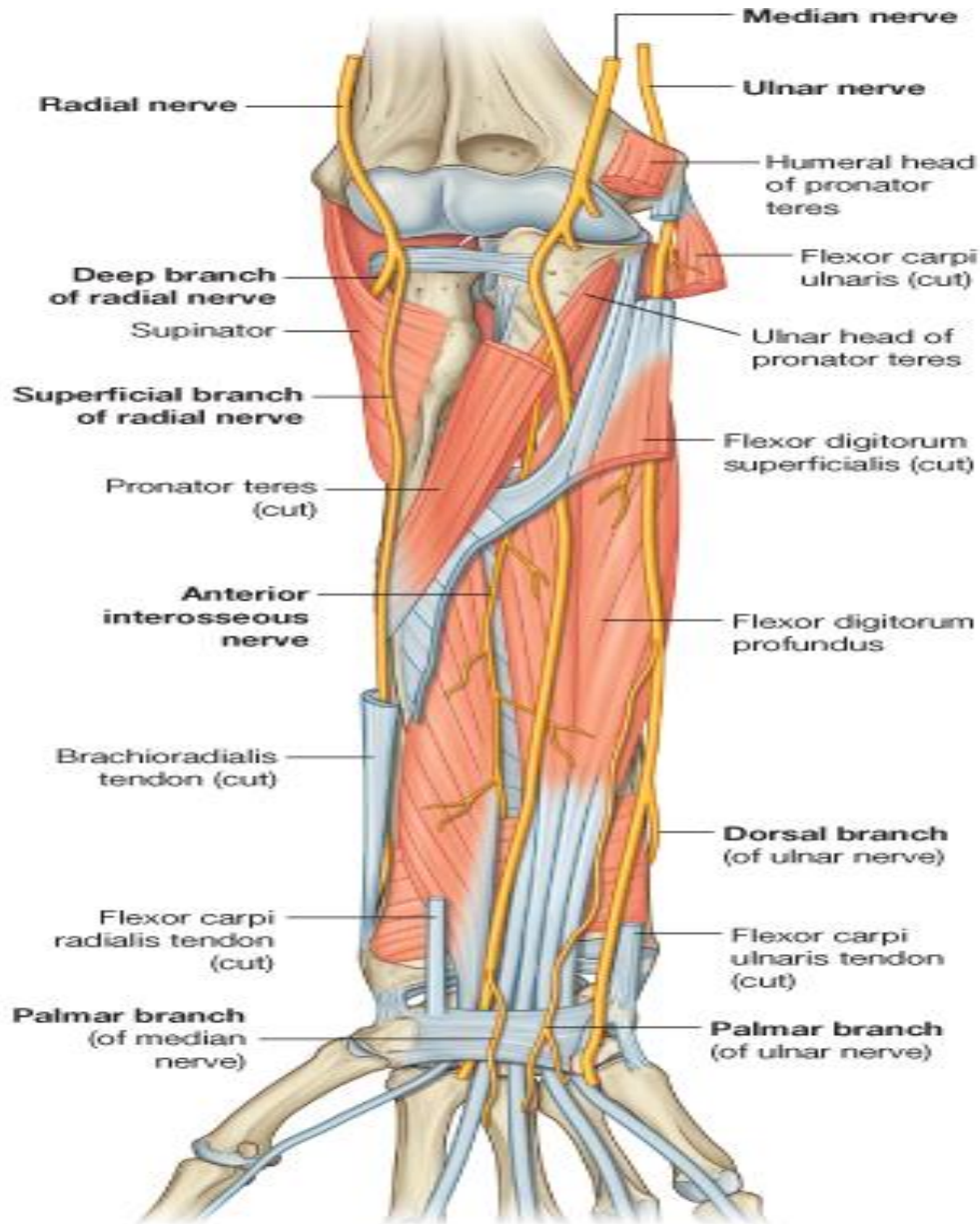
- ⊕ **Muscular branch** It innervates; flexor carpi ulnaris, medial 1/2FDP, hypothenar muscles, all Interossei, 3rd and 4th lumbricals, adductor pollicis
- ⊕ **Cutaneous branch innervates** ulnar side of both aspects of the hand
 - both surfaces of the ulnar 1 & 1/2 fingers

The nerves of the forearm,,,

❖ Radial nerve:

- ✚ At the level of the lateral epicondyle its posterior interosseous nerve is given off, which winds round the radius within the supinator muscle then sprays out to be distributed to the extensor muscles of the forearm
- ✚ The radial nerve itself continues as the superficial radial nerve, lying deep to brachioradialis
- ✚ Above the wrist, it emerges posteriorly from beneath this muscle to end by dividing into cutaneous nerves to the posterior aspects of the radial 3 & 1/2 digits

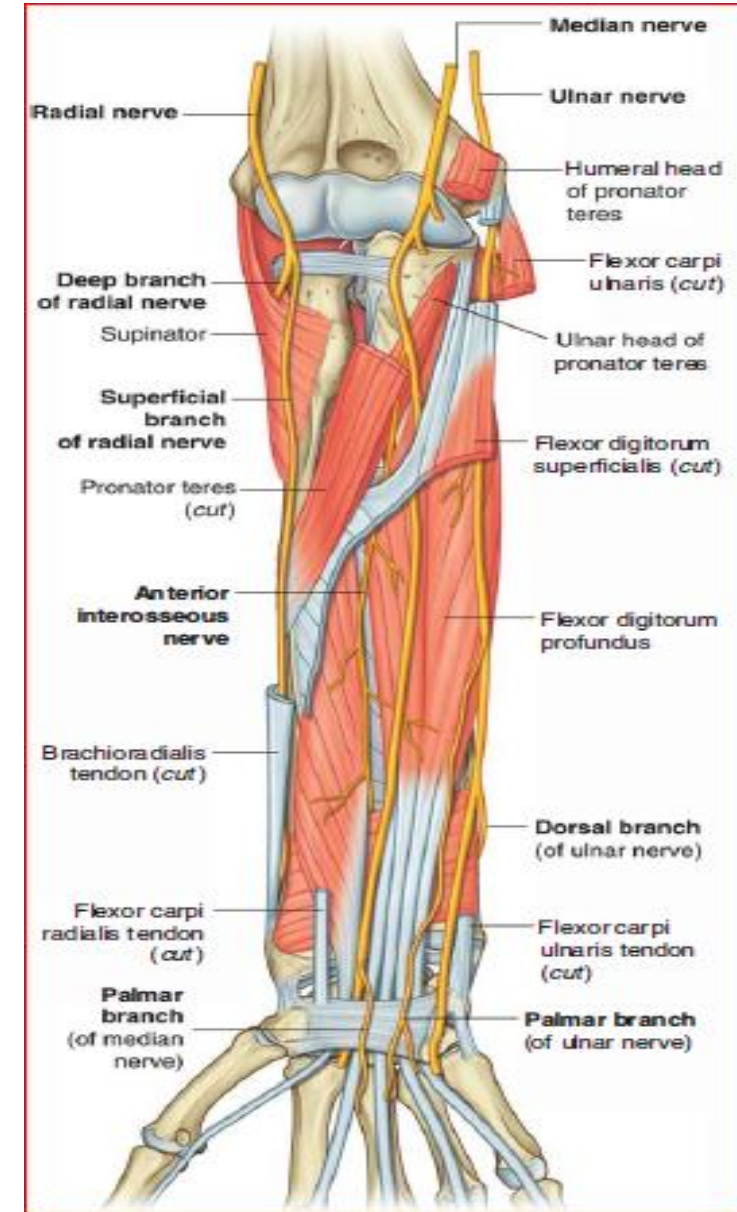
The nerves of the forearm,,,



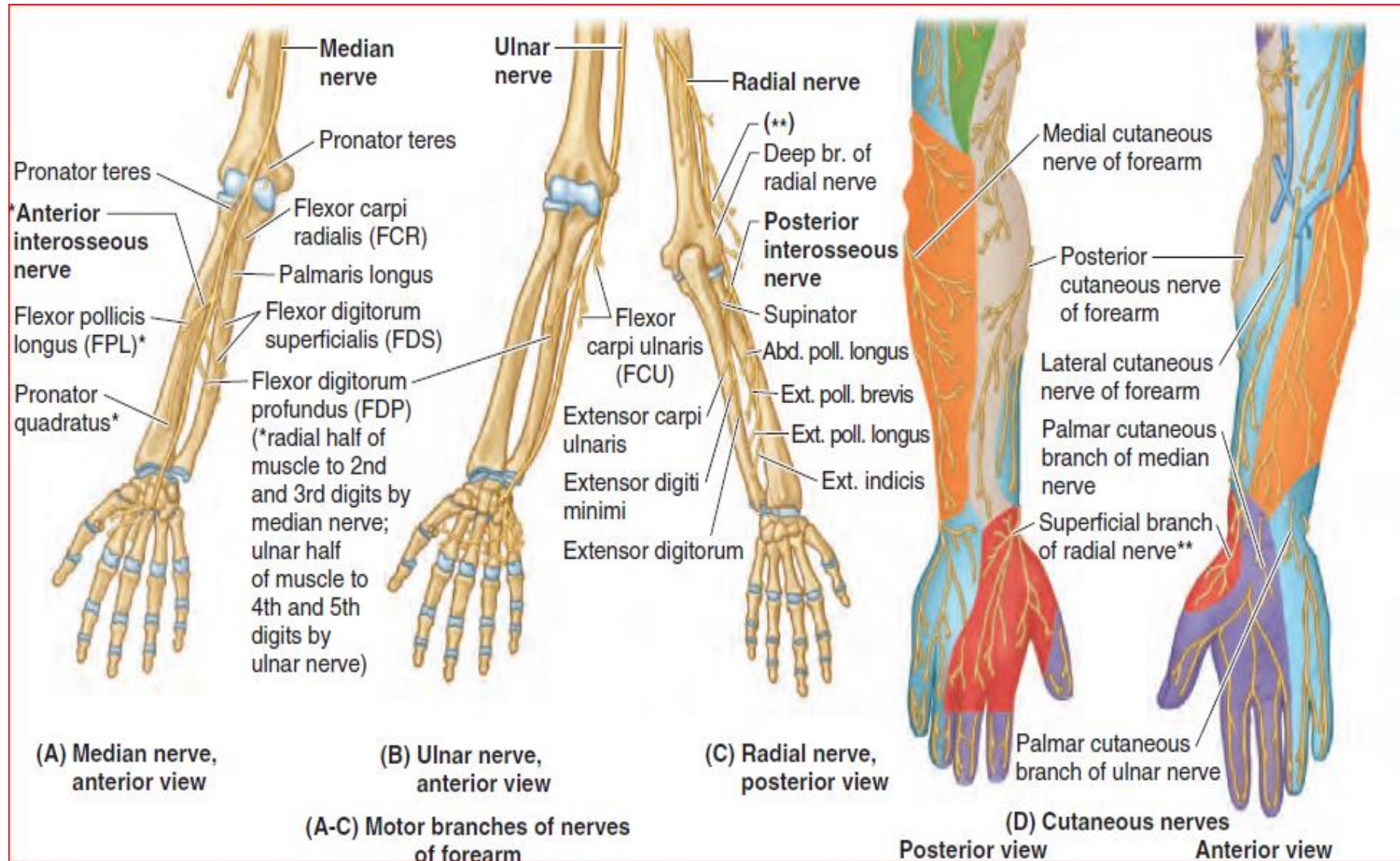
The nerves of the forearm,,,

❖ Nerve of the Posterior Compartment of the Forearm

- Unlike the medial and ulnar nerves, **the radial nerve** serves motor and sensory functions in both the arm and the forearm (but only sensory functions in the hand).
- The *posterior cutaneous nerve of the forearm* arises from the radial nerve in the posterior compartment of the arm, as it runs along the radial groove of the humerus.



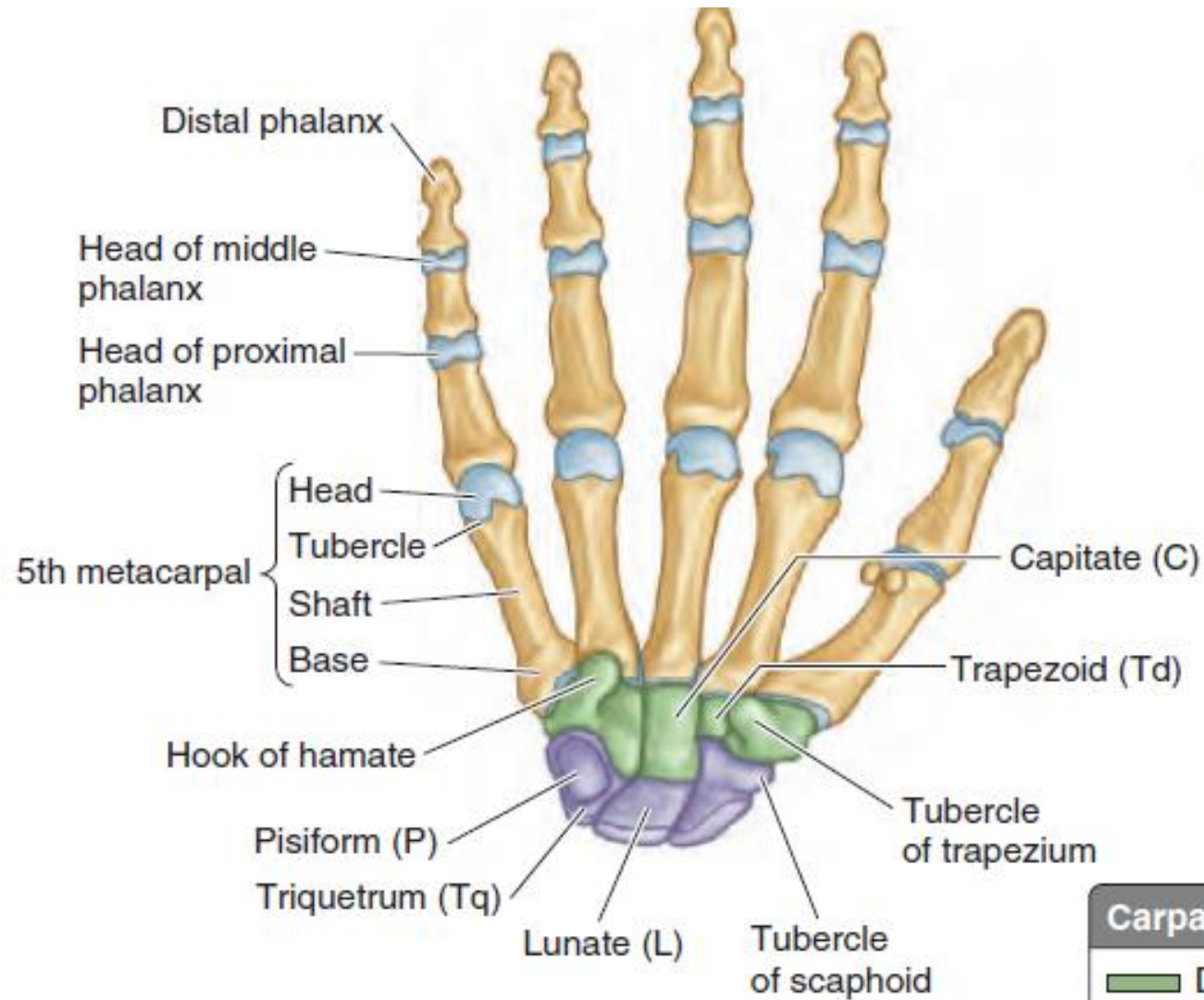
The nerves of the forearm,,



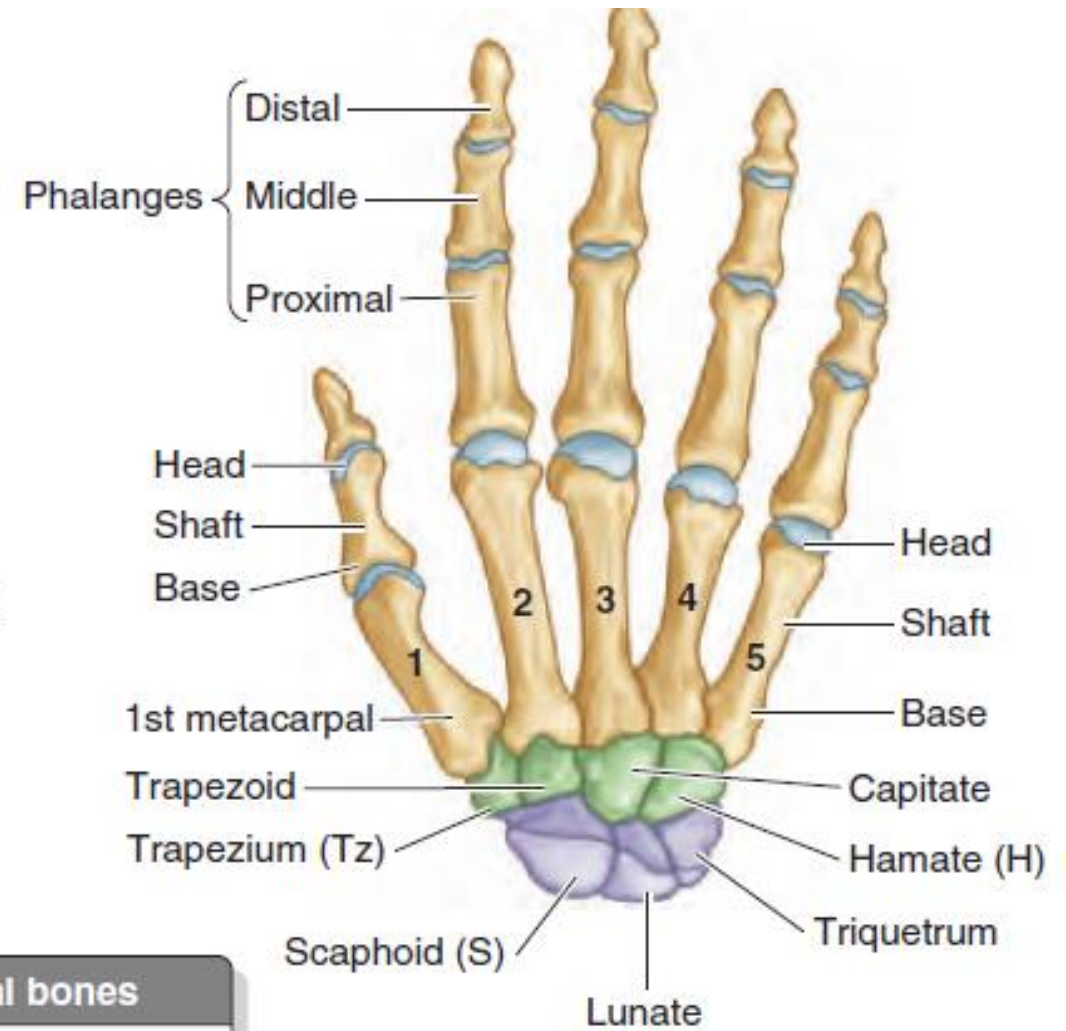
The Hand

- The **hand** is the manual part of the upper limb distal to the forearm.
 - The ***hand*** is one of the most useful structures in the human body
 - Is subdivided into three parts: **the wrist, the hand proper and the digits**
- ➔ injuries to the hand should be taken care of carefully & immediately.
- The skeleton of the hand consists of *carpals* in the wrist, *metacarpals* in the hand proper, and *phalanges* in the digits (fingers).
 - ➔ The bones, consists of **27 bones**, divide the hand into ***palmar & dorsal components***.

The Hand,,



(A) Anterior (palmar) view



(B) Posterior (dorsal) view

Carpal bones

- Distal row
- Proximal row

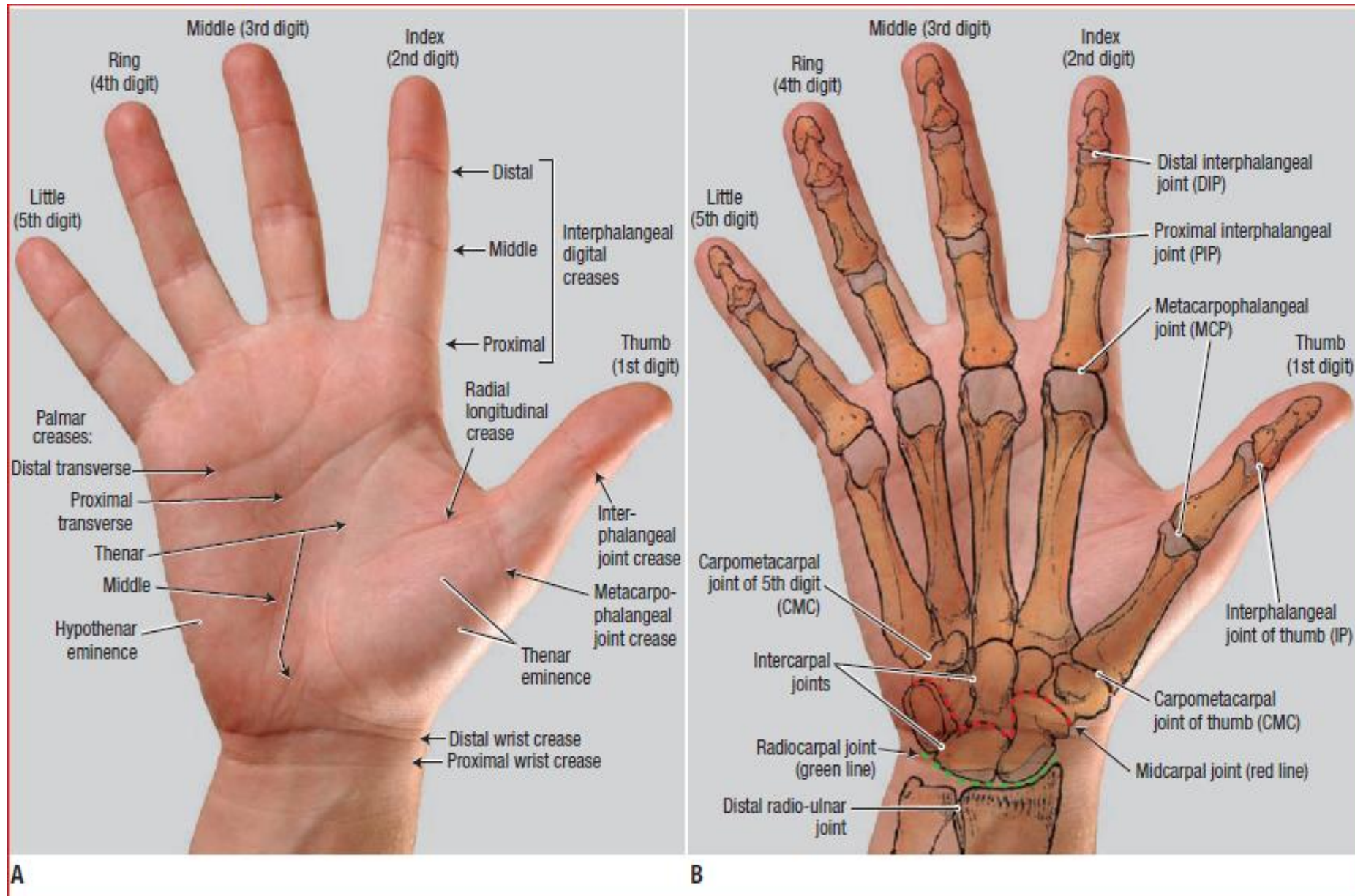
The Hand,,

- The hand has an anterior surface (**palm**) and a dorsal surface (**dorsum of hand**)
- **The hand is a mechanical and sensory tool.**
- Many of the features of the upper limb are designed to facilitate positioning the hand in space.
- The skeleton of the hand consists of carpals in the wrist, metacarpals in the hand proper, and phalanges in the digits.
- The digits are numbered from one to five, beginning with the thumb and ending with the little finger.

The Palm of the Hand,,,

- The skin on the palm is thick because it must withstand the wear and tear of work and play.
- It is richly supplied with sweat glands but contains no hair or sebaceous glands.
- The palmar skin presents several more or less constant flexion creases, where the skin is firmly bound to the deep fascia, that help locate palmar wounds and underlying structures
- **Wrist creases:**
 - proximal, middle, distal
 - The distal wrist crease indicates the proximal border of the flexor retinaculum

The Palm of the Hand,,



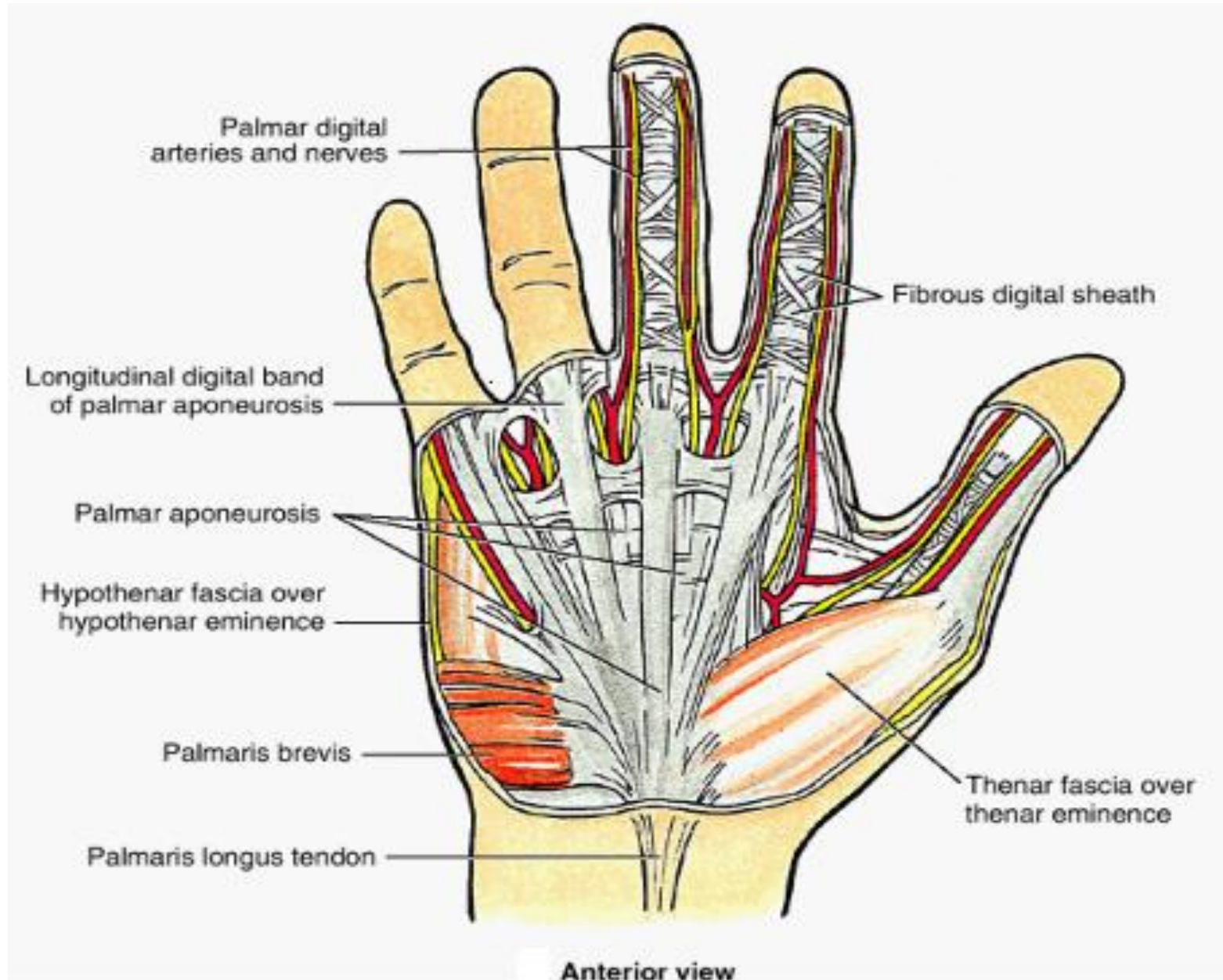
Fascia of the Palm

❑ Is continuous with the antebrachial fascia & the fascia of the dorsum of the hand is thin over the **thenar and hypothenar eminences**, but it is thick centrally where it forms the **fibrous palmar aponeurosis** and in the fingers where it forms the digital sheaths.

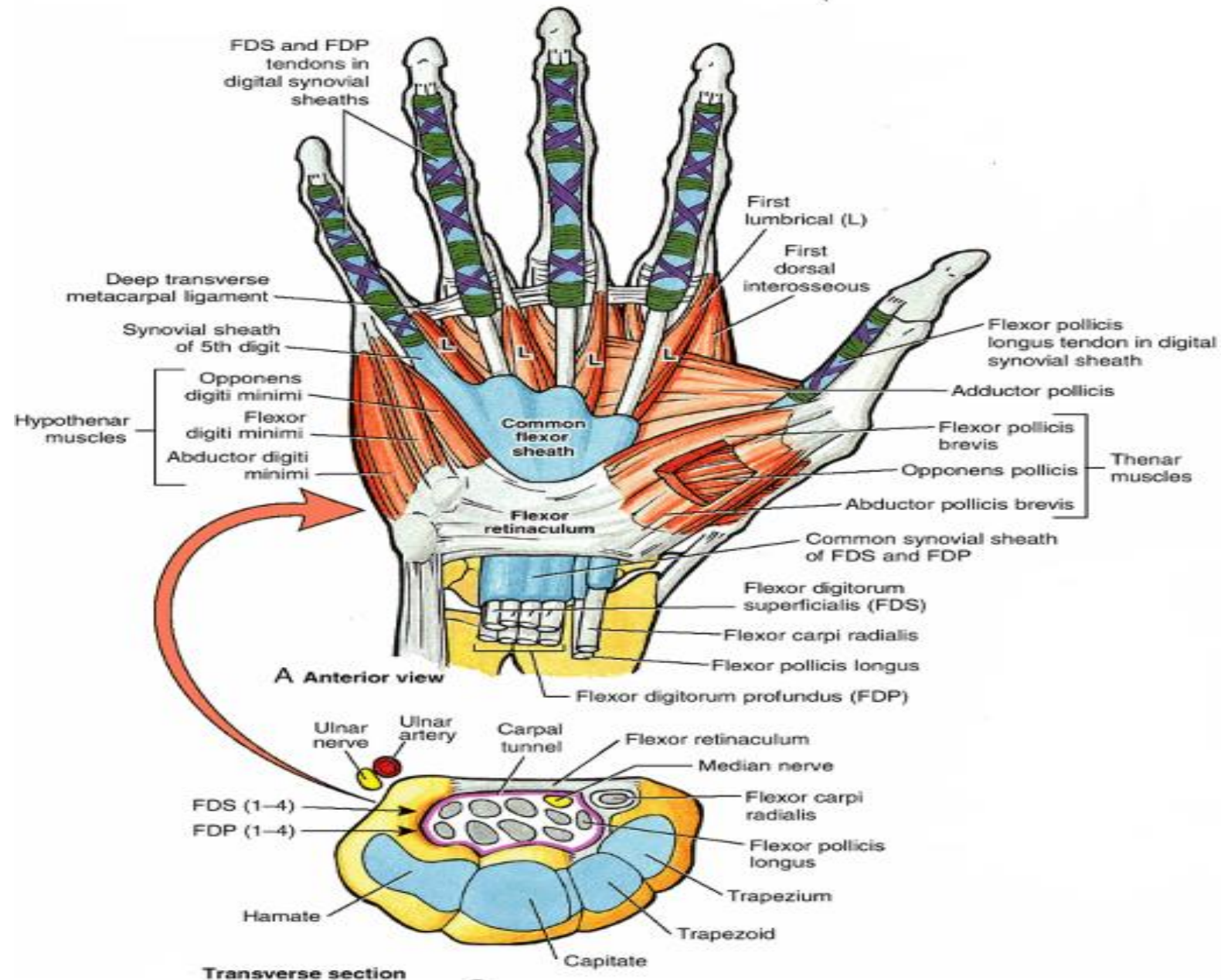
❑ **The palmar aponeurosis:** is a strong, part of the deep fascia of the palm covers the soft tissues and overlies the long flexor tendons.

- The proximal end or apex of the triangular palmar aponeurosis is continuous with the flexor retinaculum and the palmaris longus tendon.

Fascia of the Palm,,,



Fascia of the Palm,,



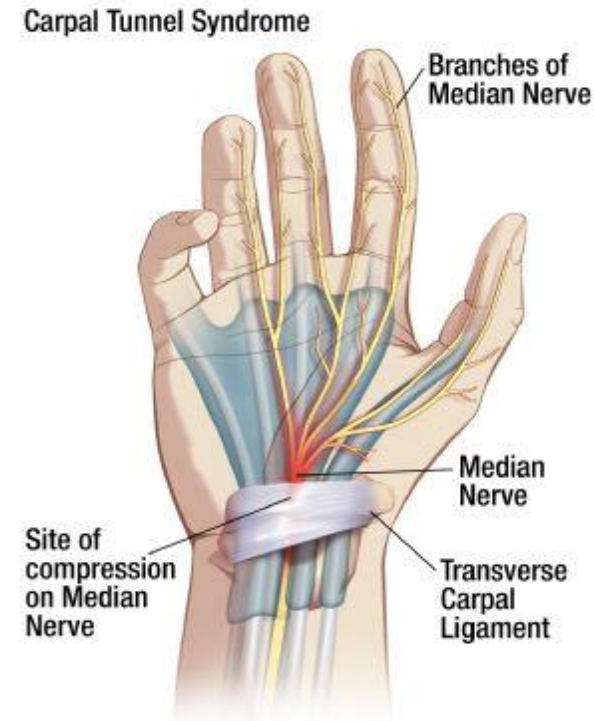
Flexor Retinaculum:

- a thick connective tissue ligament that bridges the space between the medial and lateral sides of the base of the arch and converts the carpal arch into the carpal tunnel.
- The flexor retinaculum holds the tendons to the bony plane at the wrist and prevents them from 'bowing'.

Attachment:

Medially: to pisiformis and
hook of hamate

Laterally: tubercle of scaphoid and
crest of trapezium



Flexor Retinaculum:,,,

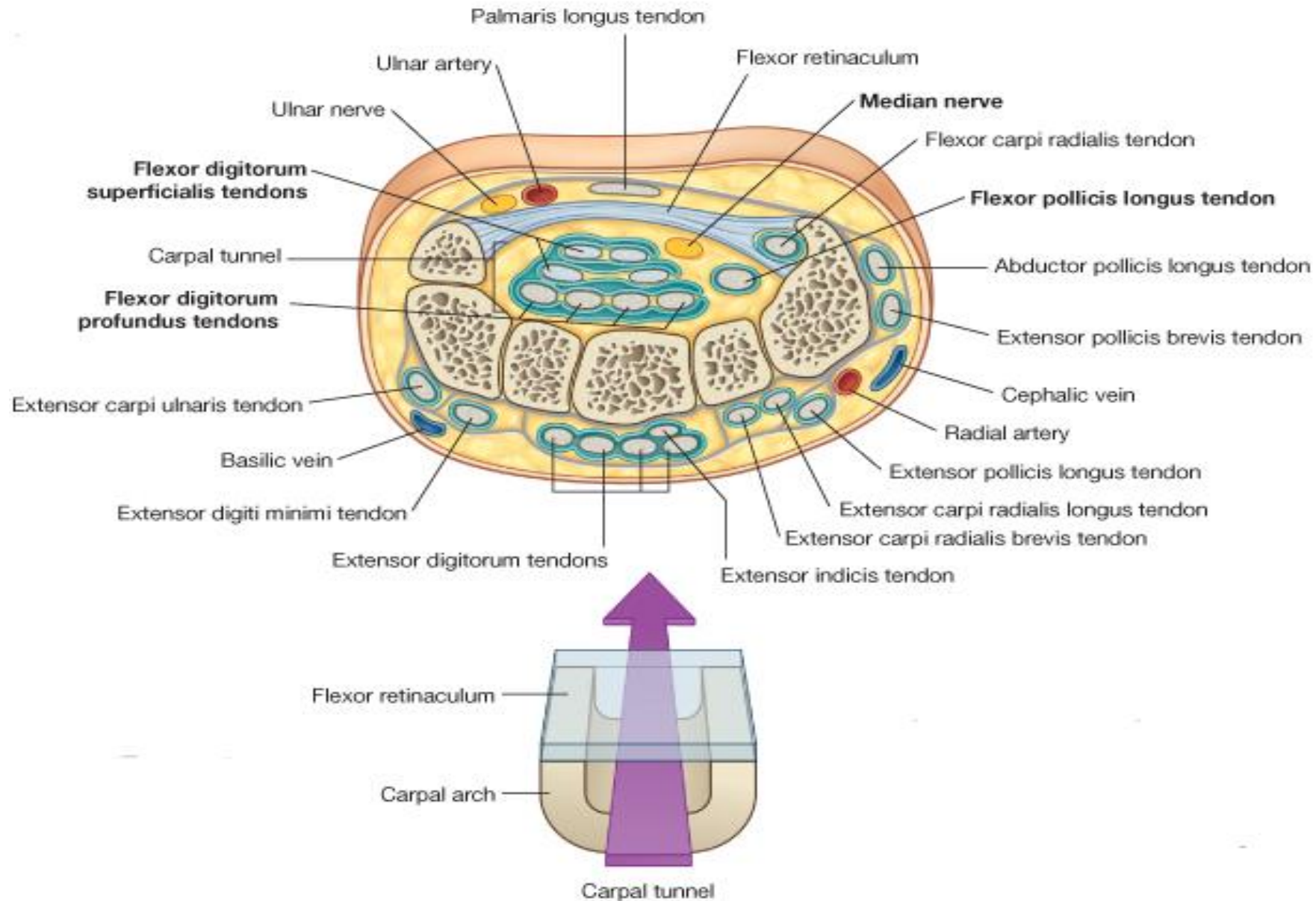
Structures passing **Superficial to flexor retinaculum:**

- Ulnar nerve
- Ulnar artery
- Palmar cutaneous branch of ulnar nerve
- Tendon of palmaris longus
- Palmar cutaneous branch of median nerve

Structures passing **Deep to flexor retinaculum:**(carpal tunnel transmits)

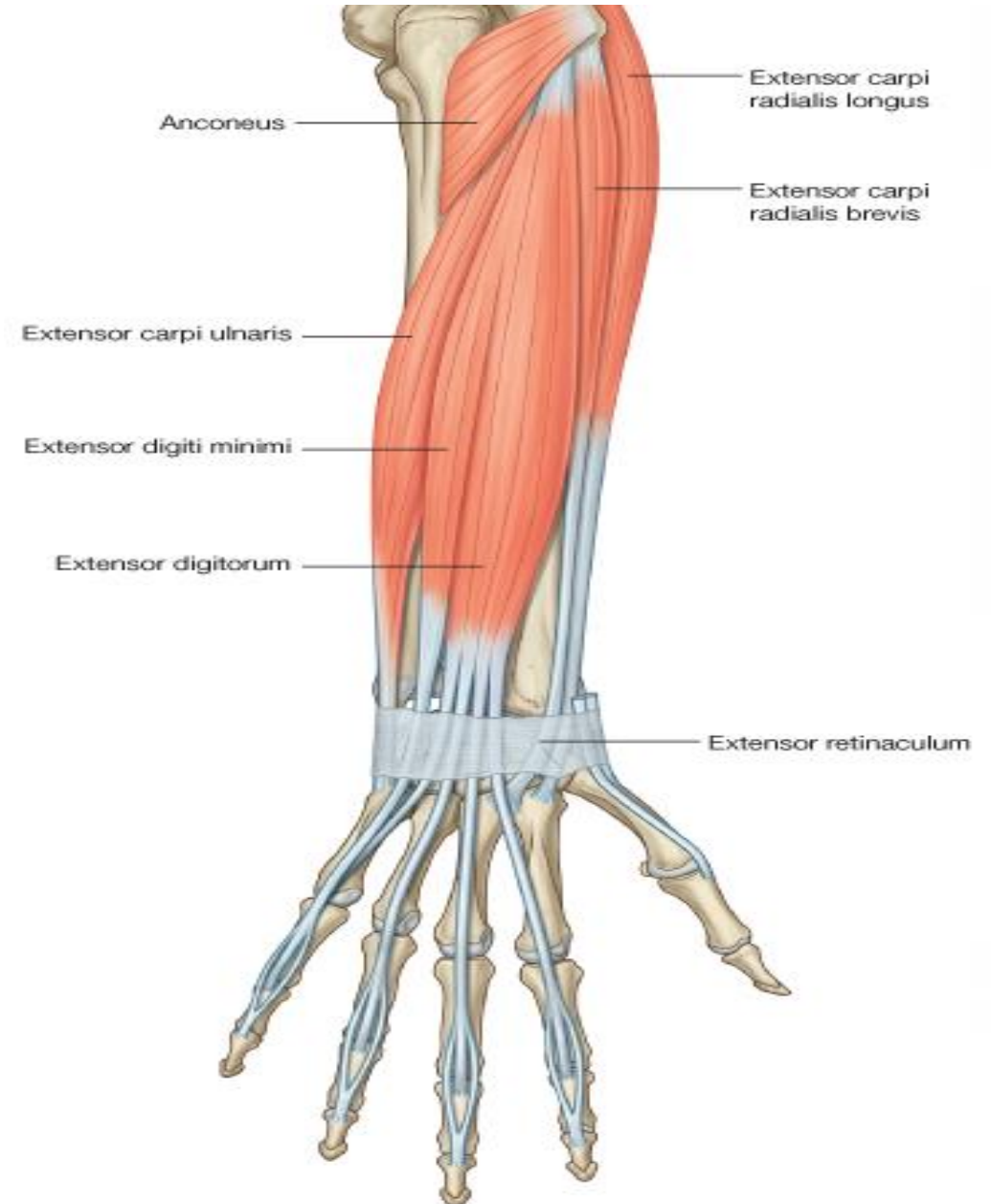
- Tendon of FCR and its synovial sheath in special compartment in front of trapezium
- Tendon of FDS and FDP
- Tendon of FPL
- Median nerve
- Synovial sheath of the tendon of FPL
- Common flexor sheath

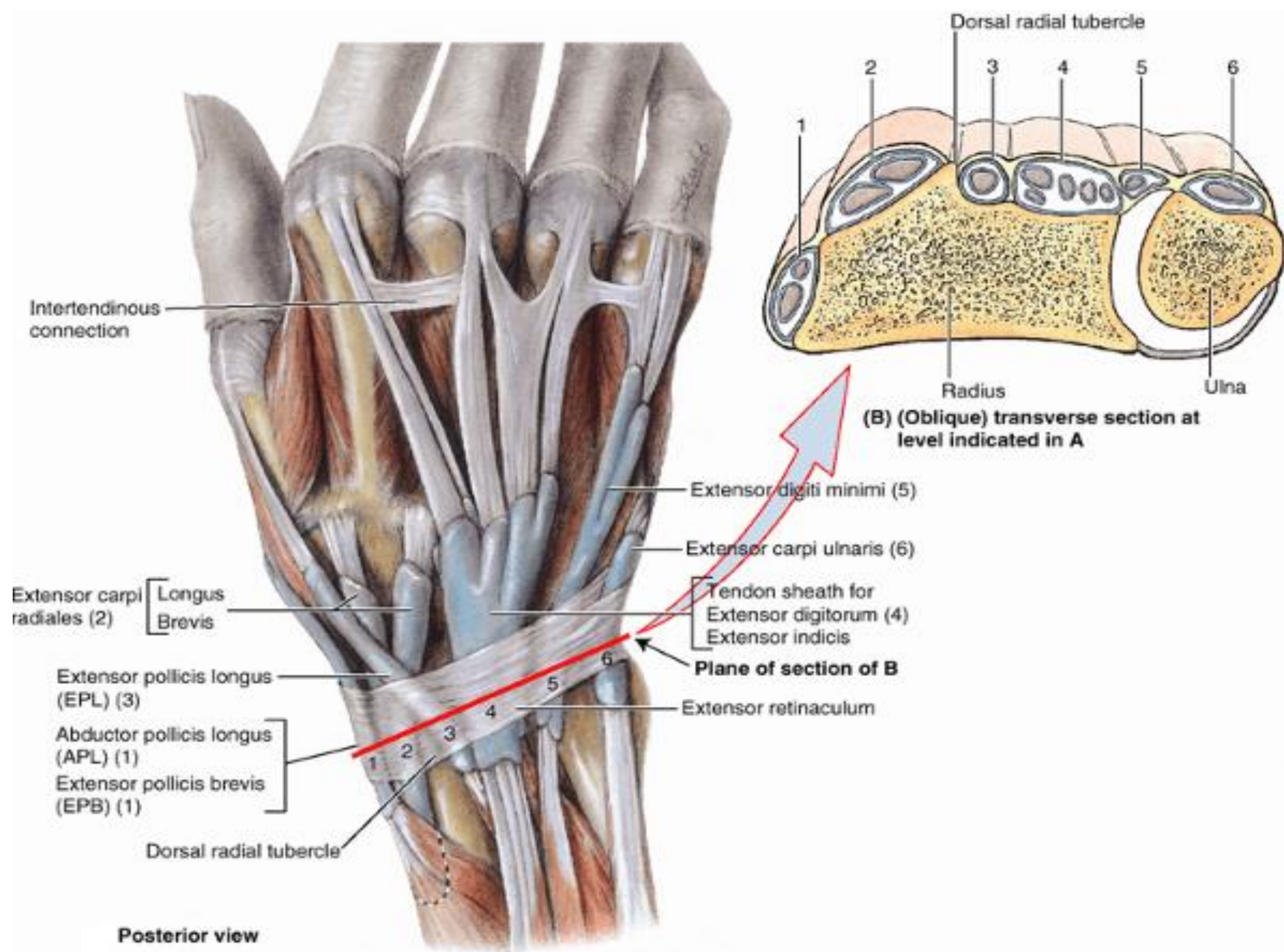
Carpal tunnel



Extensor retinaculum

- **Attachment:**
- **Medial:** to the triquetral and pisiform bones
- **Lateral:** to the lower part of anterior aspects of radius
- Structures passing superficial to extensor retinaculum:
 - Beginning of cepalic vein laterally
 - Beginning of basalic vein medially
 - Terminal branch of radial nerve





Muscles of the Hand

❖ **The intrinsic muscles of the hand are located in five compartments:**

❑ **The thenar compartment:** abductor pollicis brevis, flexor pollicis brevis and opponens pollicis.

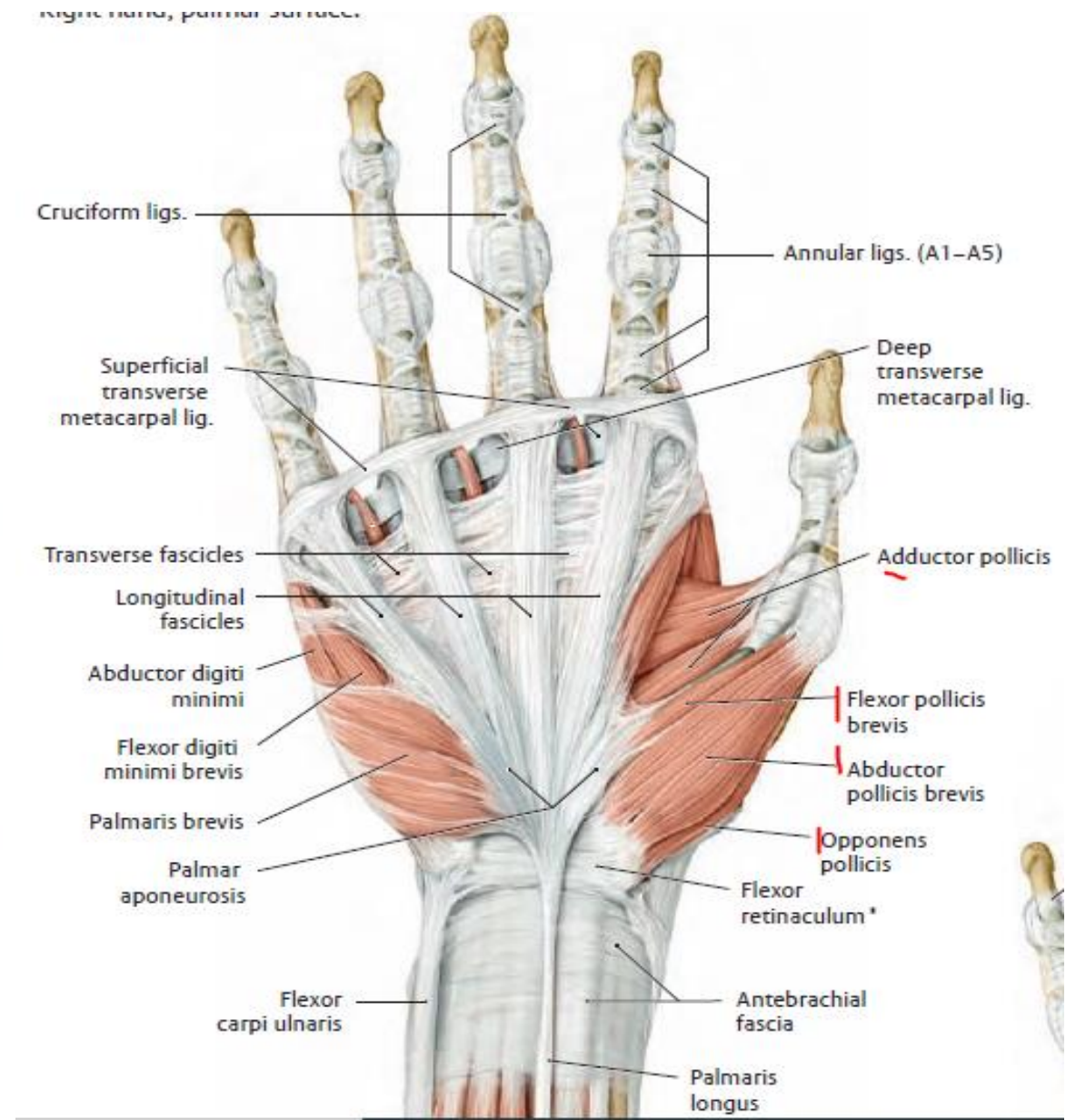
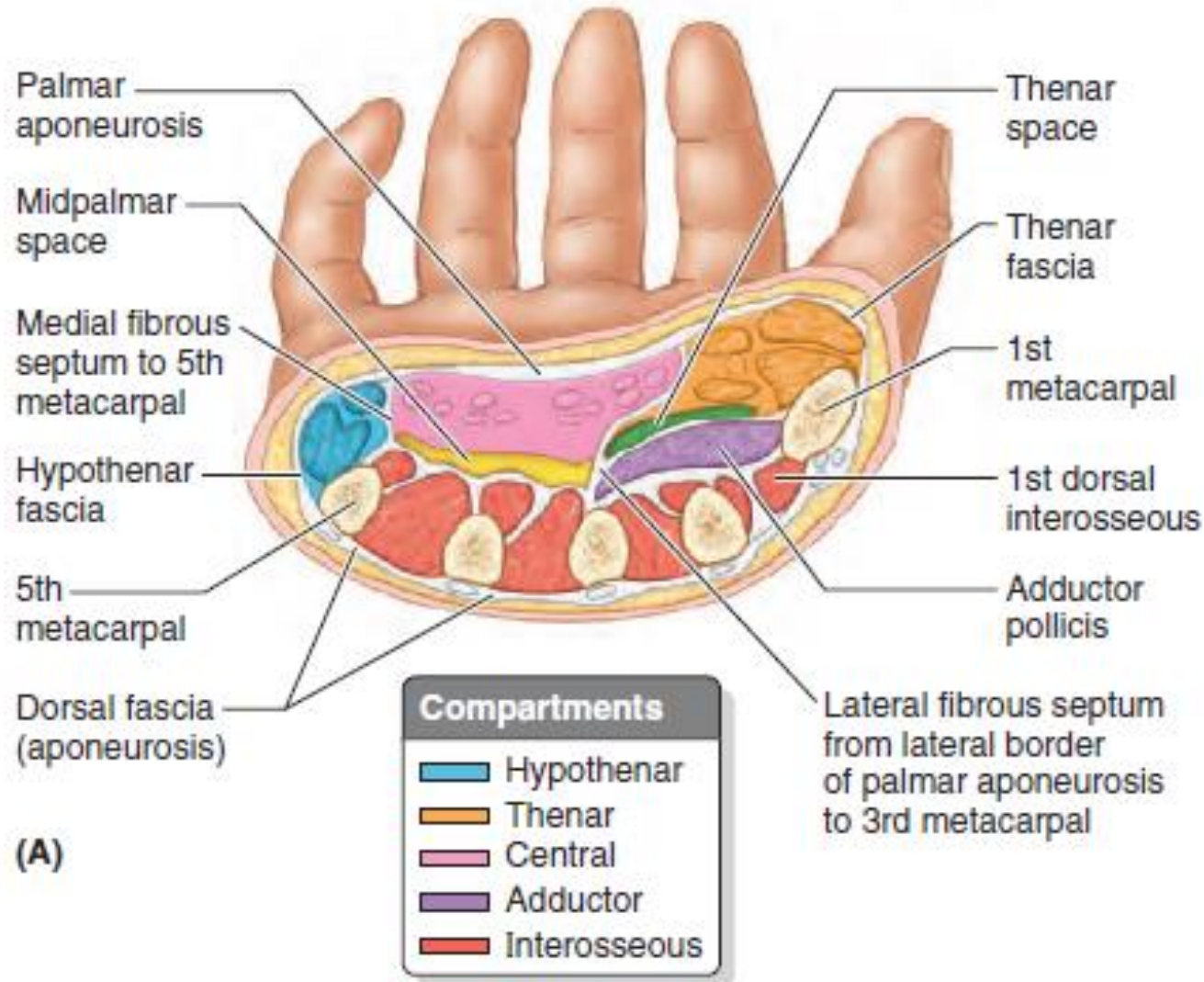
❑ **The adductor compartment:** adductor pollicis muscle

❑ **The hypothenar compartment:** abductor digiti minimi, flexor digiti minimi brevis and opponens digiti minimi.

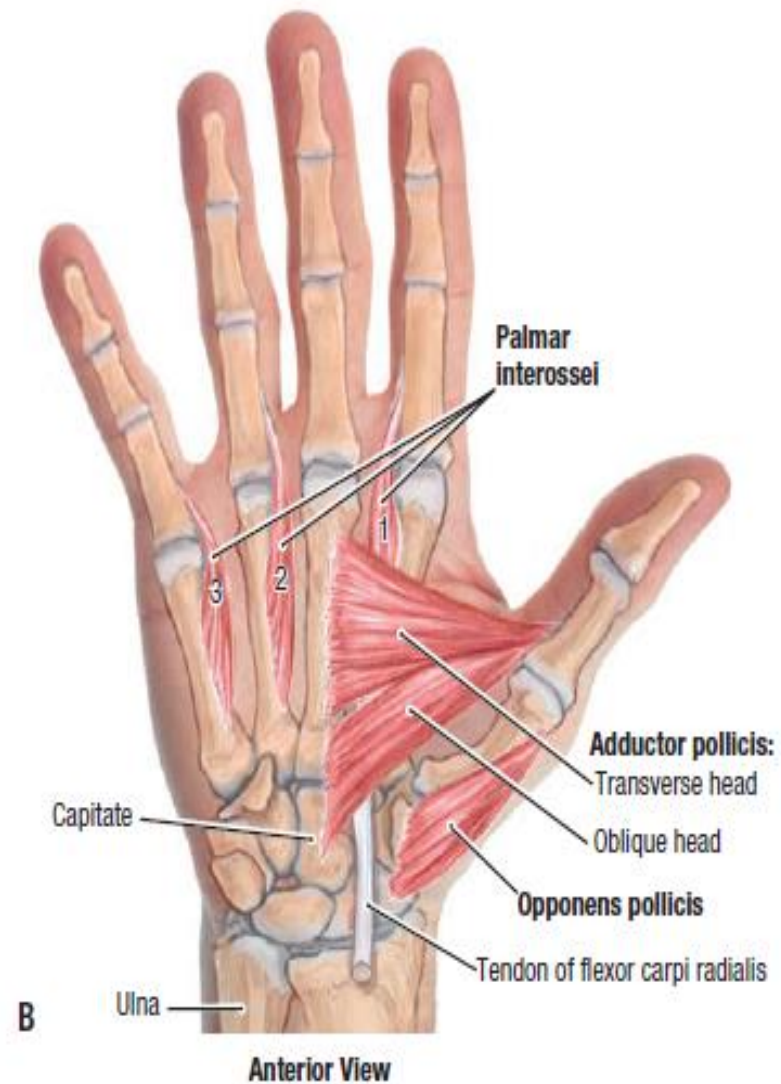
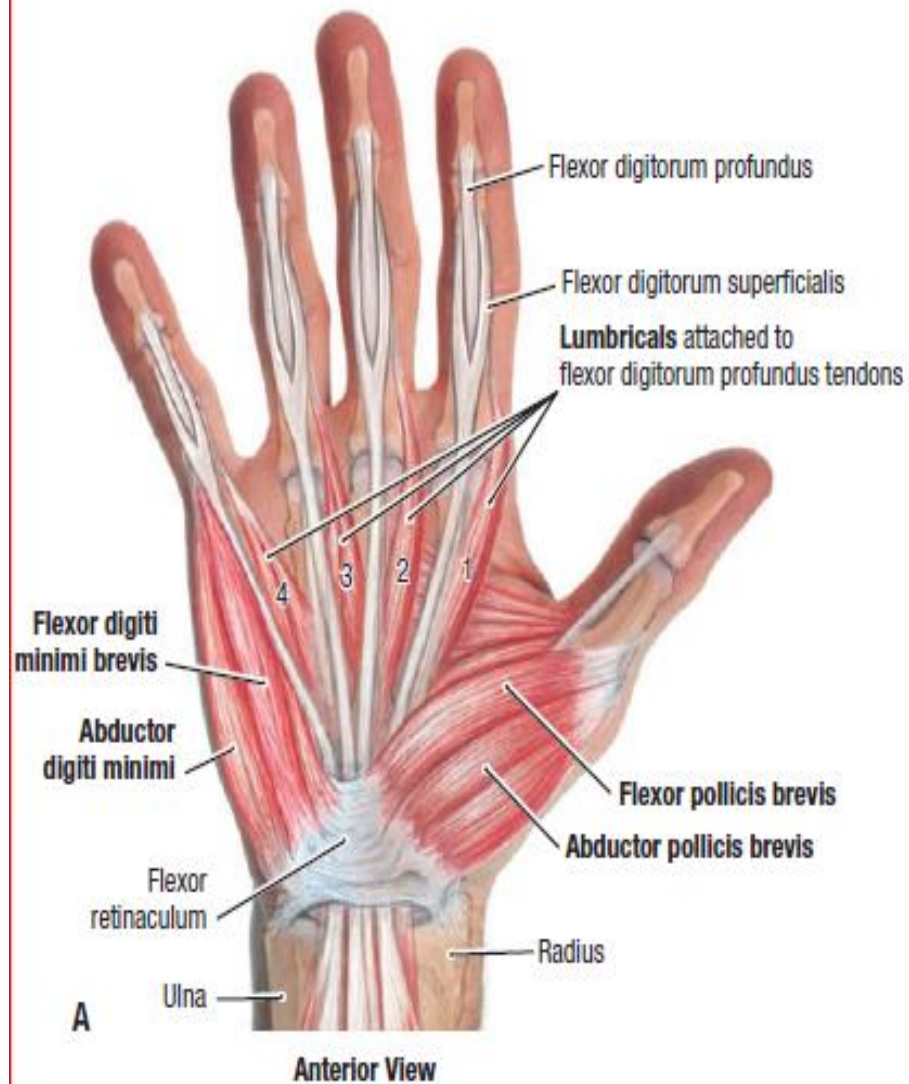
• **The central compartment:** lumbricals, in the *central compartment* with the long flexor tendons

❑ **Interosseous compartments :** The interossei lie in separate b/n metacarpals

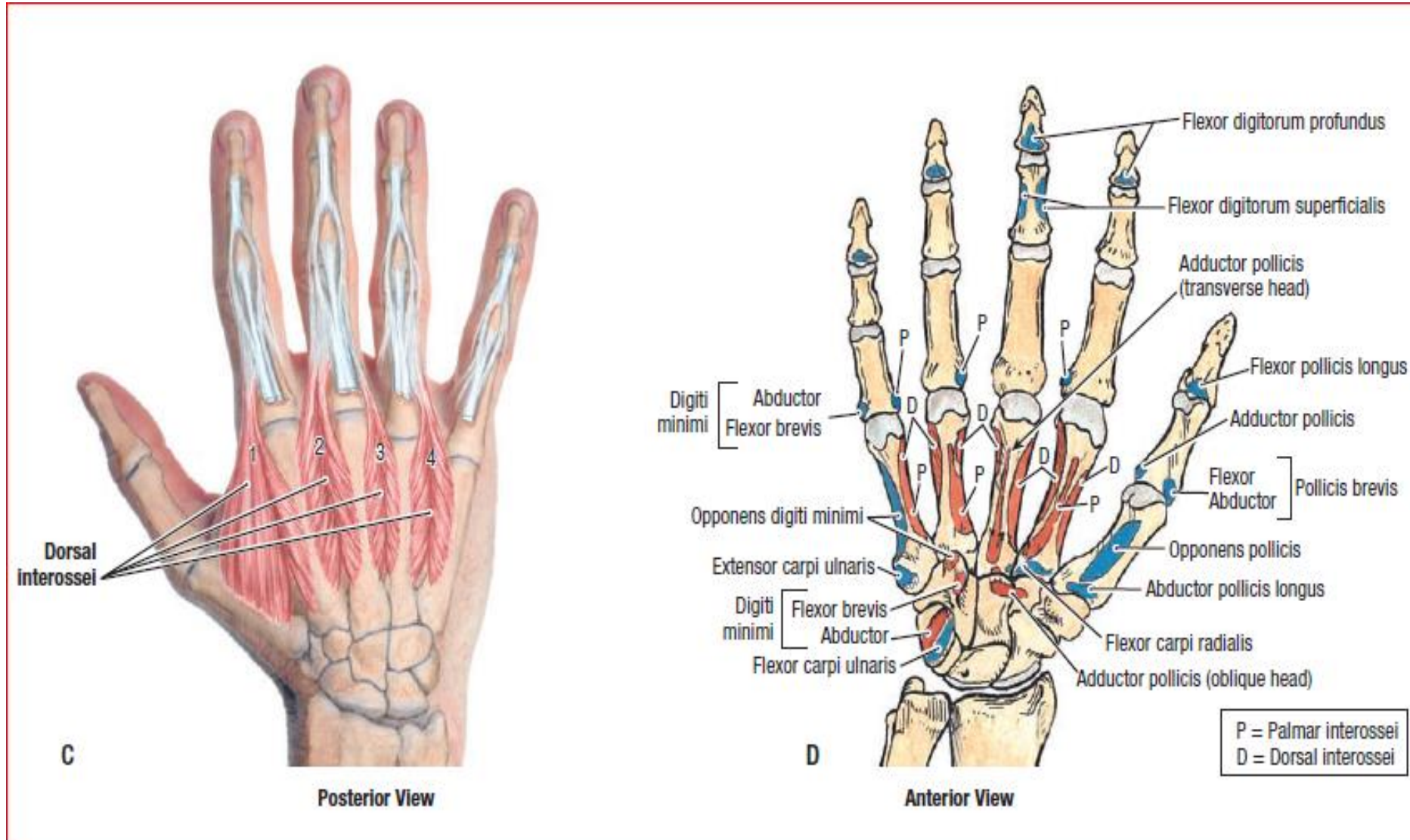
Muscles of the Hand,,



Muscles of the Hand,,



Muscles of the Hand,,



Muscles of the Hand,,

- Movement of the thumb is important for the precise activities of the hand.
- The high degree of freedom of the movements results from the 1st metacarpal being independent, with mobile joints at both ends.

❑ **Several muscles are required to control the freedom of thumb movements:**

- ✓ ***Extension***: extensor pollicis longus, extensor pollicis brevis, and abductor pollicis longus.
- ✓ ***Flexion***: flexor pollicis longus and flexor pollicis brevis.
- ✓ ***Abduction***: abductor pollicis longus and abductor pollicis brevis.
- ✓ ***Adduction***: adductor pollicis and 1st dorsal interosseous.
- ✓ ***Opposition***: opponens pollicis. This movement occurs at the carpometacarpal joint and results in a “cupping” of the palm.

Thenar Muscles of Hand

Muscle	Origin	Insertion	Nerve supply	Action
abductor pollicis brevis (thenar)	flexor retinaculum of wrist scaphoid & trapezium	proximal phalanx of thumb	median	abducts the thumb
flexor pollicis brevis (thenar)	flexor retinaculum trapezium	proximal phalanx of thumb	median	flexes the thumb
opponens pollicis (thenar)	trapezium & flexor retinaculum	lateral border of 1st metacarpal	median	opposes thumb to other digits
adductor pollicis	capitate 2nd & 3rd metacarpals	proximal phalanx of thumb	ulnar	adducts the thumb

Hypothenar Muscles of Hand

Muscle	Origin	Insertion	Nerve supply	Action
abductor digiti minimi	pisiform	proximal phalanx of little finger	ulnar	abducts little finger
flexor digiti minimi	flexor retinaculum hook of hamate	proximal phalanx of little finger	ulnar	flexes little finger
opponens digiti minimi	flexor retinaculum hook of hamate	5th metacarpal	ulnar	opposes little finger to other digits

Short Muscles of the Hand

- Include: the **lumbricals and interossei**

1. Lumbricals The four slender lumbrical muscles were named because of their worm-like form (L. lumbricus, earthworm)

- Are unique because they link flexor tendons with extensor tendons

Origin: Arise from the lateral side of the tendon of the flexor digitorum profundus

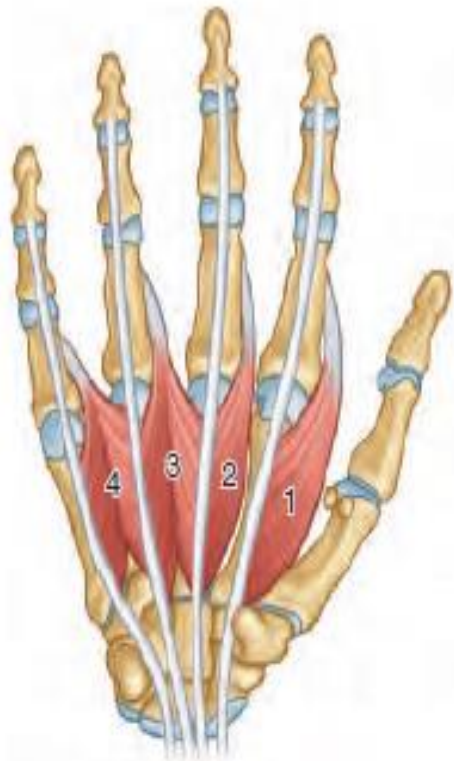
Insertion: their slender tendons pass across the lateral side of the corresponding metacarpophalangeal joint to join the extension expansion

Innervations: the lateral two (1st & 2nd) by median nerve

- The medial two (3rd & 4th) by ulnar nerve

Action: The lumbricals flex the fingers at the metacarpophalangeal joints and extend the interphalangeal joints.

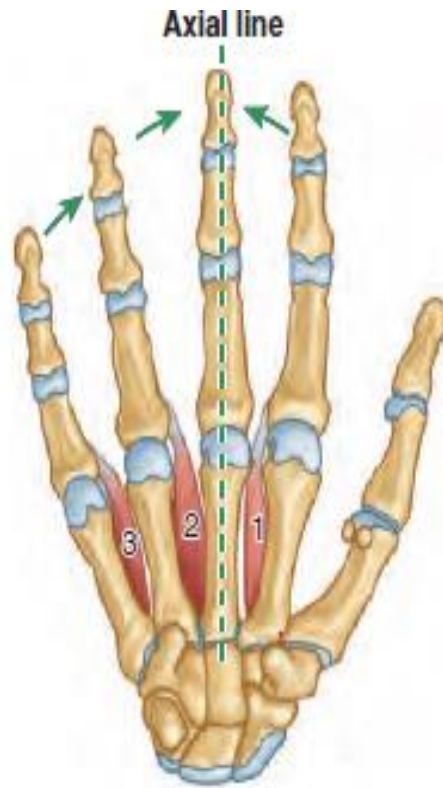
Short Muscles of the Hand,,



(C) Lumbricals (1-4)

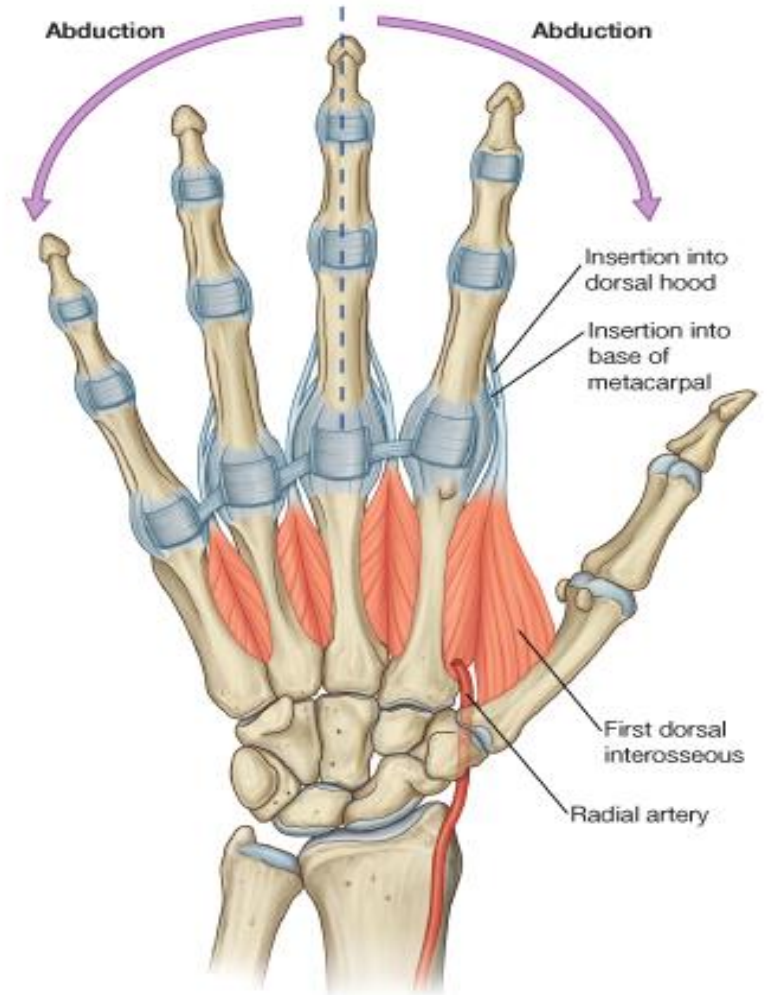


(D) Dorsal interossei (1-4)

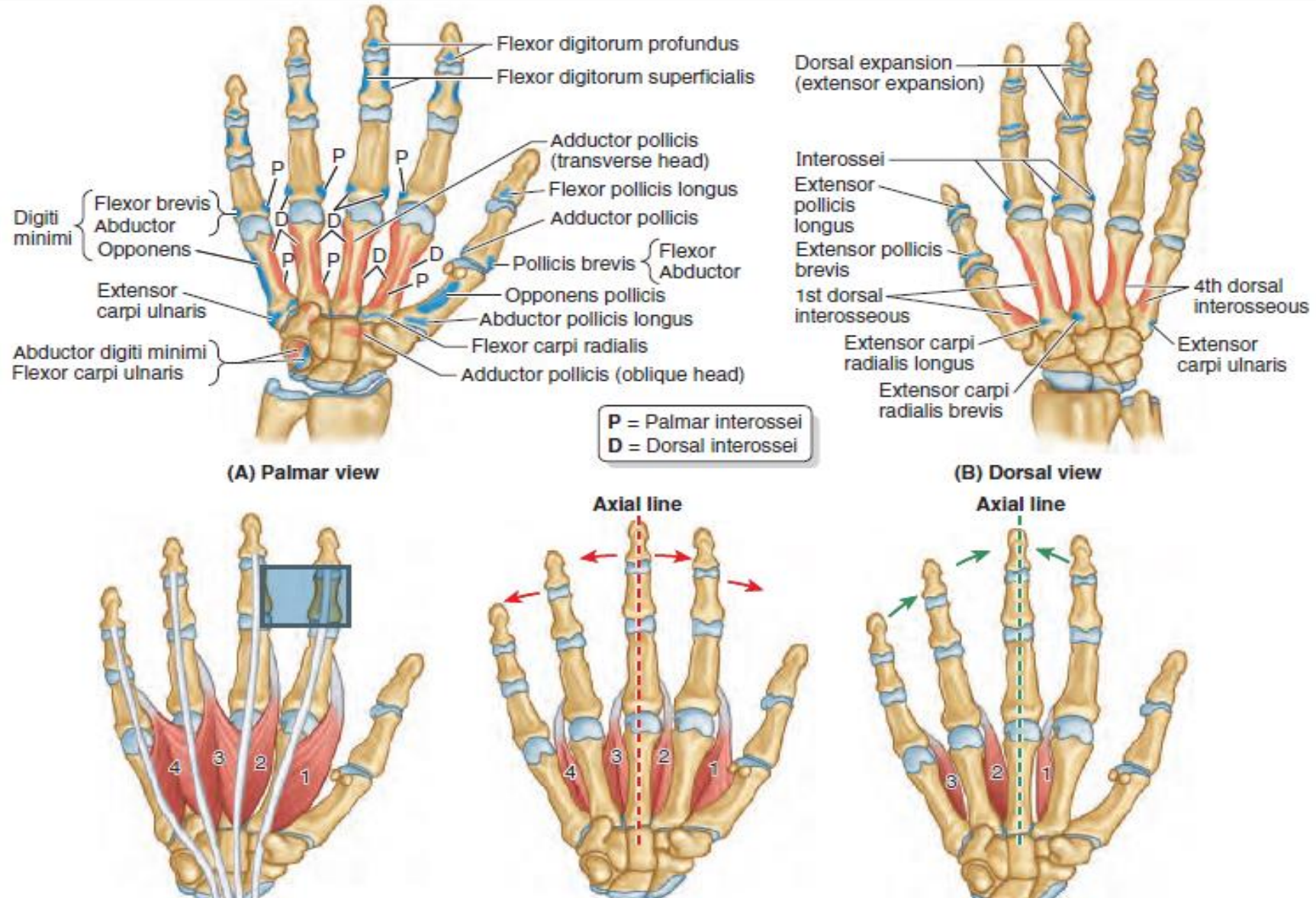


(E) Palmar interossei (1-3)

Palmar views



Short Muscles of the Hand,,



Short Muscles of the Hand,,

2. Interossei

a. Palmar interossei (unipennate)

- are four in number

□ **Origin:** Arise from metacarpal bone of the finger on which it acts

- The 1st is the smallest or abesent
- The 2nd arise from medial side of 2nd metacarpal bone and join the extension expansion of the index finger
- The 3rd arise from the lateral side of 4th metacarpal bone and join the extension expansion of the ring finger
- The 4th arise from lateral side of 5th metacarpal bone and join the extension expansion of the little finger.

Short Muscles of the Hand,,

Action:

The palmar interossei adduct the finger to the midline of the middle finger

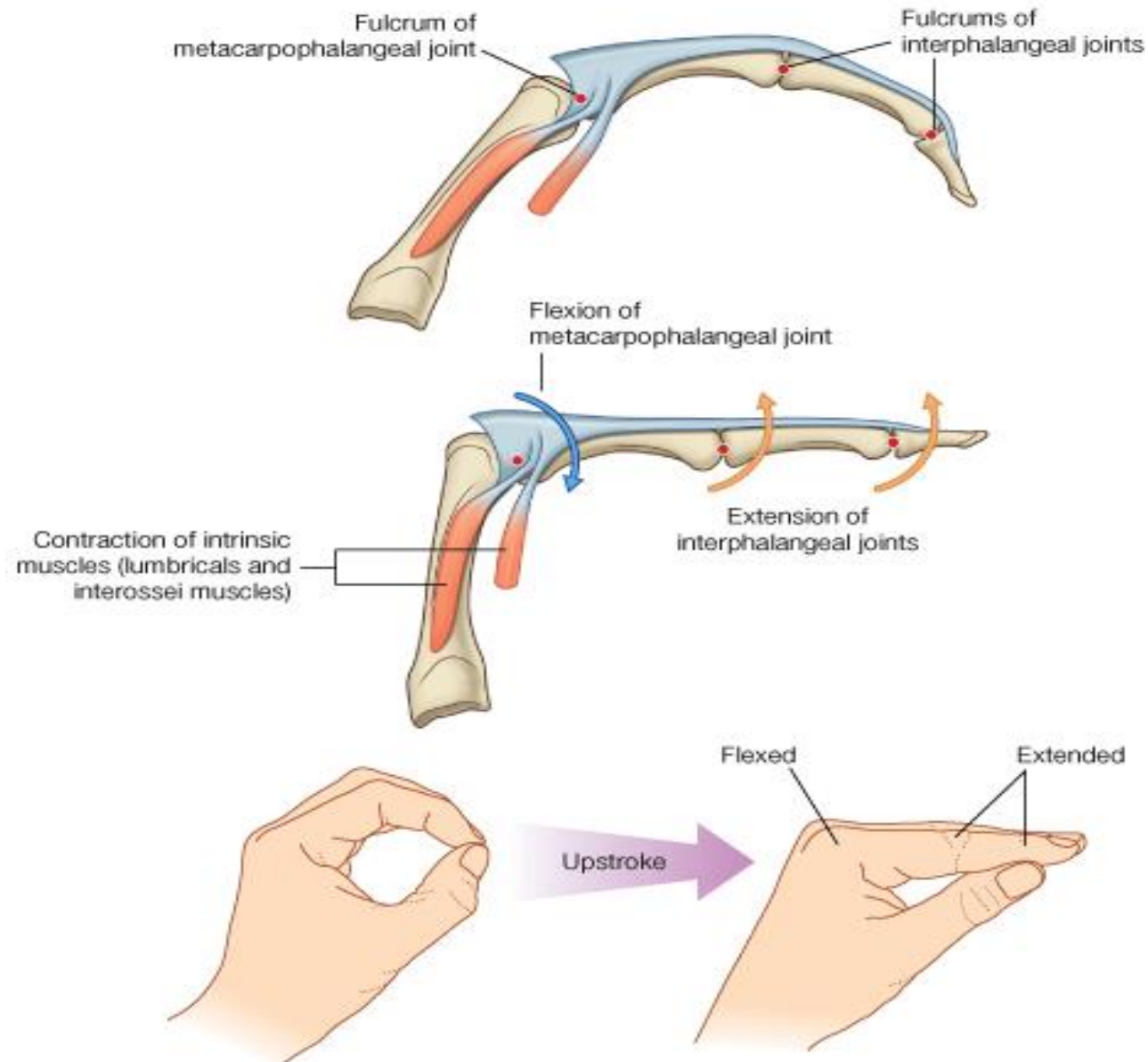
The dorsal interossei abduct the finger from the midline of the middle finger

Action of lumbricals and interossei

- Both flex the metacarpophalangeal and extend the interphalangeal joint
- Thus the action of these muscles put the hand in the **writing position**

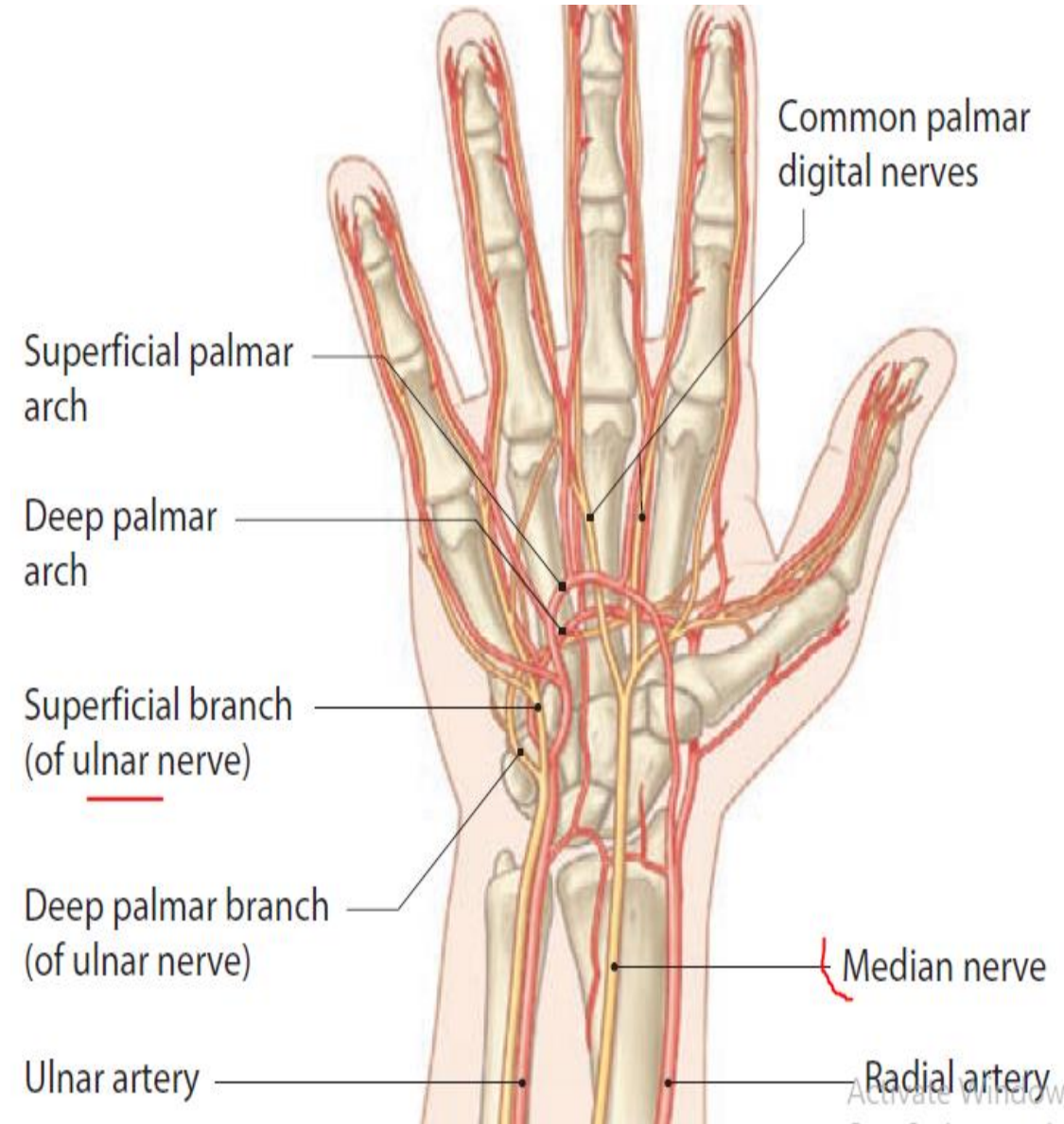


Short Muscles of the Hand,,,



Nerves of the Hand

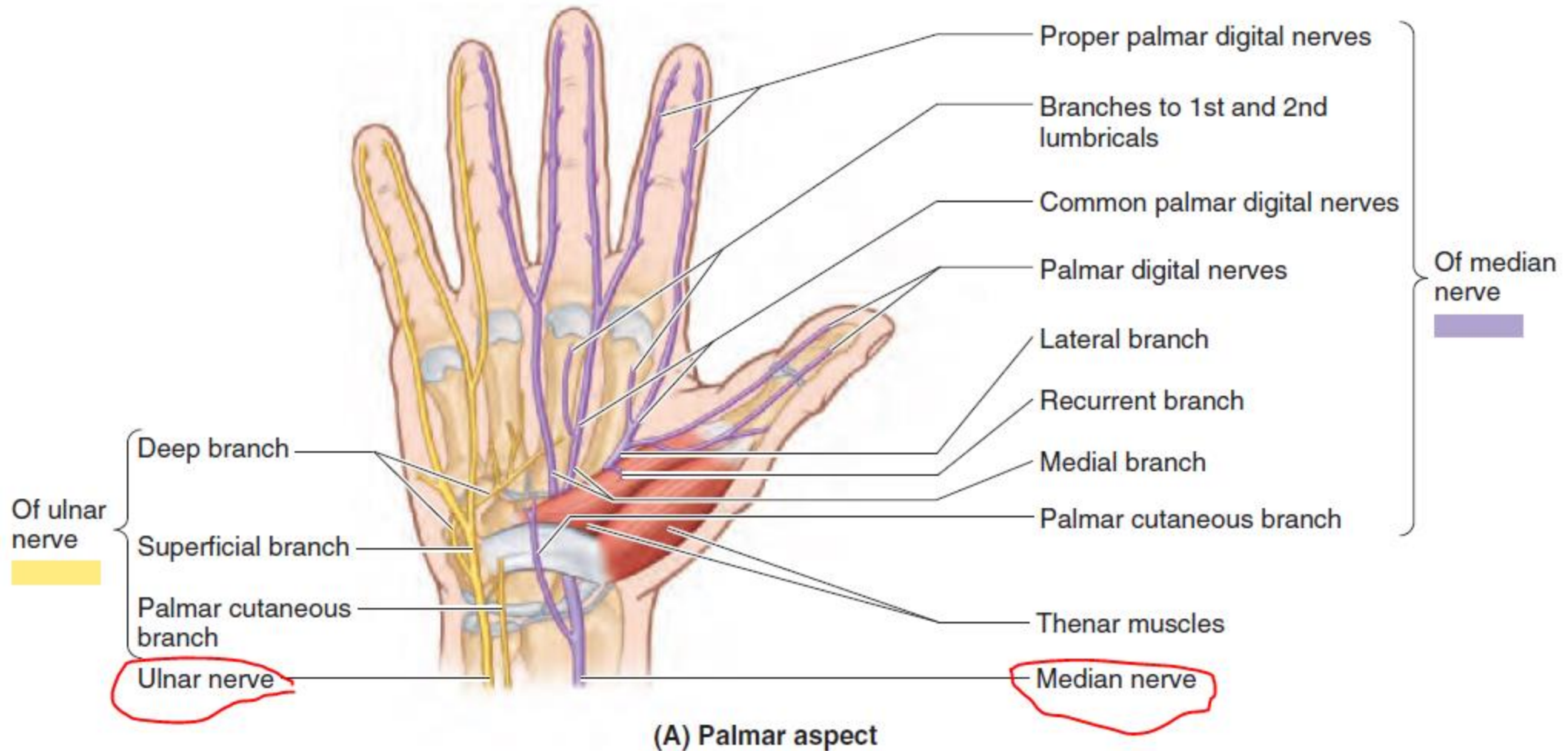
- The **Median**, **Ulnar**, and **Radial** nerves supply the hand
- In addition, branches or communications from the lateral and posterior cutaneous nerves may contribute some fibers that supply the skin of the dorsum of the hand.



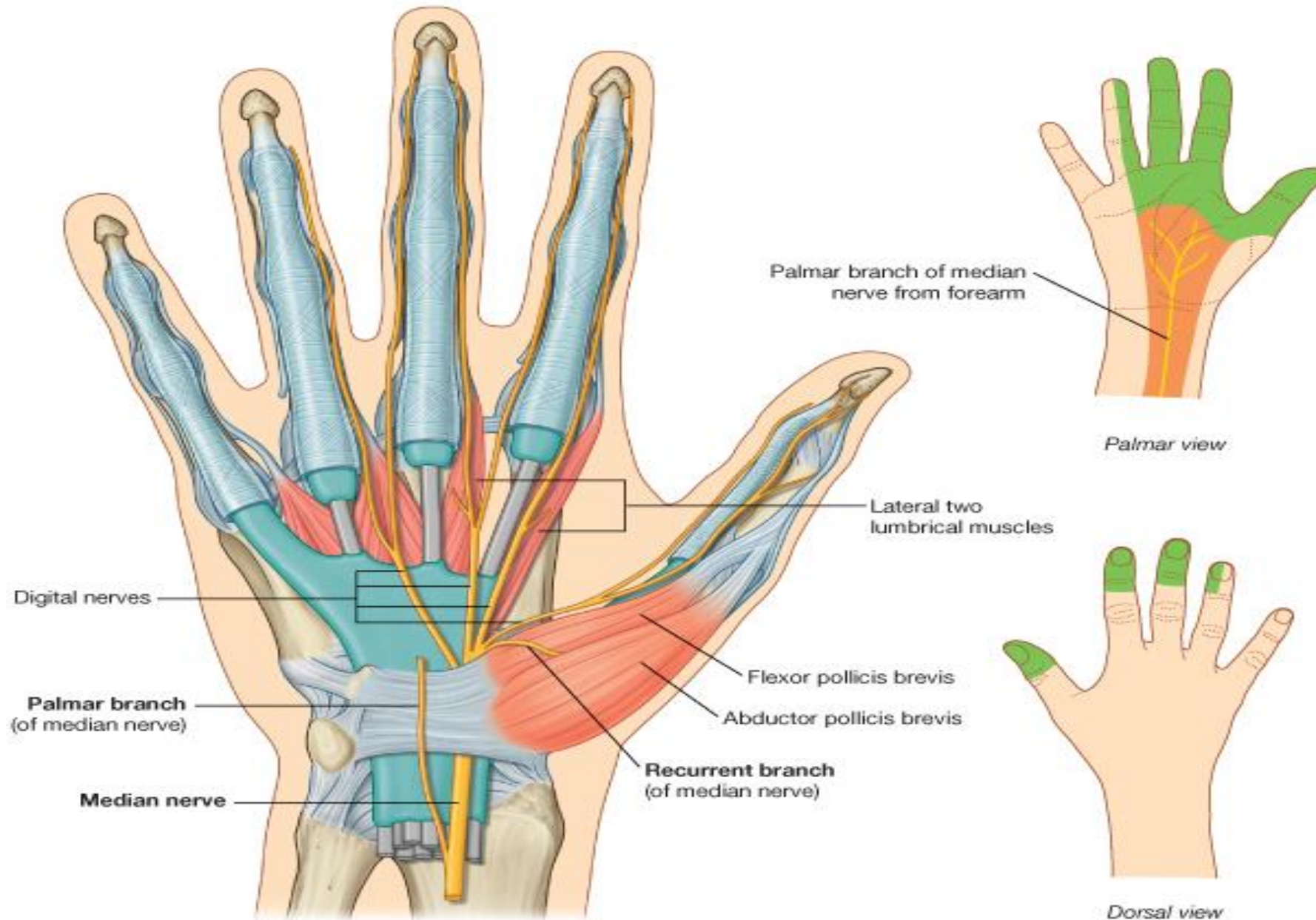
The Median Nerve in the Hand

- The median nerve enters the hand through the carpal tunnel, deep to the flexor retinaculum, along with the nine tendons of the FDS, FDP, and FPL
- Distal to the carpal tunnel, the median nerve supplies two and a half thenar muscles and the 1st and 2nd lumbricals.
- It also sends sensory fibers to the skin on the entire palmar surface, the sides of the first three digits, the lateral half of the 4th digit & the dorsum of the distal halves of these digits
- Note, however, that the palmar cutaneous branch of the median nerve, which supplies the central palm, arises proximal to the flexor retinaculum and passes superficial to it.

The Median Nerve in the Hand,,

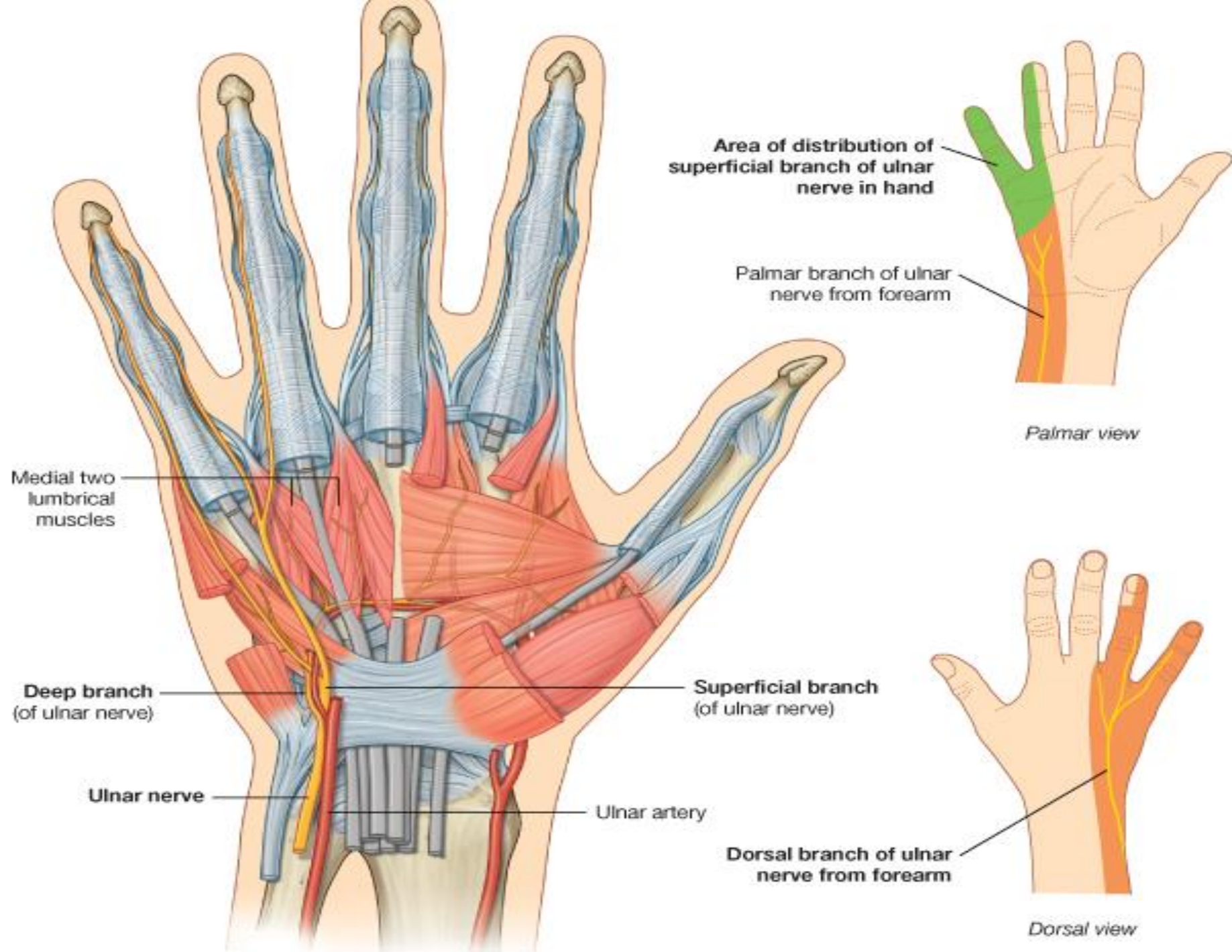


The Median Nerve in the Hand,,



The Ulnar Nerve in the Hand

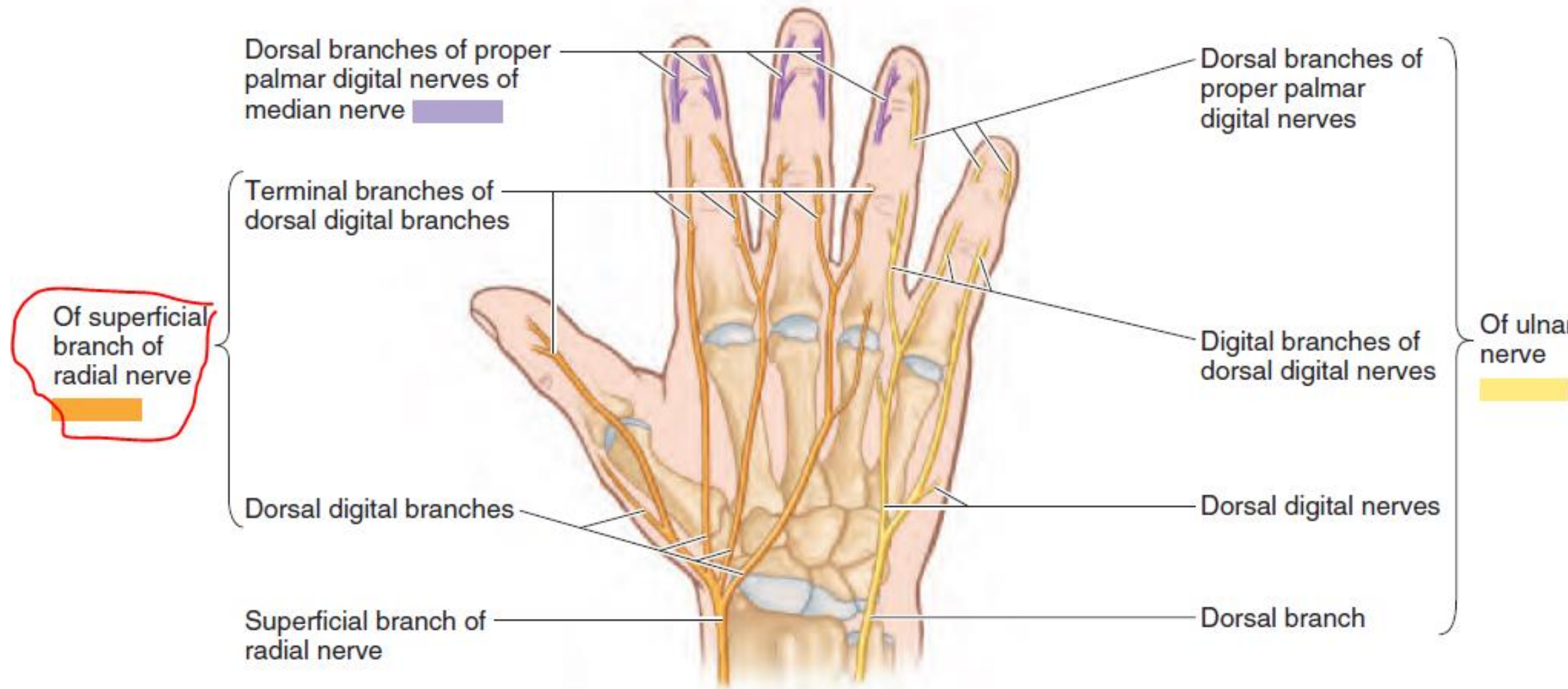
- The ulnar nerve ends at the distal border of the flexor retinaculum by dividing into: **superficial and deep branches**
- **The superficial branch-** supplies cutaneous branches to the anterior surfaces of the medial one and a half fingers
- **The deep branch** – supplies the hypothenar muscles, the medial two lumbricals, the adductor pollicis, the deep head of the FPB, and all the interossei
- The ulnar nerve is often referred to as **the nerve of fine movements** because it innervates most of the intrinsic muscles that are concerned with intricate hand movements.



Radial Nerve in the Hand

- The radial nerve supplies no hand muscles
- The superficial branch of the radial nerve is entirely sensory
- Supplies the skin and fascia over the lateral two thirds of the dorsum of the hand, the dorsum of the thumb, and proximal parts of the lateral one and a half digits.
- The **radial nerve** crosses the dorsal surface of the wrist into the hand and conveys sensation to the skin over most of the medial portion of the dorsum of the hand.
- **It does not innervate any hand muscles.**

Radial Nerve in the Hand,,

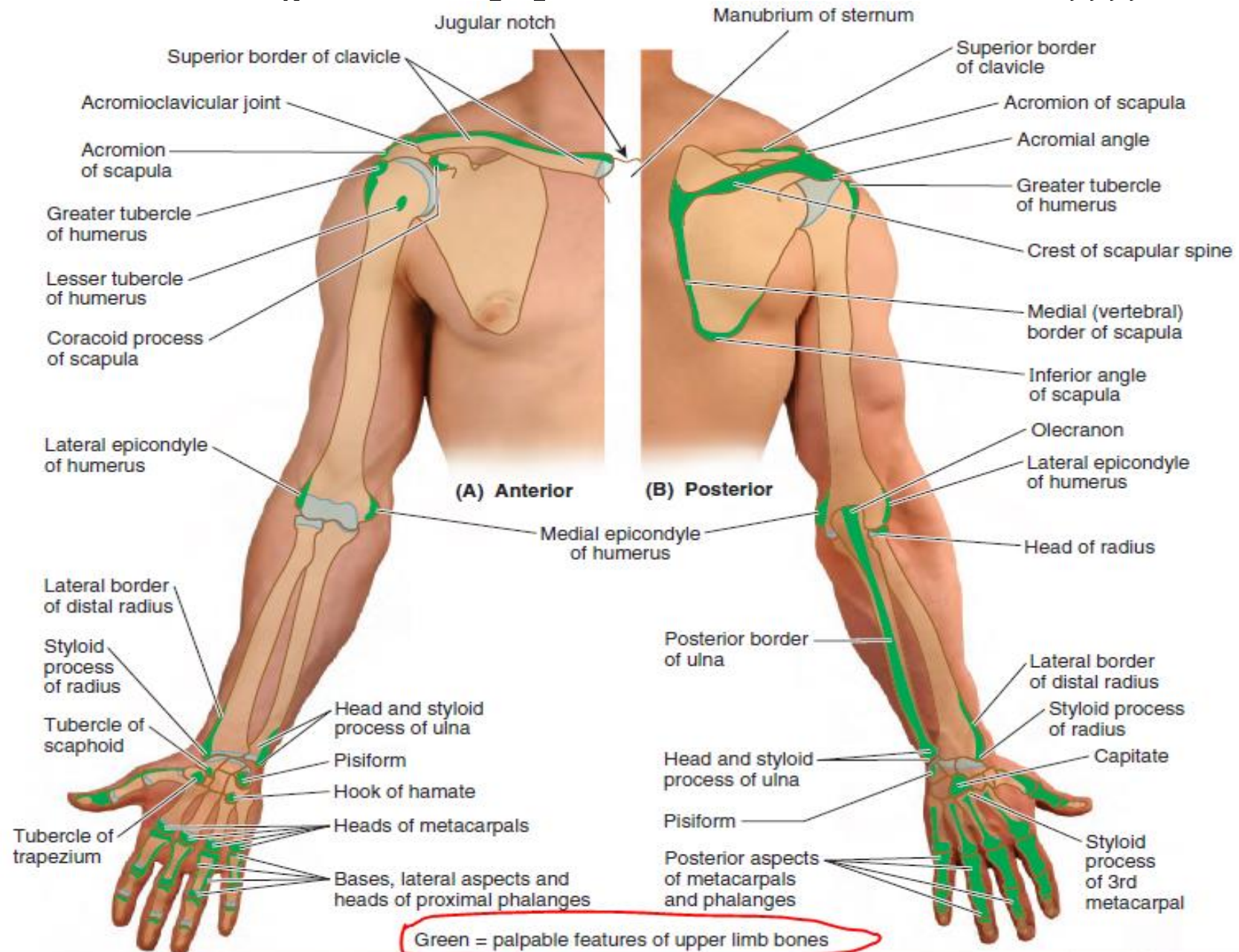


(B) Dorsal aspect

Surface Anatomy of Upper Limb Bones

- Most bones of the upper limb offer a palpable segment or surface.
- ❖ **The *clavicle*** is subcutaneous and can be easily palpated throughout its length.
Its sternal end projects superior to the manubrium.
- Between the elevated sternal ends of the clavicles is the **jugular notch** (suprasternal notch).
- The acromial end of the clavicle often rises higher than the acromion, forming a palpable elevation at the *acromioclavicular (AC) joint*.
- The acromial end can be palpated 2–3 cm medial to the lateral border of the acromion, particularly when the arm is alternately flexed and extended.

Surface Anatomy of Upper Limb Bones,,,



Surface Anatomy of Upper Limb Bones,,,

- The *acromion* of the scapula is easily felt and often visible, especially when the deltoid contracts against resistance.
- The superior surface of the acromion is subcutaneous and may be traced medially to the AC joint.
- The lateral & posterior borders of the acromion meet to form the **acromial angle**.
- The humerus in the glenoid cavity and the *deltoid muscle* form the rounded curve of the shoulder.
- The **crest of the scapular spine** is subcutaneous throughout and easily palpated.

Surface Anatomy of Upper Limb Bones,,,

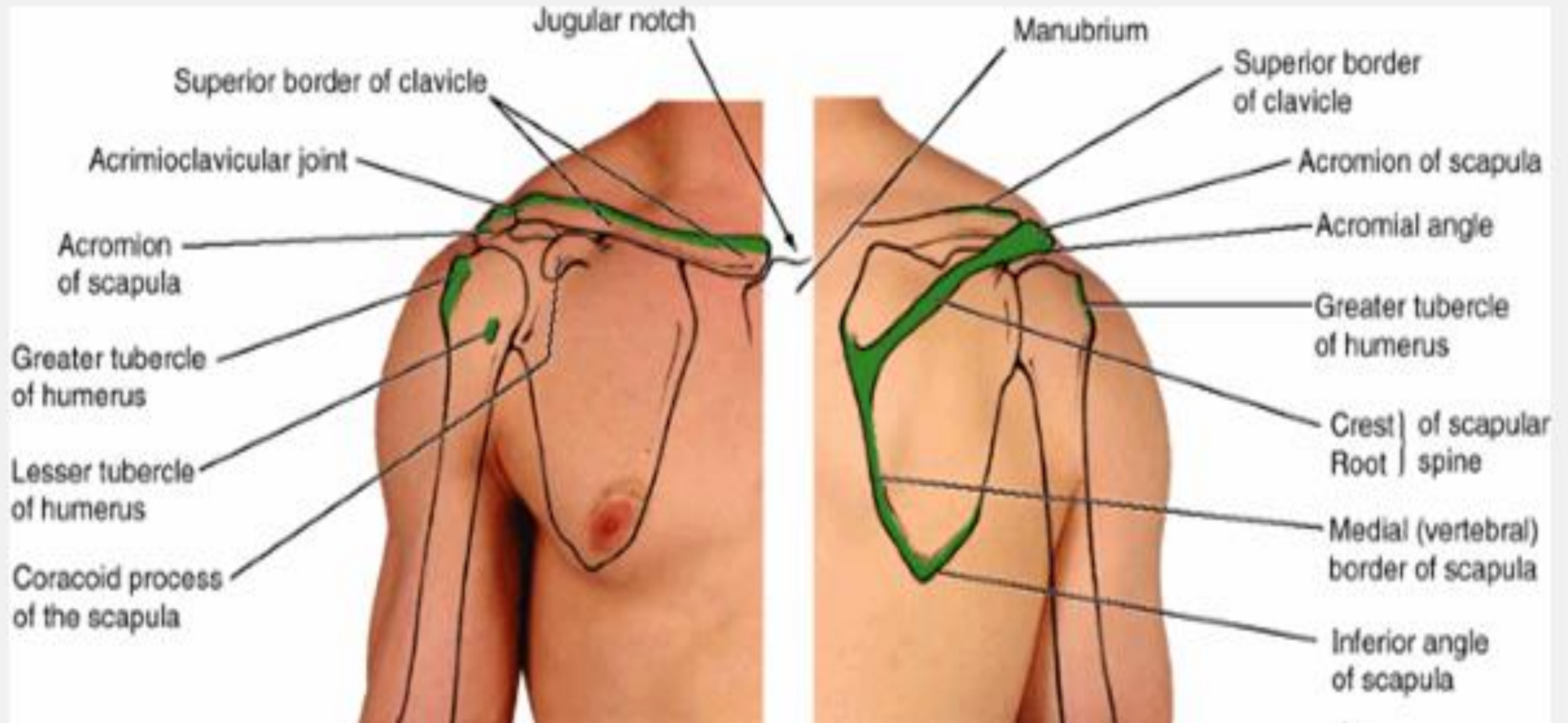
❑ When the upper limb is in the anatomical position, the:

- **Superior angle** of the scapula lies at the level of the **T2** vertebra.
- **Medial end** of the root of the scapular spine is **opposite** the spinous process of the **T3** vertebra.
- **Inferior angle** of scapula: lies at level of **T7** vertebra, near inferior border of **7th** rib & **7th** intercostal space. is palpated when we test movements of the **glenohumeral joint** to immobilize the scapula.

❖ The *medial border of the scapula* is palpable inferior to the root of the spine of the scapula as it crosses the 3rd–7th ribs.

❖ The **lateral border of the scapula** is not easily palpated because it is covered by the **teres major and minor muscles**.

Surface Anatomy of Upper Limb Bones,,,



Surface Anatomy of Upper Limb Bones,,,

- The *head of the humerus* is surrounded by muscles, except inferiorly; consequently, it can be palpated only by pushing the fingers well up into the *axillary fossa* (armpit).
- The *greater tubercle of the humerus* may be felt with the person's arm by the side on deep palpation through the deltoid.
- The *lesser tubercle of the humerus* may be felt with difficulty by deep palpation through the deltoid on the anterior aspect of the arm.
- The *shaft of the humerus* may be felt with varying distinctness through the muscles surrounding it.

Surface Anatomy of Upper Limb Bones,,,,

- The medial and lateral epicondyles of the humerus are subcutaneous and easily palpated on the medial and lateral aspects of the elbow region.
- The *olecranon* of the ulna can be easily palpated
- The *posterior border of the ulna*, palpable throughout the length of the forearm, demarcates the posteromedial boundary between the flexor–pronator and the extensor–supinator compartments of the forearm.
- The *head of the ulna* forms a large, rounded subcutaneous prominence that can be easily seen and palpated on the medial side of the dorsal aspect of the wrist, especially when the hand is pronated.

Surface Anatomy of Upper Limb Bones,,,

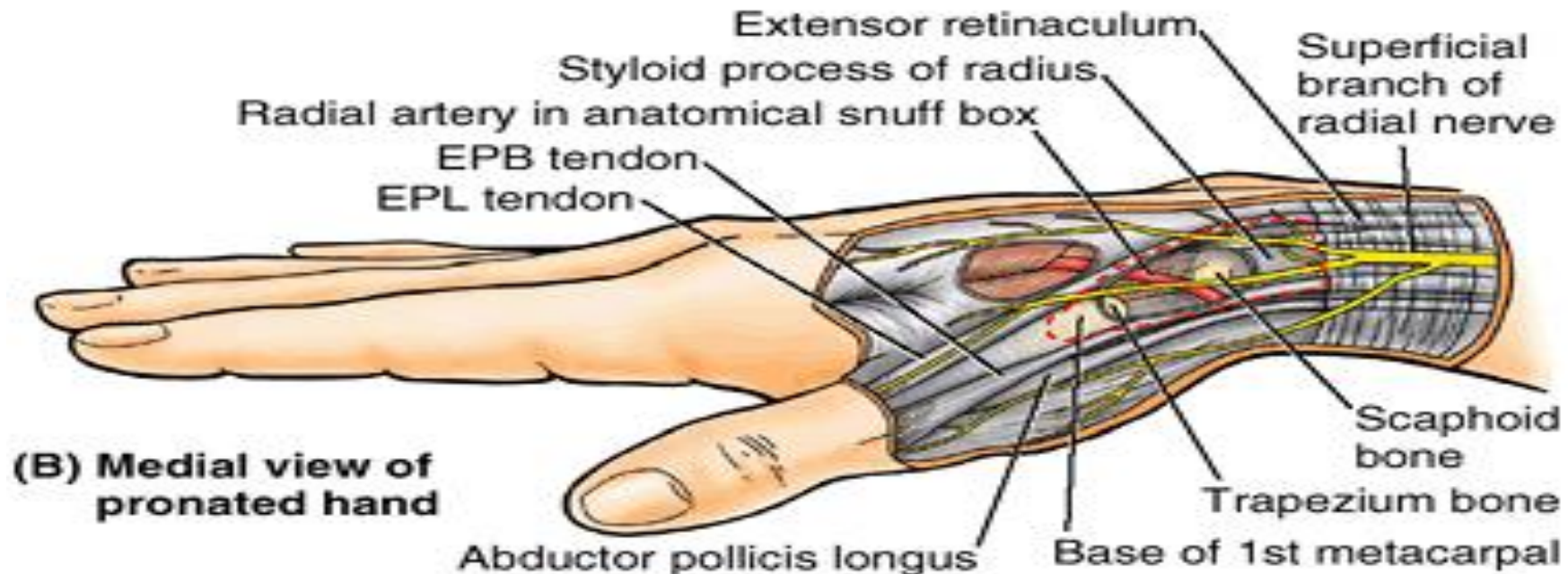
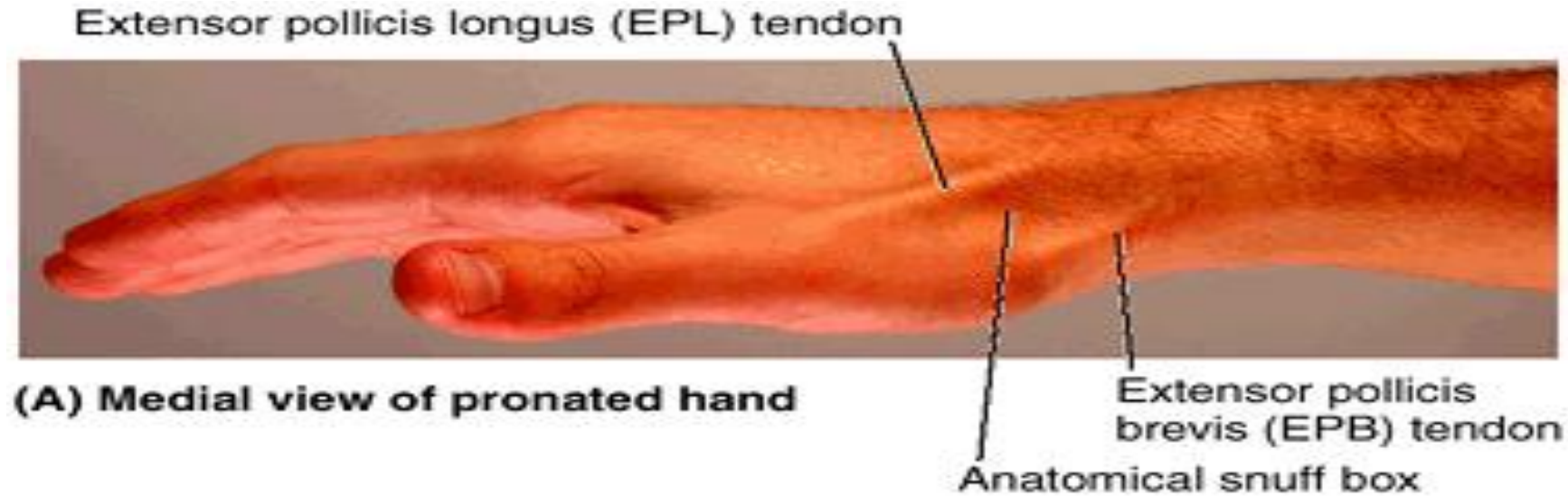
- The pointed subcutaneous *ulnar styloid process* may be felt slightly distal to the rounded ulnar head when the hand is supinated.
- The *head of the radius* can be palpated and felt to rotate in the depression on the posterolateral aspect of the extended elbow joint, just distal to the lateral epicondyle of the humerus.
- The radial head can also be palpated as it rotates during pro nation and supination of the forearm.
- The *ulnar nerve* feels like a thick cord where it passes posterior to the medial epicondyle of the humerus; pressing the nerve here evokes an unpleasant “funny bone” sensation.

Surface Anatomy of Upper Limb Bones,,,

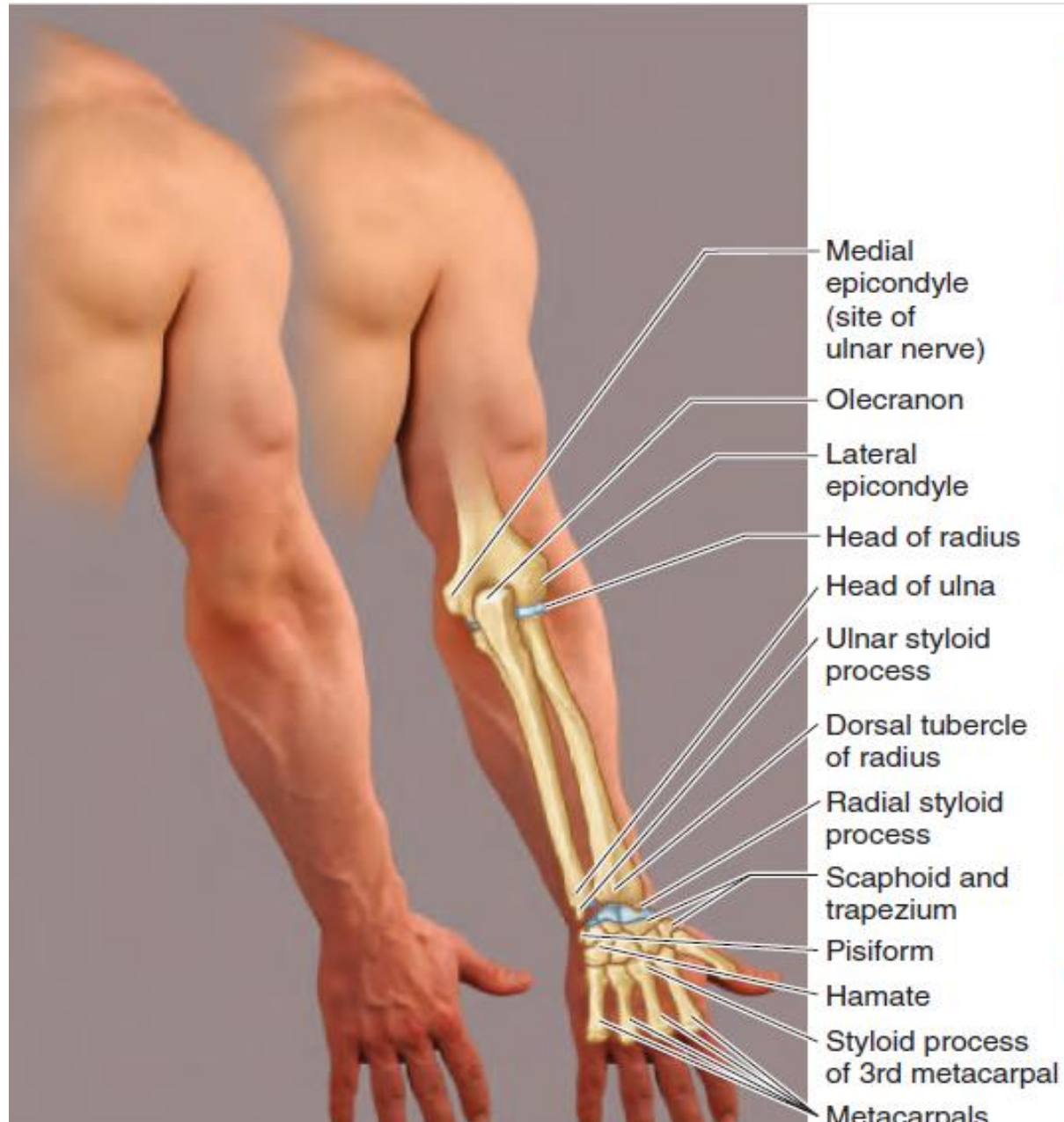
❑ **Anatomical snuff box:** When the thumb is extended, a triangular hollow appears between the tendon of the extensor pollicis longus (EPL) medially and the tendons of the extensor pollicis brevis (EPB) and abductor pollicis longus (APL) laterally

- The floor of the snuff box, formed by the scaphoid and trapezium bones, is crossed by the radial artery as it passes diagonally from the anterior surface of the radius to the dorsal surface of the hand.
- The *radial styloid process* can be easily palpated in the *anatomical snuff box* on the lateral side of the wrist
- It is larger and approximately 1 cm more distal than the ulnar styloid process
- The radial styloid process is easiest to palpate when the thumb is abducted

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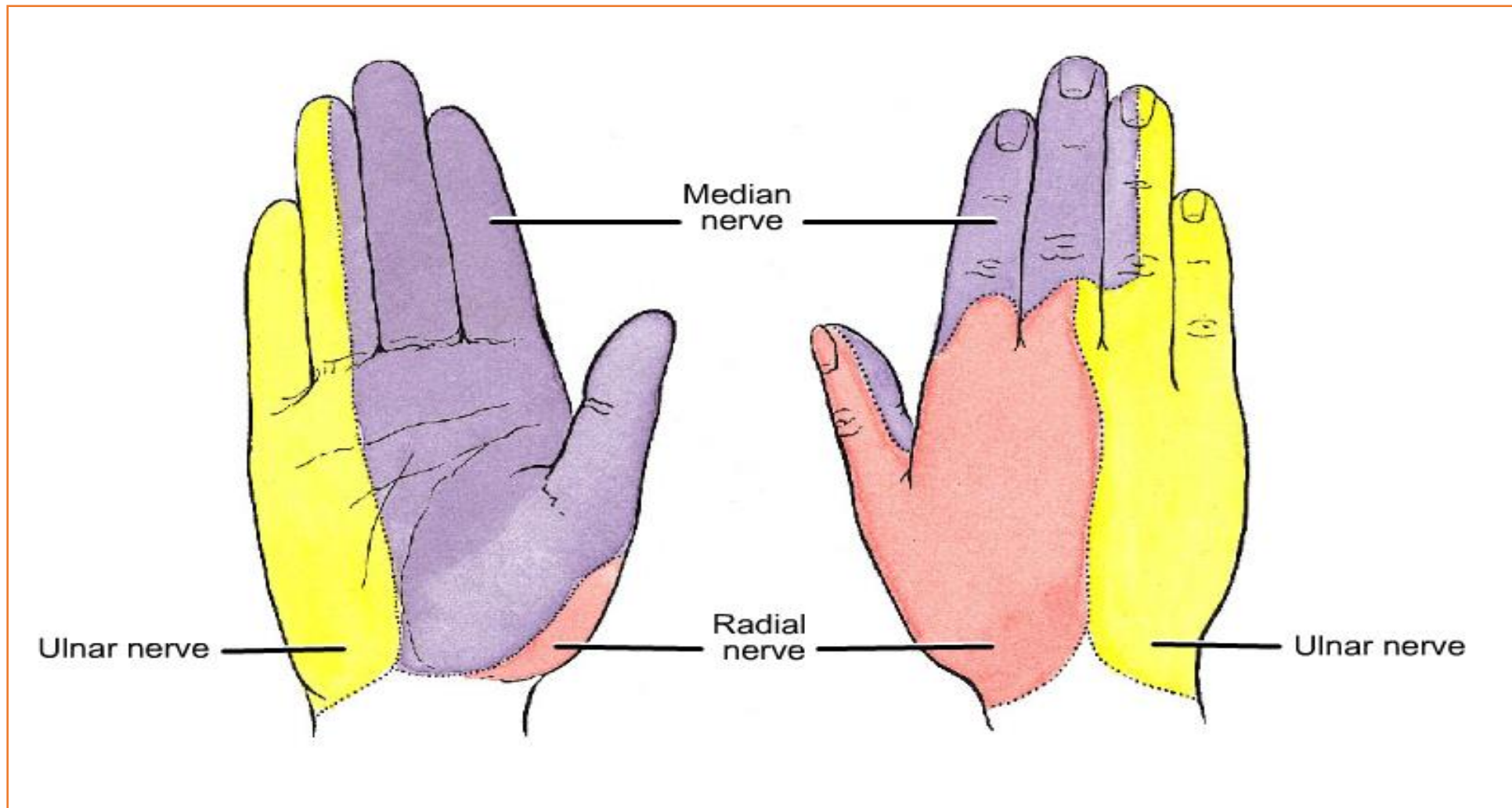
Surface Anatomy of Upper Limb Bones,,,,



Posterior view

(B) Extension: epicondyles and olecranon aligned during extension





Thank you!!